Bridging Perception and Reasoning: A Comprehensive Survey of Causal Representation Learning and Discovery in Deep Learning

1 Foundations, Taxonomies, and Theoretical Underpinnings

1.1 Formal Definitions and Extensions of Structural Causal Models

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Structural Causal Models (SCMs), also widely recognized as structural equation models (SEMs), constitute the predominant mathematical framework for formalizing causal relationships in artificial intelligence and statistics. At their core, SCMs define a system of variables where each endogenous variable is determined by a structural function of its direct causes and an exogenous noise term. Traditionally, the literature has focused heavily on recursive or acyclic SCMs, which generalize causal Bayesian networks to allow for latent confounders while maintaining a Directed Acyclic Graph (DAG) structure. In these standard formulations, the absence of cycles ensures that the model induces unique observational, interventional, and counterfactual distributions, satisfying convenient Markov properties that facilitate efficient inference and discovery. However, the restrictive assumption of acyclicity often fails to capture the complexity of real-world dynamical systems characterized by feedback loops, equilibrium states, and deterministic dependencies. These limitations necessitate a broader theoretical foundation to support the rigorous reasoning required by the Causal Hierarchy discussed in the following section.

To address the limitations of acyclic models, recent theoretical advancements have extended the SCM framework to accommodate cyclic structures. It has been demonstrated that in the presence of cycles, many standard properties of acyclic SCMs do not hold universally; specifically, cyclic models may lack solutions, fail to induce unique distributions, or violate the Markov property relative to their graphs. To resolve these issues, researchers have introduced the class of "simple SCMs," which extends the acyclic setting to the cyclic domain while preserving essential causal semantics under specific solvability conditions. This generalization provides the foundations for a comprehensive theory of statistical causal modeling that includes both latent confounders and feedback mechanisms, ensuring that marginalization and intervention operations remain well-defined even in complex non-recursive systems [1]. Furthermore, the development of input/output SCMs (ioSCMs) has enabled the proof of main rules of causal calculus, including generalizations of the backdoor and selection-backdoor criteria, for systems containing cycles, latent confounders, and arbitrary probability distributions [2].

A critical extension involves the modeling of deterministic relations within causal structures, a factor that significantly impacts identifiability—a central theme of the subsequent hierarchical analysis. Standard causal discovery algorithms often assume that every observed variable is associated with a distinct source of noise with non-zero variance, thereby precluding deterministic dependencies. However, realistic systems frequently exhibit variables that are deterministic functions of other observed variables or latent confounders. Extensions to linear SCMs have been proposed to handle such scenarios, deriving necessary and sufficient conditions for the unique identifiability of causal structures even when deterministic relations and latent confounding coexist. These advancements allow for more accurate modeling of information propagation and influence in systems where strict stochastic independence does not hold [3].

Beyond static equilibrium systems, the formal definition of SCMs has been expanded to explicitly incorporate time-dependence through Structural Dynamical Causal Models (SDCMs). SDCMs represent dynamical systems as collections of stochastic processes governed by structured systems of random differential equations of arbitrary order. In this framework, the static random variables of traditional SCMs are replaced by dynamic stochastic processes and their derivatives, allowing for a rigorous causal interpretation of systems evolving over time. The theory of SDCMs establishes conditions under which these dynamic models equilibrate to static SCMs as time tends to infinity, thereby bridging the gap between dynamic process modeling and static causal inference. This correspondence enables the application of established statistical tools for SCMs to a broad class of stochastic dynamical systems, provided the appropriate graphical representations and Markov properties for initial conditions are respected [4]. Related work has also explored the representation of dynamical systems with infinitely many variables, introducing Generalized SEMs (GSEMs) that naturally specify the results of interventions in systems described by differential equations, overcoming the limitations of naively extending finite-variable frameworks [5].

For systems operating at equilibrium, particularly those described by differential equations or functional laws like the ideal gas law, standard SCMs may still prove insufficiently flexible. To address this, the concept of Causal Constraints Models (CCMs) has been proposed as a generalization of SCMs. CCMs are designed to capture the causal semantics of dynamical systems at equilibrium by utilizing constraints rather than solely functional assignments. This framework allows for the construction of causal models directly from differential equations and initial conditions, providing a sensible and intuitive way to model functional laws and ubiquitous biochemical reactions that defy standard acyclic or even simple cyclic SCM representations [6]. Complementary research has elucidated the conditions under which the equilibrium states of first-order Ordinary Differential Equation (ODE) systems can be described by deterministic SCMs, shedding light on the concept of causality within cyclic and deterministic contexts [7].

Finally, the theoretical landscape includes efforts to unify different causal frameworks and handle selection bias, further solidifying the groundwork for assessing identifiability limits. Recent work has formalized conditioning operations on SCMs to model latent selection, demonstrating that such operations can transform an SCM with explicit selection mechanisms into one without, while preserving simplicity and commutativity with marginalization. This allows classical causal inference results to be generalized to include selection bias, a ubiquitous challenge in real-world data [8]. Additionally, comparative analyses have clarified the relationship between the Rubin Causal Model (Potential Outcomes) and SCMs, showing that every Potential Outcomes model can be viewed as an abstraction of a representable SCM, thus providing a conciliatory perspective that leverages the algebraic constraints of graphs to substantiate comparisons between frameworks [9]. Together, these extensions form a robust and versatile theoretical foundation, enabling causal reasoning across a spectrum of complexities ranging from static acyclic networks to dynamic, cyclic, and deterministic systems, thereby setting the stage for understanding the specific boundaries of what can be inferred at each rung of the causal ladder.

1.2 The Causal Hierarchy and Identifiability Limits

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Building upon the versatile theoretical foundations of Structural Causal Models (SCMs) detailed in the previous section—ranging from acyclic networks to dynamic, cyclic, and deterministic systems—the conceptual framework governing modern causal inference is best articulated through Pearl's Ladder of Causation. This hierarchy delineates three distinct levels of reasoning: association (seeing), intervention (doing), and counterfactuals (imagining). It represents a fundamental stratification of inferential capabilities, where each rung requires assumptions and data types that cannot be derived solely from the levels below. The first rung, association, deals with passive observation and the estimation of conditional probabilities, such as $P(Y|X)$, characterizing statistical dependencies found in observational data. While powerful for prediction, this level is inherently limited in its ability to answer questions about the effects of actions or hypothetical scenarios. Moving to the second rung, intervention, involves reasoning about the consequences of external manipulations, formalized by the *do*-operator as $P(Y|do(X))$. This level allows agents to predict the outcome of policies and experiments, distinguishing causal effects from spurious correlations induced by confounding variables. The third and highest rung, counterfactuals, addresses retrospective reasoning about what would have happened had circumstances been different, represented as $P(Y_x|X = x', Y = y')$. This level is crucial for understanding individual responsibility, fairness, and root cause analysis, yet it poses the most stringent requirements for identification.

A central tenet of this framework is the Causal Hierarchy Theorem, which rigorously establishes that data collected at a lower rung of the ladder is insufficient to answer queries belonging to a higher rung without additional structural assumptions. Specifically, observational data alone cannot uniquely determine interventional distributions, and interventional data alone cannot uniquely determine counterfactual probabilities. This theoretical boundary implies that crossing from one level to the next necessitates the incorporation of an SCM or equivalent graphical knowledge that encodes the underlying mechanisms of the data-generating process, leveraging the extended definitions discussed previously. Recent literature has emphasized the intricacies and challenges of applying this hierarchy to complex, high-dimensional domains such as computer vision, where the lack of ground truth benchmarks often hinders the validation of causal claims across these layers [10]. Furthermore, the formalization of these three tiers has been extended into probabilistic logical languages of increasing expressivity, demonstrating that while satisfiability and validity are decidable for each language, the computational resources required grow with the complexity of the causal query [11].

The limits of identifiability are particularly acute when attempting to infer Layer 3 counterfactuals from Layer 1 or Layer 2 data. Traditional approaches often rely on the consistency rule, which posits that the potential outcome under the actual treatment equals the observed outcome. However, recent theoretical advancements have identified a "degenerative counterfactual problem" arising from this rule within standard Potential Outcomes and SCM frameworks, limiting their capacity to model heterogeneity and personalized incentives effectively [12]. To address these limitations, novel frameworks such as Distribution-consistency Structural Causal Models (DiscoSCMs) have been proposed. These models embrace heterogeneity by incorporating unit selection variables and independent counterfactual noises, thereby resolving degeneracy issues and enabling the derivation of heterogeneous counterfactual bounds that are inaccessible to conventional methods [13].

Despite these theoretical hurdles, significant progress has been made in developing algorithms capable of navigating the hierarchy under specific conditions. For instance, the identification of nested counterfactuals—essential for mediation analysis and fairness metrics—has been advanced through the Counterfactual Unnesting Theorem (CUT). This theorem allows arbitrary nested counterfactuals to be mapped to unnested forms, accompanied by a sufficient and necessary graphical condition for identification from arbitrary combinations of observational and experimental distributions [14]. Moreover, the computational complexity of counterfactual reasoning has been shown to be tractable under certain structural constraints; specifically, if associational or interventional reasoning is tractable on a fully specified SCM, then counterfactual reasoning is also tractable, provided the causal treewidth of the twin network remains bounded [15].

In practical applications involving high-dimensional data, such as medical imaging, the assumption of accurately estimating the full probability distribution is often impractical. To bridge this gap, modular learning strategies have been introduced that leverage pre-trained deep generative models. These approaches, such as Modular-DCM, utilize adversarial training to construct deep causal generative models capable of sampling from identifiable interventional and counterfactual distributions, even in the presence of latent confounders [16]. Similarly, diffusion models have emerged as a powerful tool for causal representation learning, where latent encodings serve as proxies for exogenous noise, enabling direct sampling under interventions and abduction for counterfactuals in causally sufficient settings [17].

However, the challenge of identifiability persists when the underlying structural equations are unknown or only partially specified. In scenarios involving categorical variables, the non-uniqueness of causal mechanisms from observational data raises concerns about the trustworthiness of counterfactual inference. Methods like deep twin networks have been developed to enforce counterfactual ordering principles, ensuring that the learned mechanisms satisfy desirable functional constraints and yield reliable counterfactual probabilities [18]. Furthermore, when full identification is impossible, researchers have turned to bounding techniques. For example, causal Expectation-Maximization algorithms can reconstruct uncertainty about latent variables to provide credible intervals and bounds for counterfactual queries, acknowledging that precise point estimates may be unattainable without stronger assumptions [19].

The distinction between correlation and causation becomes increasingly critical as one ascends the ladder. While observational studies can reveal associations, they often fail to distinguish genuine causal processes from temporal correlations or homophily, as seen in social influence analysis. Principled causal approaches that partition data to avoid Simpson's paradox and apply constrained maximum likelihood estimation are necessary to retrieve genuine causal arcs [20]. Ultimately, the Causal Hierarchy serves as a rigorous guide for determining what can be learned from available data, highlighting that while deep learning offers powerful tools for approximating distributions, the leap to causal reasoning requires explicit modeling of mechanisms. This necessity sets the stage for the following section, which introduces the principle of Independent Causal Mechanisms (ICM) as the constructive assumption required to overcome these inherent identifiability limits and enable the discovery of causal structure from data.

1.3 Key Principles: Independent Causal Mechanisms and Invariance

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While the Causal Hierarchy outlined in the previous section establishes the theoretical limits of what can be inferred from data, the principle of Independent Causal Mechanisms (ICM) provides the constructive assumptions necessary to overcome these limits. Standing as a cornerstone postulate in modern causal inference, ICM asserts that the generative processes of real-world data consist of autonomous modules that do not inform or influence one another. Specifically, the mechanism generating the cause is assumed to be independent of the mechanism mapping the cause to the effect. This independence implies that local modifications to one part of the causal system—such as an intervention on a specific variable—should not alter the functional forms or distributions governing other independent mechanisms. Although this principle has driven significant algorithmic advancements, its theoretical underpinnings have historically lacked rigorous statistical formalization, often relying on intuitive arguments regarding algorithmic independence. Recent literature has begun to bridge this gap by providing robust mathematical frameworks that ground ICM in exchangeability, group theory, and information theory, thereby transforming it from a heuristic assumption into a testable and exploitable property for causal discovery.

A pivotal advancement in the statistical formalization of ICM is the development of causal de Finetti theorems. Traditionally, learning causal structure from observational data assumes independent and identically distributed (i.i.d.) samples, a setting where identifiability is fundamentally limited to Markov equivalence classes. To overcome these limits, researchers have turned to data originating from different but related environments. The work titled [21] presents new causal de Finetti theorems that offer the first rigorous statistical formalization of the ICM principle within the context of exchangeable data. These theorems demonstrate that when data is grouped by environment, the independence of causal mechanisms manifests as a specific form of conditional independence across groups. This theoretical breakthrough justifies a broad class of multi-environment learning techniques, showing how causal structures richer than those identifiable from i.i.d. data alone can be recovered by leveraging the variations induced by independent mechanism changes across environments.

Complementing the probabilistic perspective, group-theoretic formulations provide a powerful lens for understanding and generalizing the independence of cause and mechanism. The paper [22] proposes a unified framework where the cause-mechanism relationship is assessed by comparing it against a null hypothesis through the application of random generic group transformations. By framing ICM through group invariance, this approach generalizes various existing methods that interpret independence differently, offering a versatile tool to study the structure of data-generating mechanisms. This geometric perspective suggests that true causal directions exhibit specific invariance properties under transformations that leave the marginal distribution of the cause unchanged, whereas anti-causal directions fail to maintain such invariance. This insight is further operationalized in methods like the Kernel Intrinsic Invariance Measure (KIIM), described in [23], which captures higher-order statistics of conditional distributions to infer causal direction. Unlike earlier norm-based approaches that sacrificed information about the shape of density functions, KIIM leverages the intrinsic deviance of mechanisms, demonstrating that the causal direction is characterized by a stability in the mechanism's form despite variations in the cause's distribution.

The implications of ICM for distinguishing causal direction extend deeply into the domain of algorithmic complexity and Minimum Description Length (MDL). The principle of algorithmic independence posits that the shortest description of the joint probability distribution factorizes most efficiently in the true causal direction. The study [24] closes a critical theoretical gap by deriving a two-part formulation of this postulate in terms of Kolmogorov complexity, directly linking abstract algorithmic independence to practical MDL encodings. The authors prove that this formulation leads, in expectation, to the same inference results as the original postulate, thereby validating the use of MDL as a proxy for causal discovery. Similarly, [25] argues that predicting the effect from the cause should be computationally "simpler" than the reverse, proposing fast criteria based on this complexity asymmetry for both discrete and continuous variables. These complexity-based views reinforce the notion that nature likely chooses causes and mechanisms independently, resulting in an asymmetry that can be detected through compression or modeling efficiency.

Furthermore, the ICM principle has profound implications for robustness and generalization, particularly in the face of distribution shifts. As highlighted in [26], standard supervised learning often fails when test distributions differ from training data because it captures spurious correlations rather than invariant causal mechanisms. By deriving objective functions directly from the ICM principle, neural networks can be trained to focus on relations that remain invariant across environments, effectively ignoring unstable, non-causal features. This approach ensures that the recovered stable relations correspond to true causal mechanisms under specific conditions, enabling models to generalize to unseen scenarios where traditional predictors fail. The utility of ICM is also evident in natural language processing, where [27] demonstrates that the causal direction of data collection processes explains performance disparities in semi-supervised learning and domain adaptation. By categorizing NLP tasks according to their causal direction and validating the ICM principle via minimum description length, the authors show that causal insights can guide more effective modeling choices.

Finally, the principle extends to specialized domains such as time-series analysis and deterministic systems. In [28], the authors postulate that the power spectrum of a cause is uncorrelated with the square of the transfer function of the linear filter generating the effect. This spectral asymmetry, rooted in the independence of cause and mechanism, allows for causal identification even in deterministic systems where noise-based methods fail. Similarly, [29] reinterprets causal inference without relying solely on graphs, using information fields to handle systems with cycles and spurious edges, further illustrating the versatility of independence principles beyond standard DAG-based assumptions. Collectively, these diverse theoretical strands—from de Finetti theorems to group invariance and complexity measures—solidify the ICM principle as a fundamental law governing causal structure. By establishing *how* causal mechanisms can be identified through invariance and independence, this section lays the theoretical groundwork for the subsequent taxonomy, which categorizes the specific variable paradigms and data regimes where these principles are applied.

1.4 Taxonomy of Variable Paradigms and Data Regimes

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Having established the theoretical foundations of Independent Causal Mechanisms (ICM) and invariance in the preceding section, we now turn to the practical landscape where these principles are operationalized. The field of causal discovery at the intersection with deep learning is vast and fragmented, necessitating a unified taxonomy to navigate the diverse methodologies that have emerged. While the previous section elucidated *why* causal structures can be identified through invariance, this section categorizes *how* modern algorithms disentangle cause from effect across different complexities. A rigorous classification system must account for the intrinsic nature of the variables under study, the temporal and environmental structure of the data regimes, and the extent to which prior domain knowledge is integrated. This tripartite framework—spanning variable paradigms, data regimes, and knowledge availability—provides the essential scaffolding for understanding the specific challenges addressed in the subsequent discussion on multi-level modeling.

First, the taxonomy distinguishes methods based on the **variable paradigm**, specifically the distinction between fully observed systems and those obscured by latent confounders or unmeasured factors. Traditional approaches often assume causal sufficiency, yet real-world applications frequently violate this assumption, necessitating the robust mechanisms discussed in Section 1.3 to handle hidden drivers. Recent advancements have shifted focus toward discovering causal structures in the presence of latent variables, where dependencies are generated by unobserved common causes. For instance, specific frameworks have been developed to systematically search for dependency patterns that reveal the necessity of latent variables, ensuring that models do not erroneously attribute correlations to direct causation when a hidden driver exists [30]. Furthermore, the challenge extends to discrete and mixed data types, where nonparametric methods are required to identify globally confounded structures without restrictive parametric assumptions [31]. In scenarios involving high-dimensional observations, such as images or text, the paradigm shifts toward causal representation learning, where the goal is to recover high-level causal variables from low-level, potentially non-invertible mixes [32]. This includes distinguishing between observed causal effects and latent confounders using method-of-moments approaches that leverage independent component analysis [33]. These efforts to recover latent variables form a critical precursor to the hierarchical reasoning explored in Section 1.5, where the relationship between micro-variables and macro-concepts is formalized.

Second, the taxonomy categorizes methods by **data regimes**, differentiating between independent and identically distributed (i.i.d.) data, time-series data, and multi-environment settings. While i.i.d. data remains a standard benchmark, the dynamic nature of many scientific domains necessitates specialized treatment for temporal data, where the invariance of mechanisms over time becomes a key identifiability constraint. Time-series causal discovery is further subdivided into multivariate time series and event sequence analysis, each requiring distinct algorithmic strategies to handle autocorrelation, non-stationarity, and varying time lags [34]. Specific algorithms have been designed to address the complexities of non-stationary and autocorrelated data, optimizing conditioning sets to detect changing modules and lagged relationships that static methods miss [35]. Moreover, the regime of "indefinite data"—such as dialogues or video sources where sample utilization is low and distributional assumptions fail—represents a frontier where traditional causal data definitions break down, requiring novel variational models to learn representations under latent confounding [36]. Beyond temporal dynamics, multi-environment data regimes exploit distribution shifts across different contexts to achieve identifiability, directly leveraging the exchangeability principles described in Section 1.3. Methods leveraging multi-environment data can recover latent causal variables even when intervention targets are unknown, provided there is sufficient diversity in the intervention coverage [37]. This includes amortized approaches that learn to infer causal graphs across samples sharing underlying dynamics, thereby improving efficiency over fitting new models for every new graph [38].

Finally, the taxonomy delineates the spectrum of **prior knowledge availability**, ranging from purely data-driven discovery to knowledge-guided approaches. Purely data-driven methods rely solely on statistical properties, such as conditional independence or score-based optimization, but often struggle with ill-posed problems and assumption violations in real-world datasets [39]. In contrast, knowledge-guided approaches integrate expert constraints, structural priors, or textual information to restrict the search space and enhance accuracy, effectively injecting the kind of structural inductive biases that mimic the independence assumptions of ICM. For example, greedy equivalence search algorithms can be significantly accelerated and improved by incorporating prior knowledge regarding the presence or absence of specific causal edges [40]. Similarly, in temporal domains, large language models (LLMs) are increasingly employed to extract meta-knowledge from textual system descriptions, guiding causal discovery in industrial scenarios where interventional targets are unavailable [41]. This integration allows for the combination of data-driven pattern recognition with human expertise, addressing gaps where purely observational studies fail due to high dimensionality or lack of hypotheses [42]. The spectrum also includes semi-supervised and active learning frameworks where limited interventional data or user feedback refines the causal model, bridging the gap between passive observation and active experimentation [43].

This comprehensive taxonomy underscores that no single algorithm dominates all regimes. Instead, the choice of method depends critically on whether variables are latent, whether data is static or dynamic, and how much domain knowledge can be encoded. By mapping existing literature onto these axes, researchers can better identify suitable methodologies for specific applications, from healthcare and neuroscience to industrial process control. Crucially, this classification highlights the limitations of flat, single-level modeling, particularly when dealing with latent variables and complex data regimes. These limitations motivate the transition to the next section, which explores causal abstraction and multi-level modeling as a means to bridge fine-grained observations with coarse-grained conceptual understanding, ensuring that the causal mechanisms identified are not only statistically sound but also semantically meaningful across different scales of granularity.

1.5 Causal Abstraction and Multi-Level Modeling

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Building upon the taxonomy of variable paradigms and data regimes outlined previously, a critical challenge arises when the causal variables of interest do not exist at a single, fixed level of granularity. Real-world systems are inherently hierarchical, often requiring reasoning that bridges fine-grained micro-variables (such as pixel intensities or neuronal spikes) and coarse-grained macro-concepts (such as object categories or cognitive states). Causal abstraction provides the rigorous theoretical framework necessary to relate Structural Causal Models (SCMs) operating at these different resolutions, ensuring that high-level explanations remain faithful to the underlying low-level mechanistic details. At its core, causal abstraction involves defining a mapping between a low-level SCM and a high-level SCM such that the interventional behavior of the system is preserved across scales. This preservation of interventional consistency is paramount; without it, conclusions drawn from an abstract model may contradict the physical or logical realities of the base system. Early formalizations introduced concepts such as exact transformations and uniform transformations to strictly define when one probabilistic or deterministic causal model can be considered an abstraction of another [44]. These definitions establish a hierarchy where strong abstraction takes into account all potential interventions, ensuring that the macro-model is not merely a statistical approximation but a functionally valid simplification [44].

However, the requirement for exact consistency is often too restrictive for practical applications involving noisy data or approximate scientific laws, particularly in the complex data regimes discussed earlier. Consequently, recent research has extended these theories to accommodate approximate causal abstraction, where the high-level model offers a close but not perfect representation of the low-level dynamics [45]. This extension allows for the quantification of discrepancies between levels, providing a mechanism to assess how uncertainty propagates from microscopic details to macroscopic descriptions. To systematically evaluate these trade-offs, researchers have introduced families of interventional measures designed to quantify both the consistency of the abstraction and the associated information loss [46]. These measures enable practitioners to navigate the tension between model simplicity and fidelity, selecting abstractions that maximize explanatory power while minimizing the distortion of causal effects. Furthermore, the drive to optimize this balance has led to formulations that explicitly incorporate information loss terms into the learning objective, moving beyond simple consistency metrics to find optimal abstractions that are both computationally tractable and scientifically meaningful [47].

A critical challenge in multi-level modeling is the operationalization of these theoretical maps, particularly when the underlying SCMs are unknown or only partially observed—a scenario common in the latent variable paradigms identified in Section 1.4. Innovative approaches have emerged to learn abstraction maps directly from observational and interventional data without assuming complete knowledge of the generative processes. For instance, the Causal Optimal Transport of Abstractions (COTA) framework utilizes a multi-marginal Optimal Transport formulation to enforce do-calculus constraints, effectively learning the alignment between different levels of granularity by minimizing a cost function grounded in interventional information [48]. Similarly, differentiable programming solutions have been developed to jointly learn consistent causal abstractions across multiple interventional distributions, solving the combinatorial sub-problems inherent in mapping variables and their domains simultaneously [49]. These methods represent a significant shift from static, manually defined abstractions to dynamic, data-driven discovery of hierarchical causal structures.

Beyond merely mapping variables, the process of abstraction often involves transforming the causal mechanisms themselves. Traditional marginalization techniques can inadvertently destroy the causal integrity of the resulting model, leading to spurious correlations or invalid interventional predictions—issues that directly complicate the distinction between correlation and causation explored in the subsequent section. To address this, the concept of "consolidation" has been proposed as an alternative to marginalization. Consolidation transforms large-scale SCMs by merging variables and mechanisms in a way that preserves consistent interventional behavior, thereby simplifying complex systems without sacrificing their causal validity [50]. This approach is particularly relevant for scaling causal inference to high-dimensional domains, such as those encountered in deep learning, where the sheer number of variables renders fine-grained modeling infeasible. By clustering variables and refining their domains, neural causal abstractions enable the use of deep learning toolkits to perform identification, estimation, and sampling at varying levels of granularity, effectively bridging the gap between abstract causal theory and practical representation learning [51].

The theoretical underpinnings of causal abstraction have also been enriched by category theory, which provides a natural language for describing compositional transformations between models. By introducing a category of finite interventional causal models, researchers have proven that model transformations and their associated errors possess desirable compositionality properties; specifically, the error of a composite transformation is bounded by the sum of errors from individual steps [52]. This mathematical rigor ensures that iterative abstraction processes remain stable and predictable. Furthermore, the scope of causal abstraction has been expanded to include soft interventions, which assign non-constant functions to variables rather than fixing them to constant values. This generalization allows for a more nuanced analysis of abstract models in scenarios where hard interventions are unrealistic or impossible, guaranteeing that the intervention map between levels maintains a specific and necessary explicit form [53].

Finally, the utility of causal abstraction extends profoundly into the realm of Explainable AI (XAI), where it serves as a foundation for generating faithful, human-intelligible explanations of complex neural networks. By treating a high-level causal model as an abstraction of a low-level neural net, one can rigorously assess the faithfulness of the explanation through interchange interventions [54]. This perspective unifies various XAI methods, such as LIME and causal mediation analysis, as special cases of causal abstraction analysis, providing a coherent theoretical basis for interpreting black-box systems. The ability to find alignments between interpretable causal variables and distributed neural representations, even when neurons play multiple distinct roles, further demonstrates the flexibility and power of modern abstraction techniques in decoding the internal logic of artificial intelligence [55]. Through these advancements, causal abstraction has evolved from a purely theoretical construct into a versatile toolkit for multi-level modeling. By establishing robust bridges between scales, it lays the essential groundwork for distinguishing genuine causal mechanisms from spurious correlational artifacts, a theme that is central to the analysis of complex systems in the following section.

1.6 Distinguishing Correlation from Causation in Complex Systems

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Building upon the multi-level modeling frameworks established in the preceding section, which bridge fine-grained micro-variables and coarse-grained macro-concepts, a fundamental dialectic arises: how to rigorously distinguish genuine causal mechanisms from spurious correlational artifacts within these complex hierarchies. While statistical correlation measures the degree to which variables vary together, it often fails to capture the directional, generative mechanisms that define causality, particularly in systems characterized by high dimensionality, nonlinearity, and dynamic feedback. Reliance on purely correlational patterns in such environments frequently yields brittle models that collapse under distributional shifts. The core distinction lies in the stability of the underlying mechanism: causal relationships are expected to remain invariant across different environments, whereas spurious correlations are contingent on specific data collection conditions. As noted in recent literature, understanding this distinction is crucial because the absence of correlation does not necessarily indicate a lack of causation, particularly in nonlinear dynamic systems where traditional linear metrics fail [56].

A primary obstacle in isolating true causality is the prevalence of spurious correlations, where models latch onto unreliable features that correlate with the target solely due to confounding factors or dataset-specific biases. These associations prevent generalization to unseen environments where historical correlations break down. Research indicates that while spurious correlations are a well-known drawback of data-driven learning, deriving joint indicators to identify them remains an open problem, with many standalone hypotheses failing to outperform simple empirical risk minimization baselines [57]. In natural language processing, for instance, the compositional nature of language implies that feature-label correlations can arise even when the label is invariant to interventions on the feature, necessitating the application of domain knowledge to identify genuine robustness threats [58]. To mitigate these issues, frameworks such as the Latent Causal Invariant Model (LaCIM) have been proposed to separate output-causative factors from those spuriously correlated via confounders, theoretically guaranteeing the disentanglement required for precise inference and robust out-of-distribution performance [59]. Furthermore, counterfactual data augmentation strategies have been developed to explicitly modify causal and non-causal features, forcing models to learn invariant representations rather than relying on shortcuts [60].

Compounding the challenge of confounding is the issue of selection bias, which distorts observed data distributions and can induce misleading causal conclusions. Selection bias occurs when the mechanism determining data observability depends on the variables of interest, effectively creating a collider that opens non-causal paths. Recent work has introduced conditioning operations on Structural Causal Models (SCMs) to formally model latent selection, demonstrating how such operations can transform an SCM with explicit selection mechanisms into one that encodes the causal semantics of the selected subpopulation while preserving acyclicity and linearity [8]. In practical applications like information extraction, long-tailed distributions caused by selection bias often lead to incorrect correlations between entities and labels; counterfactual frameworks have been successfully applied to uncover main causalities and debias predictions in these scenarios [61]. Similarly, in local causal discovery, specific structural patterns such as Y-Structures have been proven sound for predicting causal relations even in the presence of selection bias and potential cycles, offering a robust alternative to standard algorithms that may fail under these conditions [62].

Beyond confounding and selection, the phenomenon of causal emergence presents a unique complexity in distinguishing correlation from causation across different scales of abstraction, directly linking back to the hierarchical concerns of Section 1.5. Causal emergence theory posits that macroscopic scales can sometimes exhibit stronger causal relationships than microscopic scales by reducing noise, a finding validated across over a dozen distinct measures of causation spanning philosophy, statistics, and genetics [63]. This suggests that causal relationships are not merely intrinsic to the finest level of detail but can emerge at higher levels of abstraction, challenging the reductionist assumption that micro-level correlations fully explain macro-level behaviors. Quantitative studies further elucidate that causal emergence arises from the optimization of uncertainty and asymmetry within the causal structure, where coarse-graining strategies can preserve or even enhance causal information compared to fine-grained models [64]. Comprehensive surveys on this topic highlight that the architectures used to identify causal emergence are shared with causal representation learning and world model-based reinforcement learning, suggesting a deep interconnection between emergence, causality, and AI [65].

To rigorously isolate true causal structures amidst these complexities, the principle of invariance across environments has emerged as a powerful methodological pillar, serving as a critical bridge to the integration of domain knowledge discussed in the following section. The premise is that while the distribution of exogenous noises or confounders may vary across different environments, the functional relations producing variables from their direct causes remain constant. Algorithms leveraging this regression invariance have been proven to identify causal structures more reliably than those relying solely on conditional independence tests, particularly in multi-environment settings [66]. This approach allows researchers to distinguish between mechanisms that are genuinely causal and those that are merely artifacts of a specific environment's data distribution. For instance, in time-series analysis, methods that test model invariance through group-level interventions or knockoff-based interventions have demonstrated superior performance in inferring causal directions compared to traditional Granger causality or correlation-based approaches [67; 68]. By rigorously testing whether a relationship holds under various perturbations and environmental shifts, researchers can filter out spurious correlations and converge on the invariant causal mechanisms that govern complex systems. This focus on invariance naturally motivates the incorporation of additional constraints and priors to further stabilize causal discovery, a topic explored in depth in the subsequent section.

1.7 Integrating Domain Knowledge and Priors into Causal Models

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Building upon the principle of invariance discussed in the previous section, which isolates causal mechanisms by testing their stability across environments, a complementary strategy for robust causal discovery involves the explicit integration of domain knowledge and prior information. While invariance leverages environmental shifts to filter spurious correlations, incorporating priors addresses the inherent limitations of purely data-driven approaches, particularly in high-dimensional regimes, sparse data settings, or scenarios where observational data alone is insufficient to identify unique causal structures. Consequently, there is a pressing need for methodological frameworks capable of systematically encoding heterogeneous prior knowledge to guide structure learning, mitigate errors inherent in imperfect priors, and enable valid inference under conditions of partial specification.

A foundational challenge in this domain is the lack of a standardized hierarchy for encoding priors with varying degrees of confidence. To address this, recent work has introduced the Causal

Knowledge Hierarchy (CKH), an abstraction inspired by the "levels of evidence" framework commonly used in medicine. The CKH provides a structured approach to encoding causal priors from diverse information sources, assigning confidence levels to different types of causal information. By combining these weighted priors, researchers can derive more accurate SCMs than those obtained through data alone. Empirical evaluations on simulated datasets have demonstrated that utilizing the CKH framework significantly improves performance relative to the ground truth causal model, highlighting the importance of sensitivity analysis when integrating priors of differing reliability [69].

Beyond hierarchical confidence levels, another significant advancement involves handling scenarios where full causal diagrams are unavailable or too complex to specify entirely. In such cases, Cluster DAGs (C-DAGs) offer a powerful graphical modeling tool that allows for the partial specification of relationships. C-DAGs define relationships between clusters of variables while leaving the internal structure within clusters unspecified, effectively representing an equivalence class of causal diagrams. This approach alleviates the stringent requirement of specifying a complete causal graph, which is often infeasible in complex domains. Theoretical foundations for C-DAGs have been rigorously established, proving the soundness and completeness of d-separation for probabilistic inference and demonstrating the validity of Pearl's do-calculus rules over these structures. Furthermore, standard identification algorithms have been shown to be sound and complete for computing causal effects from observational data given a C-DAG, extending valid inference to counterfactual reasoning about variable clusters [70].

The source of prior knowledge has also expanded significantly with the advent of Large Language Models (LLMs), which possess extensive world knowledge that can be leveraged to infer causal relationships based on textual descriptions of variables. However, directly using LLM outputs as hard constraints can lead to unreliable structures due to hallucinations or overconfidence. To mitigate this, frameworks like Iterative LLM Supervised Causal Structure Learning (ILS-CSL) have been proposed. This approach integrates LLM-based inference with data-driven causal discovery in an iterative loop, refining the causal Directed Acyclic Graph (DAG) using feedback mechanisms. This method not only utilizes LLM resources more efficiently but also generates more robust structural constraints compared to static prior incorporation [71]. Moreover, recognizing that LLM-derived priors may contain errors, specifically "order-reversed" edges, recent research has developed strategies resilient to such mistakes. By classifying prior errors and analyzing their impact on structural metrics like the Structural Hamming Distance, it has been proven that specific post-hoc strategies can identify and correct order-reversed errors by detecting the formation of "quasi-circles," thereby maintaining the integrity of the learned structure even when prior knowledge is imperfect [72].

Incorporating prior knowledge extends beyond structure learning to enhance predictive modeling and regularization. When a causal graph is available, it encodes conditional independence relations that can be exploited to improve supervised learning. A model-agnostic data augmentation method has been proposed to incorporate these conditional independence constraints directly into the training process. Theoretical justification via excess risk bounds indicates that this approach suppresses overfitting by reducing the effective complexity of the predictor hypothesis class, leading to improved prediction accuracy, particularly in small-data regimes [73]. Similarly, the concept of "collider regression" leverages probabilistic causal knowledge regarding collider structures to constrain the regression hypothesis space, providing strictly positive generalization benefits under mild assumptions [74].

Furthermore, the integration of domain knowledge can be formalized through Bayesian approaches that quantify uncertainty over causal structures. Variational inference methods have been developed to approximate the posterior distribution over DAGs, allowing models to reason about epistemic uncertainty when planning interventions. These methods utilize parametric variational families modeled by autoregressive distributions, ensuring tractability even as the number of variables grows [75]. Advanced frameworks like Tractable Uncertainty for Structure Learning (TRUST) employ probabilistic circuits to represent posterior beliefs, capturing a richer space of DAGs and enabling tractable reasoning about uncertainty through various inference queries [76].

Finally, the dynamic acquisition of knowledge offers a promising direction where algorithms actively request expert input only when necessary. Instead of requiring a complete set of priors before learning begins, novel approaches allow structure learning algorithms to dynamically identify uncertain relationships and request specific knowledge, such as arc orientations, from human experts. This human-in-the-loop strategy has been shown to offer considerable gains in structural accuracy while using human expertise more effectively than static prior integration [77]. Additionally, in federated settings where data privacy is paramount, frameworks like FedCDI integrate interventional data across distributed nodes by exchanging belief updates rather than raw samples, demonstrating how prior knowledge and interventional signals can be aggregated securely to infer causal structures [78]. Collectively, these advancements underscore that the future of causal discovery lies not in choosing between data and knowledge, but in developing sophisticated mechanisms to fuse them synergistically, setting the stage for the deployment of these robust causal models in real-world deep learning applications.

2 Identifiability Theory and Guarantees in Latent Causal Models

2.1 Nonparametric Identifiability via Unknown Interventions and Multi-Environment Data

2.1 Nonparametric Identifiability via Unknown Interventions and Multi-Environment Data

While the previous discussion highlighted the severe ill-posedness of recovering latent causal variables from purely observational data, this subsection explores how multi-environment data arising from unknown interventions can resolve these ambiguities in general nonparametric settings. The central premise is that even when the specific targets and nature of interventions are not explicitly known—a scenario reflective of complex real-world scientific data where intervention metadata is often missing—the geometric signatures left by these distributional shifts provide sufficient signal for identifiability. This section critically analyzes the conditions under which latent causal structures can be uniquely recovered, bridging the gap between theoretical impossibility in static settings and achievability in heterogeneous environments.

A cornerstone of modern identifiability theory in this domain is the utilization of score functions—the gradients of the log-density of the observed data—to detect these geometric signatures. Significant progress has been made in establishing that perfect recovery of the latent causal model is possible even when the correspondence between interventional environments and the specific nodes intervened upon is unknown. Specifically, it has been demonstrated that under a general nonparametric latent causal model and a general transformation mapping latents to observations, two hard uncoupled interventions per node are sufficient to guarantee identifiability [79]. The term "uncoupled" is critical here, implying that the learner possesses no metadata indicating which pair of environments corresponds to an intervention on the same latent variable. By exploiting score variations across these diverse environments, algorithms can estimate the inverse of the mixing transformation and subsequently recover the latent variables without requiring faithfulness assumptions that were previously deemed necessary in the presence of observational data [79].

Extending these results to broader settings, researchers have investigated the efficacy of soft interventions and stochastic changes in causal mechanisms. In scenarios where hard interventions are unavailable or impractical, the focus shifts to identifying conditions under which soft interventions suffice. It has been shown that for linear transformations, one stochastic hard intervention per node guarantees identifiability, while for general non-linear transformations, two stochastic hard interventions per node are required [80]. Crucially, these results hold even when the pairing of interventional environments is unknown, reinforcing the robustness of score-based approaches. The underlying mechanism relies on the property that a valid transformation renders the score function of the latent variables to exhibit minimal variations across different interventional environments, a characteristic that can be optimized to recover the underlying Directed Acyclic Graph (DAG) [81].

However, the landscape of identifiability is not without its intrinsic limitations, particularly when moving beyond single-node interventions to more general environmental shifts. Theoretical boundaries emerge in such cases; for instance, in linear causal models driven by general environments, while the causal graph itself may be fully recoverable, the latent variables are often identified only up to a "surrounded-node ambiguity" (SNA) [82]. This ambiguity suggests that certain entanglements among latent variables are fundamentally irresolvable from interventional data alone unless specific structural constraints are met. This finding underscores the necessity of carefully characterizing the diversity and coverage of interventions; mere distributional shifts are insufficient if they do not break the specific symmetries inherent in the causal mechanism.

Further relaxing the assumptions regarding the knowledge of intervention targets, recent work has established that latent causal graphs are nonparametrically identifiable even with unknown interventions, provided specific graphical conditions are met. Notably, it has been proven that at most one unknown intervention per hidden variable is needed to reconstruct the latent structure in measurement models without assuming linearity or Gaussianity [83]. This represents a significant relaxation of prior requirements, introducing novel graphical concepts such as "imaginary subsets" and "isolated edges" to characterize the limits of edge orientations within equivalence classes induced by unknown interventions. Similarly, in the most general setting where both the causal model and the mixing function are nonparametric, it has been demonstrated that the observational distribution combined with one perfect intervention per node suffices for identifiability, subject to a genericity condition that rules out fine-tuned spurious solutions [84].

The role of sparsity and mechanism changes has also been pivotal in advancing nonparametric identifiability. In settings where interventions are not explicitly labeled but induce sparse changes in the causal mechanisms, it is possible to recover the moralized graph of the underlying DAG and, in some cases, the latent variables up to component-wise transformations [85]. This approach leverages the sparsity constraint on the recovered graph and sufficient change conditions on causal influences to distinguish true causal relations from mere statistical correlations. Furthermore, the concept of mechanism sparsity regularization has been introduced to facilitate partial disentanglement, where latent factors are recovered up to a novel equivalence relation called "consistency," allowing for the handling of multi-node interventions with unknown targets [86].

Finally, the integration of these theoretical guarantees into practical algorithms has led to the development of methods capable of handling entirely latent interventions, where neither the identity of the intervened variable nor the correspondence of samples to distributions is known. By framing the problem as learning a shared causal graph among an infinite mixture of intervention structural causal models, variational inference approaches have shown promise in discovering causal relations in such challenging scenarios [87]. Collectively, these works establish a rigorous foundation for CRL, demonstrating that while observational data alone is insufficient, the strategic accumulation of multi-environment data—even with unknown intervention targets—provides a powerful signal for disentangling the causal fabric of complex systems. The convergence of score-based analysis, sparsity principles, and graphical characterizations marks a paradigm shift from relying on strong parametric assumptions to leveraging the rich structural information embedded in heterogeneous data distributions, setting the stage for a deeper examination of identifiability across specific linear and non-linear mixing regimes.

2.2 Identifiability in Linear and Non-Linear Mixing Regimes

2.2 Identifiability in Linear and Non-Linear Mixing Regimes

While Section 2.1 established that multi-environment data with unknown interventions can resolve identifiability in general nonparametric settings through score-based geometric signatures, this subsection delves deeper into how specific structural assumptions on the mixing function—ranging from linear to general non-linear regimes—refine these guarantees and dictate the necessary intervention diversity. The fundamental challenge in Causal Representation Learning (CRL) remains the recovery of latent causal variables and their structural relationships from high-dimensional observations that are mixtures of these latents. Without additional constraints, this problem is severely ill-posed due to the infinite number of transformations mapping latents to observations. Consequently, recent theoretical advancements have focused on establishing precise identifiability boundaries by coupling specific mixing regimes with interventional data to break inherent symmetries.

In the regime of linear mixing, where observed variables are generated via a linear transformation of latent causal variables, identifiability has been rigorously characterized through the lens of intervention diversity. A pivotal finding in this domain is that observational data alone is insufficient; interventions are strictly necessary to disentangle the latent structure. Specifically, it has been proven that if even a single latent variable lacks an intervention, distinct causal models exist that cannot be distinguished from one another based on the available data. Conversely, the application of a single intervention on each latent variable is sufficient to guarantee identifiability. This result relies on a novel generalization of the RQ decomposition, which incorporates a partial order determined by the latent causal graph to replace standard orthogonal conditions, thereby uniquely recovering the latent model and the mixing matrix [88]. Furthermore, when the data involves heterogeneous environments resulting from unknown interventions, methods have been developed to recover exogenous noises and match them to endogenous variables. In scenarios with causal sufficiency, such approaches can uniquely identify intervention targets, while in the presence of latent confounders, they provide a refined candidate set that improves upon previous state-of-the-art bounds without requiring restrictive linearity assumptions or explicit invariance tests [89].

Transitioning to more complex scenarios, the field has made significant strides in handling general non-linear mixing functions, where the relationship between latents and observations is governed by arbitrary, potentially deep neural network-based transformations. In these non-parametric settings, the geometric signatures left by interventions become the primary signal for identification, extending the principles discussed in Section 2.1 to specific functional forms. A landmark contribution in this area demonstrates that even with completely unknown intervention targets, strong identifiability results can be achieved if the latent distribution is Gaussian but the mixing function is general. The proof strategy involves analyzing the high-dimensional geometric structure of the data distribution post-transformation, specifically by examining quadratic forms of precision matrices across different interventional environments. This approach allows for the unique recovery of latent variables from non-paired interventions, a significant generalization over prior works that required paired counterfactual data or linear maps [90].

Building on the utility of geometric signatures, score-based methods have emerged as a unifying framework for achieving identifiability across both linear and non-linear regimes. These methods exploit the variations in score functions (the gradient of the log-density) across different interventional environments, offering a cohesive theoretical bridge between the regimes. For linear transformations, it has been shown that a single stochastic hard intervention per node suffices for identifiability, while soft interventions offer partial guarantees depending on the non-linearity of the causal mechanisms. In the case of general non-linear transformations, the theory establishes that two stochastic hard interventions per node are sufficient to guarantee the perfect recovery of the latent causal model. Crucially, these score-based algorithms do not require knowledge of which specific pairs of environments correspond to interventions on the same node, effectively handling "uncoupled" interventions. The core insight is that any valid transformation rendering the latent variables independent will minimize the variation of their score functions across environments, providing a tractable objective for learning [80] [79].

The power of interventional data is further elucidated by its ability to reveal the support structure of latent factors. Interventions can decouple the support of an intervened latent variable from its ancestors, creating distinct geometric patterns in the data distribution. Leveraging this property, it has been proven that latent causal factors can be identified up to permutation and scaling given data from perfect do-interventions, even without assumptions about the specific distribution or dependency structure of the latents. For imperfect or soft interventions, this approach yields block affine identification, ensuring that estimated factors are only entangled with a limited subset of other latents [91]. Additionally, the concept of mechanism sparsity regularization offers another pathway to identifiability in non-linear settings. By enforcing sparsity in the learned causal graph relating latent factors to past factors or auxiliary variables, one can recover latent variables up to permutation. This principle connects nonlinear Independent Component Analysis (ICA) with causality, showing that unknown-target interventions can serve as the auxiliary signal needed to disentangle factors when the underlying mechanisms are sparse [92].

Despite these advances, intrinsic ambiguities remain when interventions are less diverse or when the setting involves general environments rather than targeted node interventions. Recent work highlights the "surrounded-node ambiguity" (SNA), demonstrating that in linear causal models derived from general environments, the causal graph may be fully recoverable, but the latent variables themselves are only identifiable up to this specific ambiguity. This limitation appears unavoidable without stronger assumptions, such as access to groups of soft single-node interventions, underscoring the delicate balance between the diversity of interventional data and the strength of identifiability guarantees [82]. Moreover, extensions to polynomial causal models and non-Gaussian noise distributions have broadened the scope beyond linear Gaussian assumptions, providing partial identifiability results even when not all causal parameters change across environments [93]. Together, these theoretical developments delineate a clear map of what is achievable in causal disentanglement under known mixing constraints. However, the assumptions of causal sufficiency and fully observed mechanisms often break down in practice; the subsequent section addresses how these identifiability frameworks must be adapted when latent confounders and dense unobserved variables obscure the causal fabric.

2.3 Handling Latent Confounders and Dense Unobserved Variables

2.3 Handling Latent Confounders and Dense Unobserved Variables

While Section 2.2 established identifiability guarantees through diverse interventions and geometric signatures, real-world applications often preclude such extensive experimental control, leaving researchers to contend with purely observational data plagued by latent confounders. The presence of unobserved common causes represents a formidable challenge, as they induce spurious correlations that can fundamentally mislead structure learning algorithms. In these scenarios, the assumption of causal sufficiency is violated, leading to biased causal effect estimates and incorrect graphical structures. Consequently, recent theoretical advancements have shifted focus from merely detecting the existence of hidden variables to rigorously identifying their locations, cardinalities, and specific causal relationships with observed variables. A pivotal development in this domain is the Generalized Independent Noise (GIN) condition, which extends the classical independent noise assumption to settings involving latent hierarchies. The GIN condition posits that for two observed random vectors, a specific linear combination of one vector is statistically independent of the other if and only if certain latent variables d-separate them within the causal graph. This theoretical insight enables researchers to not only detect latent confounders but also to distinguish between distinct latent structural patterns, such as those found in Linear, Non-Gaussian Latent Hierarchical Models (LiNGLaHs). By leveraging the GIN condition, it becomes possible to efficiently estimate the causal structure among both latent and observed variables, effectively generalizing the independent noise condition which was previously limited to fully observed systems [94] [95].

Beyond conditions based on independence, score-based methods have been adapted to handle dense unobserved confounding, where latent variables exert widespread influence across many observed nodes. In such "dense" regimes, traditional conditional independence tests often fail because almost all variable pairs remain dependent given any subset of others. To address this, the deconfounded score method utilizes the principle of independent mechanisms to identify a statistical footprint left by latent confounding in the observed data distribution. This approach demonstrates that even under dense confounding, a sparse linear Gaussian directed acyclic graph among observed variables can be approximately recovered by adjusting the scoring function to disentangle spurious correlations from true causal effects. The robustness of this method lies in its ability to scale to high-dimensional problems while maintaining performance despite deviations from strict model assumptions [96]. Complementing these score-based approaches are frameworks that explicitly model the latent space using deep generative models. For instance, neural Acyclic Directed Mixed Graphs (ADMGs) have been developed to learn non-linear functional relations in the presence of latent confounding. By employing autoregressive flows within a gradient-based optimization framework, these models can simultaneously determine complex causal structures and estimate functional relationships, overcoming the computational intractability of discrete search methods often associated with ADMG learning [97].

Instrumental variable (IV) techniques offer another powerful avenue for identification when latent confounders are present, though they traditionally rely on strong, often untestable assumptions regarding the validity of the instrument. Recent work has advanced this field by developing data-driven methods to discover ancestral instrumental variables (AIVs) directly from observational data without requiring complete prior knowledge of the causal graph. By leveraging maximal ancestral graphs (MAGs), these methods can identify valid conditioning sets for AIVs, thereby enabling unbiased causal effect estimation even when the full causal structure is unknown [98] [99]. Furthermore, the concept of conditional instrumental variables has been expanded through deep generative models, which learn representations of both the instrument and its required conditioning set. This approach, such as the DVAE.CIV method, disentangles these representations to facilitate causal effect estimation in high-dimensional settings where manual selection of instruments is infeasible [100] [101]. The utility of IVs is further enhanced by ensemble methods that can provide consistent estimates even when a portion of the candidate instruments are invalid, relying on the modal relationship among multiple candidates to isolate the true causal effect [102].

In scenarios involving multiple causes, the "blessings of multiple causes" paradigm suggests that the correlation structure among several treatment variables can serve as indirect evidence for unobserved confounding. The deconfounder algorithm capitalizes on this by inferring a latent substitute variable that renders the causes conditionally independent, effectively acting as a proxy for the unmeasured confounders. Theoretical justifications from a causal graphical view confirm that this multiplicity allows for the identification of intervention distributions in specific graph classes, expanding the applicability of causal inference to settings previously deemed intractable due to hidden confounding [103] [104]. Additionally, causal reduction methods provide a principled way to simplify complex latent structures by replacing an arbitrary number of high-dimensional latent confounders with a single latent variable that preserves both observational and interventional distributions. This reduction facilitates the combination of interventional and observational data, improving estimation accuracy particularly when interventional samples are scarce [105].

Finally, specialized techniques have been developed for dynamic systems and specific data modalities. For time-series data, methods like the Time Series Deconfounder utilize recurrent neural networks to infer latent factors that account for multi-cause hidden confounders over time, enabling unbiased estimation of time-varying treatment effects [106]. Similarly, in the context of Granger causality, deep latent-variable recurrent neural networks have been proposed to recover unobserved confounders by modeling the relationship between observed variables and proxy signals, offering a robust alternative to traditional proxy-based approaches that often suffer from bias [107] [108]. Collectively, these diverse methodologies—ranging from algebraic conditions like GIN to deep generative modeling and advanced instrumental variable strategies—form a comprehensive toolkit for tackling the pervasive issue of latent confounding. While these methods successfully identify latent variables and their immediate effects, the subsequent section explores how these insights extend to recovering intricate hierarchical architectures and mixed graphical representations where latent factors themselves form deep causal chains.

2.4 Hierarchical and Graphical Structures in Latent Variable Models

2.4 Hierarchical and Graphical Structures in Latent Variable Models

While Section 2.3 established methods for detecting the presence and immediate effects of latent confounders, real-world systems often exhibit far more intricate architectures where latent factors themselves form deep causal chains or hierarchical dependencies. Moving beyond flat models with isolated hidden variables, this subsection addresses the challenge of recovering complex latent hierarchies and mixed graphical representations. In domains ranging from biological systems to unstructured data analysis, observed variables are frequently generated by causally related latent variables, which may in turn be influenced by higher-level latent factors. This necessitates rigorous identification criteria for nonlinear latent hierarchical models, aiming to recover both the deep causal structure and the latent variables themselves. Recent theoretical advancements have demonstrated that identifiability in such nonlinear settings is achievable under mild assumptions, even when multiple paths exist between variable pairs, thereby relaxing the restrictive latent tree assumptions prevalent in earlier literature. Specifically, by establishing novel identification criteria for elementary latent variable models, researchers have shown that both the causal structures and latent variables of hierarchical models can be identified asymptotically through explicit estimation procedures, marking a significant departure from strict parametric constraints [109].

A critical avenue for recovering these deep causal hierarchies involves leveraging the concept of mixture oracles. The problem of reconstructing a causal graphical model in the presence of latent variables often reduces to learning the bipartite structure between observables and unobservables. By drawing a connection between learning the order of a mixture model and recovering latent causal graphs, it has been proven that both latent representations and the underlying causal model are identifiable under specific conditions. This approach is particularly powerful because it allows for general, potentially nonlinear dependencies between variables, focusing on high-level latent features such as concepts or objects rather than raw observational dependencies like image pixels. The constructive nature of these proofs has led to efficient algorithms capable of explicitly reconstructing full graphical models, offering a robust framework for scenarios where the dependence between raw observations is less relevant than the dependence between latent concepts [110].

Complementing mixture-based approaches, the utilization of rank constraints provides a potent mechanism for discovering latent hierarchical structures where children of latent variables may also be latent, and only leaf nodes are measured. In such challenging scenarios, where the graph structure extends beyond simple trees to include multiple paths between variables, estimation procedures leveraging rank deficiency constraints over measured variables can efficiently locate latent variables and determine their cardinalities. These methods have been shown to identify the correct Markov equivalence class of the entire graph asymptotically, provided proper restrictions on the graph structure are met. This capability is essential for modeling complex systems where latent confounders form a hierarchy, enabling the recovery of deep causal layers that would otherwise remain hidden [111].

In parallel, the study of Acyclic Directed Mixed Graphs (ADMGs) offers a formalism for handling latent confounding through the inclusion of bidirected edges alongside directed ones, providing a unified view of direct and confounded relationships. Traditional approaches to ADMG learning often rely on linear functional assumptions or discrete search methods that lack scalability. However, recent developments have introduced gradient-based approaches capable of learning ADMGs with nonlinear functional relations directly from observational data. By utilizing autoregressive flows within a neural causal model framework, these methods can determine complex causal structural relationships and estimate functional treatment effects simultaneously. Crucially, it has been established that the presence of latent confounding is identifiable under the assumptions of bow-free ADMGs with nonlinear additive noise models, providing a theoretical foundation for these scalable, continuous optimization techniques [97].

Structural sparsity and the grouping of observational variables also play pivotal roles in facilitating the recovery of latent structures. In domains where observed variables naturally group into classes, such as in educational or psychological assessments, Hierarchical Latent Attribute Models (HLAMs) provide a structured approach. Despite the degeneracy introduced by attribute hierarchy graphs, sufficient and necessary identifiability conditions have been developed to sharply characterize the impact of different attribute types on model recovery. These conditions not only aid in diagnosing test designs but also serve as vital tools for assessing the validity of estimated hierarchical models [112]. Additionally, in high-dimensional settings, exploiting low-rank assumptions regarding the adjacency matrix of causal models can address the challenges of learning dense graphs. By adapting causal structure learning methods to utilize low-rank techniques, it is possible to recover interpretable graphical conditions related to hubs and scale-free networks, maintaining superior performance even when graphs are not strictly sparse [113].

Residual analysis and independent noise conditions further refine our ability to disentangle latent hierarchies by providing algebraic signatures of causal separation. The Generalized Independent Noise (GIN) condition extends the classic independent noise assumption to linear non-Gaussian models with latent variables, establishing independence between linear combinations of measured variables. This condition implies the existence of exogenous sets that d-separate variables, allowing for the efficient estimation of Linear, Non-Gaussian Latent Hierarchical Models where latent confounders are causally related. The GIN condition effectively generalizes previous methods, enabling the identification of underlying causal structures under mild assumptions and demonstrating effectiveness in both synthetic and real-world applications [94]. Moreover, in scenarios involving discrete observables and global confounders, it has been demonstrated that causal structures remain identifiable without parametric assumptions if the number of latent classes is small relative to the DAG's sparsity, highlighting the interplay between discrete structures and latent confounding [31].

Finally, the integration of these structural insights with modern deep learning architectures, such as Variational Autoencoders (VAEs), enables the learning of interpretable causal structures in fully unsupervised settings. By placing Gaussian mixture priors on latent spaces and identifying mixtures with DAG nodes, algorithms can simultaneously discover key features and their causal relationships, effectively bridging the gap between representation learning and causal discovery in multimodal data [114]. Collectively, these diverse approaches—ranging from mixture oracles and rank constraints to ADMGs and GIN conditions—form a comprehensive toolkit for unraveling the complex, hierarchical causal architectures that underpin high-dimensional observational data. While these structural mechanisms provide the theoretical backbone for untangling latent hierarchies, the fundamental challenge of Causal Representation Learning remains its inherent ill-posedness in the absence of strong supervision. The subsequent section explores how weaker, more realistic supervision signals—such as paired interventional data, multimodal constraints, and physical priors—can resolve these remaining ambiguities to guarantee the identification of causal representations.

2.5 Weak Supervision, Paired Data, and Multimodal Constraints

2.5 Weak Supervision, Paired Data, and Multimodal Constraints

While Section 2.4 detailed the structural mechanisms—such as hierarchical latent variables, rank constraints, and mixture oracles—required to untangle complex causal architectures, the fundamental challenge in Causal Representation Learning (CRL) remains its inherent ill-posedness. Without additional constraints, uniquely recovering latent causal variables and their structural relationships from purely observational, high-dimensional data is mathematically impossible. Whereas strong assumptions regarding functional forms or perfect interventional metadata have traditionally ensured identifiability, recent theoretical advancements have shifted focus toward leveraging weaker, more realistic supervision signals. This subsection critically reviews how weak supervision, specifically in the form of paired pre- and post-intervention samples, multimodal data alignment, and physics-based constraints, serves as a potent mechanism to resolve the ambiguities left by structural assumptions alone and guarantee the identification of causal representations.

A primary avenue for achieving identifiability under weak supervision involves the utilization of paired data arising from unknown interventions. In many practical scenarios, such as robotic manipulation or biological experiments, one may observe pairs of states before and after a perturbation without knowing the specific target or nature of the intervention. Theoretical works have demonstrated that such paired samples provide a sufficient learning signal to disentangle causal factors even when the intervention targets are latent. For instance, it has been proven that under mild assumptions, causal representations are identifiable in a setting where the dataset consists of paired samples before and after random, unknown interventions, effectively bypassing the need for explicit intervention labels [115]. This approach allows models to learn implicit latent causal structures without optimizing an explicit discrete graph, thereby facilitating the reliable identification of causal variables in complex image data. Furthermore, the concept of sparse perturbations offers another robust pathway for identification. When perturbations are applied to mutually exclusive blocks of latent variables, or when overlapping perturbation blocks intersect at single latent variables, the underlying latent factors can be identified up to permutation and scaling [116]. This theory enables the development of natural estimation procedures that succeed even with unknown continuous latent distributions, provided the perturbations exhibit sufficient sparsity.

Beyond paired temporal or interventional data, the integration of multimodal constraints has emerged as a powerful strategy for enhancing identifiability in fully unsupervised settings. Scientific datasets often contain multiple modalities or adhere to known physical laws, which can act as auxiliary signals to constrain the solution space. Algorithms designed to leverage these multimodal inputs, such as those incorporating physics-based constraints, can discover important features with causal relationships without relying on linear structural assumptions or explicit interventional data [114]. By utilizing a differentiable parametrization to learn a directed acyclic graph alongside a latent space within a variational autoencoder framework, these methods exploit the shared information across modalities to isolate causal mechanisms. Additionally, the challenge of aligning unpaired multimodal data, common in fields like biology where paired measurements are destructive, can be addressed through causal analogies. By drawing parallels between potential outcomes in causal inference and potential views in multimodal observations, researchers can estimate propensity scores to define distances between samples, enabling effective alignment and representation learning even in the absence of paired samples [117]. This approach underscores how causal frameworks can transform the problem of multimodal alignment into a solvable identification task.

The role of domain-specific knowledge, particularly physics-based constraints and task structures, further refines the identifiability guarantees. In dynamic systems, leveraging the structure of tasks or the distribution of tasks can significantly reduce the equivalence class of identifiable representations. When a causal structure is assumed over a distribution of tasks, maximum marginal likelihood optimization can be employed to recover canonical representations, outperforming general unsupervised models [118]. Moreover, in interactive systems where measurement rates may be slower than causal effects, leading to instantaneous dependencies, methods like iCITRIS utilize observed intervention targets (e.g., agent actions) to identify multidimensional causal variables and their graphs simultaneously [119]. This demonstrates that even partial knowledge of intervention targets, combined with temporal sequence data, can overcome the limitations imposed by instantaneous effects.

Furthermore, the application of weak supervision extends to scenarios involving partial observability and mixed observations. In settings where each measurement captures only a subset of the underlying causal state, enforcing sparsity in the inferred representation can lead to successful recovery of ground-truth latents, provided specific conditions on linear or piecewise linear mixing functions are met [120]. Similarly, in multi-view settings where different views constitute nonlinear mixtures of latent subsets, contrastive learning combined with a single encoder per view can identify shared information up to a smooth bijection, unified under a framework of "identifiability algebra" [121]. These findings collectively illustrate that the ill-posed nature of CRL can be effectively mitigated not by demanding perfect knowledge of the generative process, but by strategically incorporating weak supervision signals. Whether through paired interventional views, multimodal consistency, physical priors, or sparse perturbations, these auxiliary constraints provide the necessary geometric and statistical signatures to disentangle true causal factors from confounding variations. However, even with these sophisticated weak supervision strategies, fundamental boundaries remain; the following section explores the intrinsic ambiguities and theoretical limits that persist when intervention diversity is insufficient or when equivalence classes cannot be fully collapsed.

2.6 Intrinsic Ambiguities, Equivalence Classes, and Theoretical Limits

2.6 Intrinsic Ambiguities, Equivalence Classes, and Theoretical Limits

As highlighted in the preceding discussion on weak supervision, strategies such as paired interventional data, multimodal alignment, and physical priors have significantly narrowed the solution space for Causal Representation Learning (CRL). However, even with these sophisticated auxiliary constraints, fundamental boundaries persist. This section explores the intrinsic ambiguities and theoretical limits that remain when intervention diversity is insufficient or when equivalence classes cannot be fully collapsed. The pursuit of causal discovery is fundamentally constrained by these ambiguities, which arise when attempting to recover latent causal structures from high-dimensional observations. While significant progress has been made in developing algorithms for structure learning, it is crucial to recognize that without specific interventional conditions or strong structural assumptions, the true causal model often remains unidentifiable. Instead, learning procedures can typically only recover an equivalence class of models that are statistically indistinguishable given the available data. Understanding these theoretical limits is essential for designing robust CRL frameworks and for setting realistic expectations regarding what can be inferred from observational versus interventional datasets.

A primary source of intrinsic ambiguity in causal representation learning, particularly in settings involving multiple environments or unknown interventions, is the phenomenon known as Surrounded-Node Ambiguity (SNA). This ambiguity arises when the causal graph structure can be recovered, but the latent variables themselves are only identified up to a specific transformation that preserves the local Markov properties of the surrounded nodes. Recent theoretical work has rigorously characterized this limitation, demonstrating that for linear causal models learned from general environments, while the causal graph topology may be fully recoverable, the latent variables suffer from SNA. It has been shown that this ambiguity is essentially unavoidable in settings where interventions are not perfectly targeted or known, as the statistical signatures left by such interventions do not provide sufficient information to disentangle the specific scaling or mixing of the latent factors surrounding a node [82]. This finding underscores a critical boundary in nonparametric identifiability: even with access to diverse environmental data, certain symmetries in the data-generating process prevent unique recovery of the ground-truth latent representation without further constraints.

Beyond SNA, the concept of equivalence classes induced by unknown interventions defines another fundamental limit. In scenarios where the correspondence between samples and specific intervention targets is lost—a common occurrence in large-scale biological or physical experiments—the space of possible causal models expands significantly. When interventions are entirely latent, the learning problem transforms into identifying a shared causal graph among an infinite mixture of intervention structural causal models. In such cases, the equivalence class is defined not just by Markov equivalence but by the inability to distinguish which specific variables were perturbed in each environment. Research utilizing variational inference and Dirichlet process priors has highlighted that without knowing the intervention targets, the posterior distribution over causal structures remains broad, necessitating strong inductive biases or semi-supervised signals to narrow down the solution space [87]. Furthermore, when interventions are stochastic or "off-target," affecting random subsets of vertices rather than single nodes, the verification and search problems become computationally complex, with approximation algorithms providing only polylogarithmic guarantees relative to the optimal number of interventions [122].

Consequently, establishing the minimum requirements for intervention diversity and coverage is paramount to guaranteeing unique model recovery. Theoretical analyses have begun to quantify these requirements, revealing that the diversity of interventions across environments is a decisive factor in breaking the symmetries that lead to ambiguities. For instance, in the context of linearly mixed causal representations, it has been proven that allowing for multi-node interventions within a single environment, provided there is sufficient coverage and diversity of these interventions across different domains, can restore identifiability. The key lies in exploiting the trace that interventions leave on the variance of the ground-truth causal variables; if the intervention pattern is sufficiently sparse and diverse, it becomes possible to regularize the learning process towards the true causal representation [37]. Similarly, in nonparametric settings, it has been demonstrated that having at least one pair of distinct perfect interventional domains per node is a sufficient condition to guarantee identifiability, subject to genericity conditions that rule out fine-tuned cancellations between observational and interventional distributions [84].

However, when these diversity conditions are not met, the system inevitably falls back into broader equivalence classes. For example, in adaptive experimental design, the trade-off between the number of intervention rounds (adaptivity) and the total number of interventions required to identify the graph is governed by strict lower bounds. Algorithms operating with limited adaptivity (fewer rounds of sequential experimentation) face an approximation gap that scales with the number of variables, indicating that without sufficient sequential feedback or intervention coverage, unique recovery is theoretically impossible within a reasonable cost budget [123]. Moreover, in collaborative settings where multiple entities possess distinct but related causal graphs, the ability to recover individual graphs depends heavily on the clustering structure of the graphs and the availability of atomic interventions; without leveraging these shared structures, the number of required interventions per entity scales linearly with the number of variables, highlighting the inefficiency of isolated discovery efforts [124].

The implications of these intrinsic ambiguities extend to the evaluation of causal models as well. Traditional metrics based on structural Hamming distance often fail to capture the nuances of these equivalence classes, potentially penalizing models that are functionally equivalent despite structural differences. Consequently, there is a growing consensus that evaluation frameworks must evolve to account for these theoretical limits, perhaps by measuring distances on the ladder of causation (observational, interventional, counterfactual) rather than solely on graph topology [125]. Furthermore, in the presence of latent confounders, the identifiability landscape becomes even more treacherous, with some studies showing that while linear models with latent confounders can be identified under specific noisy-OR parameterizations or with sufficient intervention sets, general discrete or nonlinear models often remain unidentifiable without restrictive assumptions [126][31].

In summary, the theoretical limits of causal discovery are defined by a complex interplay of intervention knowledge, environmental diversity, and structural assumptions. While methods like [82] and [84] have pushed the boundaries of what is identifiable, they also clearly delineate the regions of intrinsic ambiguity such as SNA. These findings suggest that future advancements in causal representation learning must focus not only on algorithmic efficiency but also on designing intervention strategies and collecting data regimes that explicitly target these theoretical blind spots, ensuring that the diversity and coverage of interventions are sufficient to collapse equivalence classes into unique, recoverable causal models.

3 Algorithmic Paradigms for Causal Structure Discovery

3.1 Differentiable Score-Based Optimization and Continuous Relaxation

3.1 Differentiable Score-Based Optimization and Continuous Relaxation

The landscape of causal structure learning has undergone a paradigm shift with the advent of differentiable score-based optimization, marking a departure from traditional combinatorial search strategies. While classical approaches often rely on discrete space exploration that suffers from exponential complexity, this modern paradigm reformulates the discovery of Directed Acyclic Graphs (DAGs) as a continuous optimization problem. By translating the discrete acyclicity constraint into a smooth, differentiable function, these methods enable the application of efficient gradient-based solvers directly to the adjacency matrix. This subsection critically reviews the evolution of continuous relaxation techniques, tracing their trajectory from foundational linear models to sophisticated nonlinear extensions, while addressing pivotal challenges regarding robustness, scale invariance, and computational scalability that distinguish them from the constraint-based methodologies discussed subsequently.

The cornerstone of this continuous optimization revolution is the NOTEARS framework, which introduced a smooth algebraic characterization of acyclicity via the matrix exponential trace. This formulation transformed structure learning into a continuous constrained optimization task, allowing for direct optimization using standard gradient descent. However, the initial implementation, which primarily targeted linear Structural Equation Models (SEMs) with least-squares loss, has faced significant scrutiny regarding its underlying assumptions. Critics have highlighted that the reliance on least-squares loss renders the method sensitive to variable scaling and potentially susceptible to exploiting variance sortability rather than uncovering true causal mechanisms. Specifically, research demonstrates that NOTEARS lacks scale invariance; simple rescaling of data (e.g., changing units from meters to centimeters) can alter the inferred causal direction, suggesting that the original formulation may fail to identify genuine causal relationships without additional constraints or modifications [127]. Furthermore, it has been shown that graph predictions can be manipulated through targeted variance attacks, where partial manipulation of data variances controls the output [128].

To overcome the limitations of linearity and Gaussian noise assumptions inherent in early continuous methods, researchers have extended these frameworks to accommodate nonlinear structural equation models. A significant advancement in this direction involves graph autoencoder approaches that generalize gradient-based methods to handle nonlinear functional relationships and vector-valued variables. These architectures leverage the expressive power of neural networks to model complex interactions while retaining the continuous optimization objective for acyclicity. Empirical evaluations indicate that such graph autoencoder frameworks significantly outperform other gradient-based baselines, particularly when scaling to large causal graphs, demonstrating near-linear training time efficiency as the graph size increases [129]. Additionally, the integration of neural networks into the score-based framework enables the modeling of complex, nonlinear dependencies that greedy search methods might miss, providing a competitive edge in both synthetic and real-world scenarios [130].

Despite these architectural advancements, the robustness of differentiable causal discovery remains a central concern, particularly regarding heterogeneous data distributions and spurious correlations. Standard differentiable methods often optimize an average score function, which can lead to overfitting "easier" samples while neglecting those containing critical causal signals or exhibiting distribution shifts. To mitigate this, adaptive sample reweighting frameworks have been proposed. These methods employ a bilevel optimization scheme to dynamically learn weights for a reweighted score function, effectively upweighting samples that the learner fails to fit and downweighting those from which spurious information is easily extracted. This approach, known as ReScore, has been shown to consistently boost structure learning performance and generalize better to heterogeneous data distributions where the common homogeneity assumption is violated [131].

Further improvements in robustness and accuracy have been achieved by explicitly incorporating graph structure into the optimization space. The CASPER framework, for instance, introduces a dynamic causal space that integrates graph structure into the score function as a new measure, faithfully reflecting the causal distance between estimated and ground truth DAGs. By revising the learning process to include adaptive attention to "DAG-ness," CASPER addresses the vulnerability of methods that measure data fitness independently of structural constraints, yielding superior accuracy and noise robustness [132]. Moreover, the issue of numerical instability in differentiable causal discovery, which often plagues methods beyond tens of variables, has been tackled by Stable Differentiable Causal Discovery (SDCD). SDCD employs an alternative, theoretically more stable acyclicity constraint and a training procedure tailored for sparse graphs, enabling scalability to thousands of variables while maintaining high convergence speed and accuracy [133].

The quest for efficiency has also driven innovations in approximating the acyclicity constraint and optimizing computational complexity. While the original trace exponential formulation incurs a cubic complexity ($O(d^3)$), recent work has introduced low-rank additive models (LoRAM) that combine low-rank matrix factorization with sparsification mechanisms. This approach reduces the complexity to quadratic ($O(d^2)$) while preserving the DAG characteristic function, allowing for efficient scaling to thousands of nodes with only moderate accuracy loss [134]. Additionally, the reliability of learned structures has been enhanced through post-processing algorithms informed by Karush-Kuhn-Tucker (KKT) optimality conditions. These local search strategies have been proven to universally improve the structural Hamming distance of continuous optimization algorithms, often by a factor of two or more, by explicitly enforcing the absence of certain edges dictated by the optimality conditions [135].

Finally, the extension of these principles to binary adjacency matrices via smooth characterizations and Gumbel-Softmax approximations has facilitated the efficient training of causal structure learning methods that naturally produce near-binary edge weights, thereby easing thresholding and interpretation [136]. Collectively, these developments illustrate a maturing field where continuous relaxation serves as a powerful engine for causal discovery. However, while these score-based methods excel in optimization efficiency and handling nonlinearity, they often rely on specific functional assumptions. This contrasts with the constraint-based, hybrid, and information-theoretic approaches discussed in the following subsection, which prioritize statistical rigor and conditional independence testing to address challenges in high-dimensional and mixed-type data settings.

3.2 Constraint-Based, Hybrid, and Information-Theoretic Approaches

3.2 Constraint-Based, Hybrid, and Information-Theoretic Approaches

While the differentiable score-based methods discussed in the previous subsection revolutionized causal discovery by enabling continuous optimization, they often rely on specific functional assumptions and may struggle with the statistical rigor required for high-dimensional or mixed-type data. In contrast, constraint-based causal discovery represents a foundational paradigm that relies fundamentally on the principle that the causal structure of a system imposes specific conditional independence (CI) constraints on the observed joint probability distribution. Traditional algorithms within this family, such as the PC algorithm and its variants, systematically perform statistical tests to prune edges from a fully connected graph, ultimately recovering an equivalence class of causal graphs consistent with the data. However, the efficacy of these methods is intrinsically tied to the reliability of the underlying conditional independence tests, which often suffer from reduced statistical power in high-dimensional settings or when conditioning sets become large. To address these limitations, recent research has focused on developing more robust testing procedures, integrating information-theoretic measures, and synthesizing constraint-based logic with score-based optimization to create hybrid frameworks that leverage the strengths of both approaches.

A primary challenge in constraint-based learning is the accuracy of CI tests, particularly when dealing with limited sample sizes or complex data types. Standard parametric tests frequently fail to reject independence incorrectly, leading to false negatives and incomplete graph structures. In response, information-theoretic approaches have gained prominence for their ability to handle non-linear dependencies and mixed data types without strong parametric assumptions. For instance, the Stochastic Complexity based Independence (SCI) test utilizes algorithmic independence principles to provide an asymptotically unbiased estimator for conditional mutual information, demonstrating lower Type II error rates compared to conventional tests on discrete data [137]. Similarly, for high-dimensional continuous variables, the Kernel-based Conditional Independence test (KCI-test) constructs appropriate test statistics derived from reproducing kernel Hilbert spaces, offering superior performance when conditioning sets are large or sample sizes are modest [138]. Furthermore, the development of non-parametric estimators, such as the Von Mises estimator built on kernel density estimation, has provided tight sample complexity guarantees for causal discovery in continuous domains, establishing a theoretical foundation for the reliability of these tests under smoothness assumptions [139].

The versatility of constraint-based methods has also been extended to handle mixed-type datasets containing both numerical and categorical variables, a common scenario in real-world applications where traditional tests often struggle. Novel non-parametric methods leveraging k-nearest-neighbor (k-NN) estimators have been proposed to robustly detect dependencies in such heterogeneous environments by carefully defining distance metrics that respect the discrete nature of categorical features [140]. Additionally, information-theoretic frameworks have been employed not just for edge removal but for orienting causal directions. By exploiting asymmetries in additive noise models or conditional variances, researchers can distinguish parents from children even among Markov equivalent structures, thereby moving beyond undirected skeletons to fully directed acyclic graphs [141]. The "Climb" algorithm exemplifies this by using algorithmic information theory to efficiently discover causal Markov blankets and direct edges that remain ambiguous in standard PC or GES outputs, significantly improving precision and recall [142].

Despite their theoretical soundness, pure constraint-based approaches can be computationally prohibitive due to the exponential growth of required CI tests with the number of variables. To mitigate this, hybrid strategies have emerged that combine the efficiency of constraint-based pruning with the global optimization capabilities of score-based methods, effectively bridging the gap between the local statistical rigor of constraint-based tests and the global search efficiency of the score-based techniques described earlier. One such innovation involves modifying the search strategy of the PC algorithm by prioritizing CI tests with large conditioning sets as a pre-processing step. This approach, known as Pre-Processing Plus PC (P3PC), leverages modern model-based tests that maintain statistical power even with large conditioning sets, drastically reducing the total number of tests required—often to less than 10% of the original count—while preserving accuracy [143]. Another hybrid direction involves incorporating Bayesian scores into constraint-based procedures. The Bayesian Constraint-based Causal Discovery (BCCD) algorithm, for example, uses Bayesian scores to assign reliability probabilities to CI statements, processing them in decreasing order of confidence to resolve conflicts and produce a single, robust output model that outperforms standard FCI and Conservative PC algorithms [144].

Furthermore, the integration of score-based logic has led to the development of new scoring functions that explicitly account for the uncertainty inherent in CI testing. The SparsityBoost score, for instance, introduces a data-dependent complexity penalty based on the probability that a CI test correctly identifies the absence of an edge. This formulation not only ensures consistency but also becomes computationally easier to maximize as data volume increases, effectively bridging the gap between local independence tests and global structure optimization [145]. Similarly, hybrid methods have been designed to correct errors from initial CI tests through iterative MCMC schemes, allowing for full Bayesian model averaging over large networks while maintaining the computational complexity characteristic of constraint-based methods [146].

Specialized algorithms have also been developed to address specific structural complexities, such as latent confounders and selection bias. Recursive constraint-based methods have been proposed to learn causal MAGs (Maximal Ancestral Graphs) by iteratively identifying and removing specific variable types, thereby reducing the size of conditioning sets and the total number of tests required while maintaining soundness and completeness [147]. In the temporal domain, constraint-based algorithms for continuous-time Bayesian networks have been introduced, utilizing specific statistical tests to handle the nuances of continuous-time dynamics, offering a viable alternative to score-based approaches especially when variables possess more than two states [148]. Finally, to enhance robustness in small-sample regimes, the concept of k-Markov equivalence has been introduced, restricting the size of conditioning sets to an upper bound k. The resulting k-PC algorithm provides a graphical characterization of equivalence classes under this constraint, enabling more reliable causal discovery when data is scarce [149]. These advancements collectively illustrate a maturing field where constraint-based, hybrid, and information-theoretic approaches are increasingly intertwined to tackle the multifaceted challenges of causal structure discovery. However, while these passive observational methods provide strong statistical foundations, they often encounter fundamental identifiability limits regarding Markov equivalent structures and latent confounders. Addressing these barriers requires a shift from passive observation to active interventional learning, a transition facilitated by Reinforcement Learning strategies as discussed in the following subsection.

3.3 Reinforcement Learning and Active Intervention Strategies

3.3 Reinforcement Learning and Active Intervention Strategies

While the constraint-based, hybrid, and information-theoretic approaches discussed in the previous section have significantly advanced passive causal discovery from observational data, they often encounter fundamental identifiability limits. Specifically, distinguishing between Markov equivalent structures or resolving the influence of latent confounders frequently requires assumptions that cannot be verified without interventional data. The transition from passive observation to active interventional learning represents a paradigm shift to overcome these barriers. Reinforcement Learning (RL) offers a robust framework for this transition by framing causal discovery as a sequential decision-making problem. In this formulation, an agent actively navigates the space of possible interventions to maximize information gain regarding the causal graph, effectively treating the experimental design process as a policy optimization task. This approach not only accelerates convergence to the true causal structure but also minimizes experimental costs, a critical factor in domains such as healthcare and robotics where interventions may be expensive or risky.

A foundational insight driving this field is the natural synergy between the goals of RL and causal inference. As noted in recent theoretical work, online reinforcement learning is inherently causal because the agent's actions serve as explicit interventions on the environment, thereby breaking spurious correlations caused by unobserved confounders that typically plague offline learning settings [150]. By leveraging the temporal ordering of events and the ability to act, RL agents can directly estimate causal effects rather than mere associations. This capability is formalized in frameworks where the agent alternates between exploring the environment through targeted interventions to learn the causal structure and exploiting this learned structure to guide policy optimization. For instance, specific frameworks have been developed that augment policy learning with causal graphical models, creating a virtuous cycle where causal structure updating informs policy guidance, and policy execution generates new interventional data for refining the graph [151].

To tackle the challenge of generalizing causal discovery to unseen environments, Meta-Reinforcement Learning (Meta-RL) has emerged as a powerful strategy. Meta-RL agents are trained on a distribution of tasks, each with a distinct underlying causal structure, enabling them to learn a "learning algorithm" that can rapidly adapt to new causal graphs with minimal samples. Research demonstrates that recurrent meta-RL agents can implicitly perform causal reasoning, selecting informative interventions and drawing causal inferences from observational data in novel situations without explicit symbolic reasoning modules [152]. Furthermore, specialized meta-RL algorithms have been designed specifically to learn intervention strategies that construct explicit causal graphs. These agents learn to perform interventions that maximize the accuracy of the estimated graph, showing superior performance compared to state-of-the-art baselines even in previously unseen environments [153]. This ability to transfer intervention strategies across domains is crucial for scaling causal discovery to complex, dynamic systems where the causal mechanisms may shift over time.

Active intervention targeting is another critical component of RL-based causal discovery, focusing on minimizing the number of experiments required to identify the true causal structure. The CORE framework exemplifies this approach by utilizing deep reinforcement learning to sequentially reconstruct causal graphs while simultaneously learning to perform the most informative interventions. By treating the selection of intervention targets as an action space, CORE efficiently uncovers causal structures in graphs with up to 10 variables, outperforming existing methods in both structure estimation accuracy and sample efficiency [154]. Similarly, other approaches formulate causal discovery as a problem of learning to perform interventions to disambiguate between competing causal hypotheses. In settings where the action space is not naturally defined as interventions on variables, meta-learning approaches enable agents to learn sequences of ego-centric actions that correspond to targeted manipulations of the environment, effectively bridging the gap between low-level control and high-level causal reasoning [155].

The application of RL extends to navigating large-scale causality graphs and answering complex causal queries. In scenarios involving massive knowledge graphs, actor-critic agents have been successfully deployed to search through causal relations to answer binary causal questions. These agents learn to prune the search space significantly, visiting a fraction of the nodes required by naive search algorithms while providing explainable paths that detail the mechanisms linking causes to effects [156]. Moreover, in multi-agent settings, causal methods are increasingly being integrated to improve safety, interpretability, and robustness, promoting a "causality first" perspective that offers strong theoretical guarantees for emergent behaviors in complex interactive environments [157].

Despite these advances, challenges remain in defining appropriate reward structures that balance exploration costs with information gain, particularly in continuous and high-dimensional state spaces. Recent work addresses action space redundancy by proposing state-wise action refinement methods that discover causal relationships between specific action dimensions and task rewards, thereby promoting more efficient causality discovery in continuous control tasks [158]. Additionally, the integration of causal reasoning into model-based RL allows agents to leverage both online interventional data and offline observational data, using deconfounding techniques to learn latent-based causal transition models that improve generalization guarantees [159]. As the field progresses, the fusion of active inference principles with RL continues to offer promising directions for developing agents that can autonomously design experiments, reason about counterfactuals, and uncover the fundamental laws governing their environments with human-like efficiency. These advancements in active learning naturally pave the way for addressing even more complex scenarios involving temporal dynamics, where causal mechanisms evolve over time, a topic explored in the following section on temporal causal discovery in non-stationary systems.

3.4 Temporal Causal Discovery in Dynamic and Non-Stationary Systems

3.4 Temporal Causal Discovery in Dynamic and Non-Stationary Systems

The advancements in active intervention strategies discussed in the previous section provide a powerful mechanism for resolving identifiability issues; however, these methods often assume a static underlying causal structure. In reality, many complex systems are governed by mechanisms that evolve over time, introducing a distinct layer of complexity driven by non-stationarity and dynamic regime shifts. Extending causal discovery from static or actively intervened settings to temporal domains requires addressing inherent challenges such as autocorrelation, time-varying causal strengths, and distributional shifts that can obscure true causal mechanisms. Traditional approaches, which frequently rely on assumptions of stationarity and linearity, often fail to capture the intricate, evolving dependencies found in fields ranging from environmental science to economics. Consequently, recent algorithmic paradigms have shifted towards deep learning architectures capable of modeling non-linear dynamics and adapting to these distributional shifts, bridging the gap between the active learning frameworks of the prior section and the cyclic structures explored subsequently.

A critical challenge in this domain is handling non-stationarity, where the underlying causal structure or the strength of causal links changes over time. Conventional methods often struggle here due to the dataset shift problem, leading to unsatisfactory performance when the stationarity assumption is violated. Novel frameworks have emerged to tackle this by explicitly modeling time-varying causal modules. For instance, specific constraint-based approaches have been developed to identify both lagged and contemporaneous causal relationships while simultaneously detecting changing modules within autocorrelated and non-stationary data. These methods optimize conditioning sets by considering lagged parents rather than the entire past, effectively addressing high dimensionality and allowing for the detection of instantaneous adjacencies that vary over time [35]. Furthermore, efficient variations of these algorithms have been proposed to handle surrogate variables representing time dependency, demonstrating superior performance in identifying time influences compared to standard baselines [160].

Beyond constraint-based methods, score-based optimization using Graph Neural Networks (GNNs) has gained prominence for learning sparse Directed Acyclic Graphs (DAGs) in discretized time temporal graphs. These approaches leverage the expressive power of GNNs to capture causal dependencies that linear Structural Causal Models (SCMs) might miss. Research indicates that GNN-based score methods significantly outperform state-of-the-art Dynamic Bayesian Network inference techniques, providing more accurate structural causal models even in complex dynamic settings [161]. The integration of GNNs allows for the modeling of inter-temporal and inter-variable relationships that traditional neural networks struggle to disentangle, offering a robust mechanism for uncovering the complex mechanisms underlying dynamic systems [162].

Another promising direction involves transforming the causal discovery problem into the frequency domain to achieve computational efficiency and handle long-range dependencies. Recovering causal structures in the time domain for Vector Auto-Regressive (VAR) models can be computationally prohibitive, with complexity scaling poorly with the number of nodes and time lags. However, FFT-based approaches offer a significant reduction in complexity by accumulating time dependencies at every frequency. By treating state variables as random variables at any given frequency, these methods enable efficient causal inference and have demonstrated that do-calculus machinery can be realized in the frequency domain for Linear Time-Invariant (LTI) systems. This allows for the application of classical criteria like front-door and backdoor adjustments with substantial computational advantages over time-domain algorithms like the PC algorithm [163]. Extending this concept, recent architectures utilize Fourier Graph Operators to perform matrix multiplications in Fourier space, rethinking multivariate time series forecasting from a pure graph perspective that unifies spatiotemporal dynamics without the need for separate spatial and temporal networks [164]. Similarly, Edge-Varying Fourier Graph Networks have been introduced to capture high-resolution variable dependencies via efficient graph convolution in the frequency domain, effectively modeling ever-changing correlations that static graph structures fail to represent [165].

The discovery of regime-switching causal structures represents another frontier in handling non-stationary environments, addressing scenarios where real-world systems undergo abrupt changes in dynamics. Deep Switching State Space Models have been engineered to identify irregular regimes hidden within nonlinear dynamics by employing discrete latent variables to represent regimes and continuous variables for driving factors. These models meld Recurrent Neural Networks with switching state space formulations to capture nonlinear dependencies governed by Markov chains, outperforming standard models in forecasting accuracy while providing meaningful regime identifications [166]. Complementary approaches utilize Hidden Markov Models within recurrent architectures to adaptively switch between internal regimes, effectively modeling the temporally varying dynamics of nonstationary sequential data in business and economic domains [167].

Furthermore, the identification of unique causal networks from non-stationary time series has been advanced by models that consider time delays to resolve the issue of Markov equivalence classes. By leveraging higher-order causal entropy algorithms, these frameworks can identify unique network structures in a distributed manner, proving particularly effective when time series exhibit non-stationary behavior where traditional Bayesian Network-based models falter [168]. Additionally, deep recurrent modeling approaches have been proposed to infer non-linear Granger causality while directly accounting for latent confounders, utilizing dual-decoder setups to conduct rigorous causal tests on stochastic time series where latent factors influence cause and effect with varying time lags [108]. Finally, comprehensive frameworks like Rhino combine vector auto-regression, deep learning, and variational inference to model non-linear relationships with instantaneous effects and history-dependent noise, proving structural identifiability even under complex dynamic conditions [169]. Collectively, these advancements signify a paradigm shift towards algorithms that not only discover static causal graphs but also unravel the evolving, non-stationary, and regime-dependent fabric of causal reality in temporal data. While these methods effectively handle temporal evolution, they typically maintain the assumption of acyclicity within each time step or regime. The following section relaxes this constraint further, addressing the complexities of learning cyclic structures and handling feedback loops that persist even within dynamic systems.

3.5 Learning Cyclic Structures and Handling Feedback Loops

3.5 Learning Cyclic Structures and Handling Feedback Loops

While the preceding section addressed the complexities of temporal dynamics and non-stationarity, a distinct yet equally critical challenge lies in relaxing the assumption of acyclicity itself. The constraint of Directed Acyclic Graphs (DAGs), while mathematically convenient for ensuring identifiability, often fails to capture the recursive nature of real-world complex systems. In domains ranging from biological signaling networks and gene regulation to economic equilibrium models and social interaction systems, feedback loops are ubiquitous. These cyclic dependencies imply that variables can mutually influence one another, creating equilibrium states or dynamic oscillations that cannot be represented by standard DAGs. Consequently, a critical frontier in causal structure discovery is the development of algorithms capable of learning cyclic causal graphs. Recent advancements have moved beyond traditional constraint-based methods, leveraging differentiable optimization, normalizing flows, and hybrid frameworks to address the challenges posed by nonlinearity, latent confounding, and incomplete data in cyclic settings.

A primary obstacle in learning cyclic structures is the lack of a straightforward factorization of the joint probability distribution, which complicates likelihood estimation. To address this, researchers have turned to normalizing flows, which offer a powerful mechanism for modeling complex, invertible transformations. The framework known as [170] proposes a novel approach for learning nonlinear cyclic causal graphical models directly from interventional data. By employing residual normalizing flows for likelihood estimation, this method performs direct likelihood optimization without relying on the linearity assumptions that constrain many earlier cyclic discovery algorithms. This approach has demonstrated significant improvements in structure recovery and predictive performance, particularly in applications involving single-cell high-content perturbation screening where feedback loops are biologically inherent. Similarly, the theoretical underpinnings of using flows for causal reasoning have been deepened in [171], which elucidates how autoregressive normalizing flows can be adapted to capture causal data-generating processes even when the underlying graph contains cycles, provided appropriate design choices are made to implement the do-operator for interventional and counterfactual queries.

Beyond flow-based models, the treatment of interventions in equilibrium systems requires specialized mathematical frameworks. In complex systems characterized by feedback, interventions can lead to counterintuitive effects that differ fundamentally from those in acyclic settings. The work titled [172] establishes a framework for differentiable soft interventions based on Lie groups. This approach utilizes modern automatic differentiation techniques applied to implicit functions to optimize interventions within cyclic causal models. This is particularly relevant for scenarios such as the transition to sustainable economies, where policy interventions must account for the system's return to a new equilibrium rather than a transient state. Complementing this, [173] introduces a method that combines observational and interventional equilibrium data to learn continuous, nonlinear cyclic models. By approximating nonlinear mechanisms through coupled local linearizations for each experimental condition, this method successfully reconstructed cellular signaling networks, providing evidence for feedback loops that were missed by acyclic assumptions.

The difficulty of learning cycles is not merely computational but also stems from fundamental statistical properties. As highlighted in [174], cyclic causal structures exhibit qualitatively different behaviors under intervention compared to their acyclic counterparts. Theoretical observations reveal that self-referential distributions in cyclic graphs can result in zero mutual information or apparent independence between variables, causing methods based on mutual information estimation or the Independent Causal Mechanisms principle to fail. This underscores the necessity of algorithms that do not rely solely on conditional independence tests. Furthermore, the presence of cycles significantly complicates the handling of missing data, creating a compounded challenge where standard imputation strategies may distort the cyclic dependencies. The framework [175], named MissNODAGS, addresses this by alternating between imputing missing data and maximizing the expected log-likelihood of the visible data within an expectation-maximization framework. This allows for the learning of cyclic graphs from partially missing interventional data, a scenario where standard imputation followed by causal learning often yields suboptimal results, thereby bridging the gap between cyclic discovery and the data completeness issues discussed in the subsequent section.

Hybrid approaches that combine the strengths of constraint-based and noise-based methods have also emerged as robust solutions for time-series data containing cycles. While earlier hybrid methods were limited to acyclic summary graphs, recent frameworks presented in [176] explicitly relax this assumption. These new algorithms are designed to discover summary causal graphs that may contain cycles, proving robust in ecological and real-world applications where predator-prey dynamics and other feedback mechanisms create cyclic dependencies. Additionally, the challenge of relational data, where units influence each other, is addressed in [177]. This work introduces the concept of relational acyclification, enabling sound and complete discovery of cyclic relational causal models under specific faithfulness assumptions, thereby extending causal discovery to interconnected systems that were previously intractable.

Finally, the application of cyclic causal modeling extends to recommendation systems and dynamic processes prone to echo chambers. [178] designs a causal graph with loops to model the iterative process of recommendation and user feedback. By analyzing the mathematical properties of these loops using Markov processes, the authors propose a model that mitigates echo chambers through counterfactual reasoning, demonstrating the practical utility of cyclic models in preventing harmful feedback loops in AI systems. Together, these diverse methodological advancements illustrate a paradigm shift from strictly acyclic assumptions to flexible frameworks capable of uncovering the rich, recursive causal structures that govern complex dynamic systems, setting the stage for addressing further complexities such as latent confounders and missing data in the following discussion.

3.6 Causal Discovery under Latent Confounders and Missing Data

3.6 Causal Discovery under Latent Confounders and Missing Data

Building upon the relaxation of acyclicity discussed in the previous section, another critical frontier involves addressing the violations of causal sufficiency and data completeness. While Section 3.5 explored how feedback loops challenge the Directed Acyclic Graph (DAG) assumption, real-world applications frequently encounter two additional, compounding obstacles: the presence of unobserved common causes (latent confounders) and incomplete observations (missing data). When causal sufficiency fails, standard methods often misattribute causal effects or conflate direct causation with spurious correlations induced by hidden variables. Similarly, missing data creates a "chicken-and-egg" dilemma where robust imputation requires knowledge of the causal structure, yet structure recovery ideally demands complete data. Naive approaches that treat these issues separately—such as performing imputation prior to discovery or ignoring latent variables—often introduce bias that distorts the underlying data distribution. Consequently, recent advancements have shifted toward sophisticated, integrated algorithmic paradigms capable of simultaneously handling unobserved variables and incomplete records.

A primary strategy for tackling latent confounding involves deriving proxies for the unobserved space directly from the statistical properties of the observed data, moving beyond traditional constraint-based limitations. This is particularly challenging when confounders are causally related to one another or act as effects of observed variables. To address this, recent frameworks utilize rank information from the covariance matrix of observed variables to locate hidden nodes and determine their cardinalities. For instance, the Rank-based Latent Causal Discovery (RLCD) algorithm establishes necessary and sufficient conditions for identifying specific latent structural patterns, allowing for the discovery of the entire causal structure over both measured and hidden variables without relying solely on conditional independence tests [179]. Furthermore, in linear non-Gaussian latent variable models, the Generalized Independent Noise (GIN) condition provides a powerful tool for estimating latent graphs. By checking for statistical independence between linear combinations of observed vectors and other observed subsets, GIN can effectively d-separate variables via latent common causes, thereby identifying causal directions and locating latent confounders even when they are causally interconnected [95].

Complementing these algebraic approaches, generative models have emerged to explicitly model the latent space, offering flexibility in nonlinear settings. One innovative direction involves iteratively deriving proxies for the latent space from the residuals of an inferred graphical model. This method controls for the latent space during DAG estimation, improving structural inference in Gaussian graphical models and extending to nonlinear cases through autoencoder implementations [180]. Similarly, deep multi-modal structural equations leverage unstructured data, such as images or text, as proxy variables to infer latent confounders, significantly enhancing causal effect estimation in domains like genomics and healthcare where traditional tabular proxies are insufficient [181].

Parallel to the challenge of latent confounders is the pervasive issue of missing data. To overcome the suboptimality of two-step procedures, researchers have developed integrated frameworks that perform imputation and structure learning simultaneously. A notable approach is MissDAG, which operates under the expectation-maximization (EM) framework for additive noise models. By maximizing the expected likelihood of the visible parts of the observations and leveraging density transformations in the M-step, MissDAG avoids the pitfalls of separate imputation and demonstrates flexibility in incorporating various causal discovery algorithms [182]. This integration of missing data handling with structure learning is conceptually aligned with the cyclic modeling challenges discussed earlier; indeed, recent work has extended these capabilities to learn cyclic causal graphs from partially missing data. The MissNODAGS framework, for example, alternates between imputing missing values and maximizing the expected log-likelihood of the visible data, effectively relaxing the acyclicity assumption while handling incomplete observations in biological and other feedback-rich systems [175]. Alternatively, optimal transport theory offers a distinct perspective by formulating structure learning as a density fitting problem. This approach seeks a causal model that induces a distribution with minimum Wasserstein distance to the observed data distribution, providing superior scalability and avoiding the local optima often encountered in EM-based methods [183].

Deep learning architectures have further advanced this field by combining encoders with reinforcement learning (RL) to handle incomplete data without explicit imputation. In these encoder-RL frameworks, the encoder is designed to extract robust feature representations from available data points despite missing values. These representations then guide an RL agent to search the space of causal graphs, optimizing the structure directly from the incomplete input. This end-to-end approach has been shown to outperform the direct combination of standard imputation and causal discovery methods, achieving significant performance gains in both synthetic and real-world scenarios [184]. Similarly, variational autoencoders adapted for missing values, such as those in the MissDeepCausal framework, learn the distribution of latent confounders to facilitate causal inference, effectively embedding the imputation process within the causal modeling procedure itself [185].

Finally, the integration of interventional data with unknown targets offers a powerful pathway to resolve ambiguities caused by both latent confounders and missingness. Bayesian frameworks like BaCaDI enable the joint inference of causal structures and unknown intervention targets via differentiable variational inference, proving particularly effective in biological settings where intervention targets (e.g., gene knockouts) may be uncertain [186]. Moreover, when latent variables are subject to soft interventions, identifiability guarantees can still be achieved even without full observation of the causal variables, provided a generalized notion of faithfulness holds. These methods allow for the recovery of latent causal models up to an equivalence class, enabling predictions of unseen combinatorial perturbation effects [187]. Collectively, these advancements signify a shift toward more robust, integrated algorithms capable of uncovering true causal mechanisms amidst the noise of hidden variables and incomplete records. However, while these methods improve robustness to data imperfections, they often incur significant computational costs as the number of variables grows. This tension between robustness and efficiency sets the stage for the next section, which explores scalable optimization, meta-learning, and knowledge-guided strategies to make causal discovery feasible for large-scale, high-dimensional systems.

3.7 Scalability, Meta-Learning, and Knowledge-Guided Discovery

3.7 Scalability, Meta-Learning, and Knowledge-Guided Discovery

While the methodologies discussed in Section 3.6 effectively address the challenges of latent confounders and missing data, their practical deployment in real-world systems often encounters a new set of bottlenecks: computational intractability and data scarcity. As the number of variables increases, the search space for valid causal graphs grows super-exponentially, rendering exact methods infeasible for large-scale problems. Furthermore, the robust algorithms designed to handle hidden variables and incomplete records often incur significant computational overhead, limiting their applicability to high-dimensional domains. Consequently, recent algorithmic paradigms have shifted towards scalable optimization techniques, meta-learning frameworks that leverage distributional priors to mitigate data sparsity, and hybrid approaches that integrate domain knowledge to constrain the search space.

A primary avenue for achieving scalability involves reformulating causal discovery as a continuous optimization problem or utilizing efficient score-matching techniques. Traditional constraint-based methods, which were extended in previous sections to handle latent variables, often struggle with the combinatorial explosion of conditional independence tests required for high-dimensional data. In response, score-matching approaches have emerged as a powerful alternative, capable of recovering causal structures with significantly reduced computational overhead. Specifically, recent work demonstrates that discovering the entire causal graph can be achieved by analyzing the second derivative of the log-likelihood in observational non-linear additive Gaussian noise models. By leveraging scalable machine learning approaches to approximate the score function, algorithms such as Discovery At Scale (DAS) have been developed to reduce the complexity of the pruning step by a factor proportional to the graph size [188]. This approach not only achieves competitive accuracy compared to state-of-the-art methods but also operates over an order of magnitude faster, effectively lowering the computational barrier for principled causal discovery in massive datasets. Furthermore, theoretical analyses provide statistical sample complexity bounds for these score-matching methods, confirming that accurate estimation of the score function—and consequently the causal graph—is achievable using standard deep neural networks trained via stochastic gradient descent [189].

Beyond direct optimization, meta-learning strategies offer a promising direction for inferring graph structures by learning from distributions of data rather than single instances, directly addressing the data scarcity that often plagues complex causal modeling. The core hypothesis is that causal mechanisms often share underlying structural properties across different environments or tasks, allowing models to generalize to unseen domains with limited samples. Meta-reinforcement learning has been successfully applied to this end, where agents learn to perform optimal interventions to construct explicit causal graphs. These algorithms demonstrate the ability to estimate accurate causal structures even in environments with previously unseen underlying causal dynamics, highlighting the potential of meta-learning to boost performance where traditional methods fail due to finite data [153]. Additionally, structured meta-learning frameworks explicitly disentangle meta-learners into hierarchical graphs, enabling the model to construct knowledge pathways that adapt quickly to new tasks by reusing relevant knowledge blocks [190]. This is particularly relevant in scientific domains where data is sparse; for instance, meta-learning has been employed to transfer knowledge from large-scale brain network datasets to smaller, data-scarce connectome studies, achieving higher stability and performance than standard pre-training strategies [191]. The theoretical underpinnings of such approaches suggest that non-convex formulations in meta-learning can achieve constant sample complexity for new tasks, a significant improvement over convex counterparts, by effectively learning the correct subspace for data projection [192].

Integrating expert knowledge and prior constraints represents a third critical pillar for enhancing both the scalability and reliability of causal discovery, serving as a necessary complement to purely data-driven approaches that may suffer from identifiability issues. To mitigate the risk of spurious correlations when data is noisy or incomplete, researchers have developed hierarchical wrappers that recursively cluster observed variables using criteria such as normalized min-cuts before applying baseline causal discovery algorithms to local sub-graphs. This strategy significantly reduces the number of required statistical tests, leading to more accurate graphs and shorter runtimes while preserving soundness and completeness [193]. Similarly, in the realm of exact score-based methods, scalability has been improved by estimating a “super-structure” based on the support of the inverse covariance matrix, which requires assumptions strictly weaker than faithfulness. This super-structure restricts the search space for exact algorithms, allowing them to scale to hundreds of nodes with high accuracy [194].

Furthermore, the integration of domain knowledge is essential in specialized fields like materials science and healthcare, where data quality and scarcity are persistent challenges that demand more than just algorithmic efficiency. In computational materials discovery, machine learning models are increasingly used to act as “decision engines” that predict the likelihood of calculation success and detect strong correlations, thereby guiding autonomous workflows and avoiding failed experiments [195]. In healthcare, bootstrap-based algorithms have been proposed to decouple the learning of variable ordering from structural learning, improving the recovery of causal dependencies in cancer driver mutations where standard approaches struggle with identifiability [196]. The synthesis of these scalable optimization techniques, meta-learning priors, and knowledge-guided constraints forms a robust framework for next-generation causal discovery, enabling the analysis of complex, high-dimensional systems that were previously intractable. As the field advances, the convergence of these methodologies promises to bridge the gap between theoretical causal identifiability and practical, large-scale deployment in scientific and industrial applications.

4 Architectures for Learning Causal and Disentangled Representations

4.1 Variational Autoencoders and Hierarchical Structural Priors

4.1 Variational Autoencoders and Hierarchical Structural Priors

While flow-based architectures discussed in the following section offer exact likelihood computation to address identifiability, the foundational pursuit of causal representation learning has long centered on the Variational Autoencoder (VAE) framework. VAEs are prized for their probabilistic grounding and capacity for unsupervised disentanglement; however, standard architectures often fail to recover true causal factors due to their reliance on independent latent variables with simple, isotropic Gaussian priors. This assumption fundamentally mismatches real-world generative processes, where causal mechanisms typically exhibit complex, structured dependencies and hierarchical relationships. To bridge this gap, recent advancements have focused on integrating hierarchical structural priors and graphical model constraints directly into the VAE architecture. These enhancements impose inductive biases that align the latent space topology with the underlying causal graph, thereby facilitating the identification of interpretable, multimodal latent distributions while addressing persistent optimization challenges such as posterior collapse and factor over-pruning.

A primary obstacle in classical VAEs is posterior collapse, where the inference network ignores latent variables, rendering the learned representation uninformative for causal reasoning. This issue is exacerbated in deep latent hierarchies, where models may fail to utilize higher-level abstractions. Research indicates that merely stacking layers is insufficient; explicit structural constraints are required to ensure effective information flow. The Structured Variational Autoencoder (SVAE) represents a pivotal approach in this regard, composing graphical models with deep learning architectures to inherit both the interpretability of probabilistic graphical models and the flexible likelihoods of neural networks [197]. By incorporating discrete latent variables and structured priors, SVAEs effectively handle multimodal uncertainty and learn interpretable discrete data representations, which are crucial for modeling distinct causal regimes. Furthermore, optimization innovations such as memory-efficient implicit differentiation and natural gradient methods have rendered these structured models tractable, allowing them to compete with state-of-the-art time-series models while maintaining structural integrity [197].

Beyond discrete structures, continuous hierarchical priors have been explored to better model the variance and dependency structures of causal factors. The Hyperprior Induced Unsupervised Disentanglement approach posits that statistical independence in the latent space can be enforced more principledly through a hierarchical Bayesian framework rather than standard Evidence Lower Bound (ELBO) modifications [198]. By augmenting the VAE with an inverse-Wishart prior on the covariance matrix of the latent code, this method enables adaptive tuning of independence among latent dimensions. This capability is critical for causal discovery, as it allows the model to distinguish between truly independent causal mechanisms and those spuriously correlated due to environmental factors. Similarly, the Bayes-Factor-VAE introduces hyper-priors on the variances of Gaussian latent priors to mimic an infinite mixture, effectively partitioning latent dimensions into relevant causal factors and nuisance variables [199]. This distinction prevents the conflation of meaningful causal drivers with irrelevant noise, a common failure mode in standard disentanglement tasks.

Addressing the interrelated challenges of over-regularization and posterior collapse in hierarchical settings, the Hierarchical Empirical Bayes Autoencoder (HEBAE) proposes placing a hierarchical prior over the encoding distribution to adaptively balance reconstruction loss and regularization [200]. This approach avoids the discontinuities associated with deterministic moment-matching techniques, producing more robust convergence essential for stabilizing the learning of complex causal graphs. Moreover, the problem of factor over-pruning, where significant numbers of stochastic factors become inactive, has been tackled through epitomic approaches. The epitomic variational autoencoder (eVAE) groups mutually exclusive latent factors into "epitomes" that compete to explain the data, ensuring efficient use of model capacity and preventing the collapse of causal factors [201]. This concept has been successfully extended to graph-structured data via the Epitomic Variational Graph Autoencoder, which mitigates over-pruning in graph datasets and enhances the learning of diverse, interpretable representations necessary for causal analysis on non-Euclidean domains [202].

The integration of temporal dynamics further enriches the causal capabilities of hierarchical VAEs. In sequential data, causal factors often evolve over time, necessitating models that can disentangle static content from dynamic transitions. The Factorized Hierarchical Variational Autoencoder (FHVAE) formulates a hierarchical generative process specifically for sequential data, learning disentangled representations that separate session-specific factors from utterance-specific factors [203]. Although originally applied to speech, the principles of FHVAE are directly applicable to learning causal dynamics in time-series data, provided scalable training algorithms are employed. Additionally, predictive constraints have proven effective in enhancing the robustness of latent representations in time-series contexts. By modifying the VAE objective to predict future time steps, models can mitigate the learning of spurious features and recover latent factors that faithfully represent underlying causal dynamics [204].

Finally, to achieve finer control over dependency structures, recent work has incorporated graphical residual flows to specify arbitrary dependencies within the VAE prior and inference network, as seen in the SIReN-VAE [205]. By encoding conditional independence constraints directly into the flow architecture, these models approximate complex posteriors that adhere to specific causal graph structures, overcoming the limitations of mean-field assumptions. This is complemented by approaches like the Neural Decomposition method, which adapts functional ANOVA to decompose variation in Conditional VAEs, separating marginal additive effects from non-linear interactions to achieve identifiability under specific constraints [206]. Together, these architectural innovations demonstrate that imposing hierarchical structural priors and graphical constraints is not merely an optimization trick but a fundamental requirement for recovering causal representations. By moving beyond simple independent priors to structured, hierarchical, and competitive latent spaces, modern VAEs provide the necessary foundation for disentangling complex causal mechanisms, setting the stage for the more rigorous identifiability guarantees offered by the causal flow architectures discussed next.

4.2 Causal Flows and Identifiable Non-Linear Mixing

4.2 Causal Flows and Identifiable Non-Linear Mixing

Building upon the hierarchical structural priors introduced in VAEs to mitigate posterior collapse and enforce dependency constraints, the pursuit of causal representation learning has increasingly turned toward flow-based architectures to address the more fundamental challenge of identifiability in non-linear mixing regimes. While Section 4.1 demonstrated how structured priors can align latent topologies with causal graphs, standard VAEs still rely on variational approximations and the Evidence Lower Bound (ELBO), which often lead to suboptimal solutions and theoretical ambiguities regarding the uniqueness of the recovered factors. Normalizing flows offer a distinct advantage in this regard through their invertible nature and exact likelihood computation. This subsection details how flow-based models, specifically leveraging autoregressive structures and explicit causal layers, provide a rigorous framework for recovering true generative factors, ensuring theoretical identifiability, and enabling tractable counterfactual inference that surpasses mere statistical independence.

The core theoretical insight driving this paradigm is the intrinsic correspondence between autoregressive normalizing flows and identifiable causal models. Autoregressive flow architectures inherently define a strict ordering over variables, which is analogous to a causal ordering in Structural Causal Models (SCMs). By exploiting this property, researchers have demonstrated that causal models derived from both affine and additive autoregressive flows with fixed orderings are identifiable, meaning the true direction of causal influence can be recovered from observational data [207]. This finding generalizes the well-known Additive Noise Model (ANM) used in causal discovery, extending its applicability to arbitrary non-linear causal dependencies without restrictive linearity assumptions. Furthermore, the ability of flow models to estimate normalized log-densities allows for the derivation of bivariate measures of causal direction based on likelihood ratios, providing a robust mechanism for distinguishing cause from effect in complex systems [208].

Building upon these foundational identifiability results, several architectural innovations have emerged to integrate explicit causal structures into generative models, effectively bridging the gap between the flexible likelihoods of flows and the structured latent spaces discussed in the previous section. A prominent example is the Causal Flows Variational Autoencoder (CauF-VAE), which introduces a variant of autoregressive flows termed "causal flows." These flows are designed to incorporate the true causal structure of generative factors directly into the transformation process. In the CauF-VAE framework, a causal layer transforms independent exogenous factors into causally related endogenous latent variables, effectively modeling the dependencies observed in real-world data where generative factors are rarely independent [209]. Theoretical analysis of CauF-VAE, particularly when supervised by ground-truth factor information, confirms the identifiability of the learned disentangled representations. This architecture not only achieves superior performance in causal disentanglement tasks but also facilitates intervention experiments, allowing the model to simulate the effects of manipulating specific latent factors while holding others constant.

Similarly, the CausalVAE framework addresses the limitation of standard VAEs that assume independence among latent factors by incorporating a structured causal layer, ensuring that latent variables correspond to causally related concepts rather than statistically independent components [210]. The model's identifiability is analyzed under conditions where supervision signals, such as feature labels, are provided, demonstrating that the true causal Directed Acyclic Graph (DAG) can be identified with high accuracy. This capability enables the generation of counterfactual data through explicit "do-operations" on the causal factors, a critical feature for reasoning about alternative scenarios that standard generative models cannot reliably support.

Beyond VAE-hybrid approaches, purely flow-based frameworks like iFlow have been proposed to maximize marginal likelihood directly, thereby dispensing with the variational approximations that often hinder identifiability in iVAE models [211]. The iFlow framework provides an analytical optimization objective that ensures theoretical guarantees on the recovery of true latent representations. By avoiding the intractability of KL divergence calculations between approximate and true posteriors, iFlow achieves more accurate recovery of latent sources, validating the effectiveness of flow-based models in settings where exact identifiability is paramount.

The utility of these causal flow models extends significantly into the realm of counterfactual inference, which resides at the highest rung of Pearl's Ladder of Causation. The invertible nature of normalizing flows naturally facilitates the evaluation of both interventional and counterfactual queries. For interventional queries, flows allow for direct manipulation of the latent space followed by forward generation, while counterfactual queries benefit from the ability to perform abduction (inferring exogenous noise from observations), action (intervening on the structural equations), and prediction (generating the counterfactual outcome) within a single unified architecture [207]. Recent advancements have further refined this capability by developing deep structural causal models that employ normalizing flows to enable tractable inference of exogenous noise variables, a crucial step often missing in other deep causal learning methods [212]. These models have been validated on high-dimensional data, such as medical imaging, demonstrating the capacity to answer complex causal questions involving association, intervention, and counterfactuals.

Moreover, the flexibility of causal flows allows them to handle complex data types and constraints, extending their applicability to diverse domains. For instance, causal-Graphical Normalizing Flows (c-GNFs) have been applied to large-scale real-world observational data to perform personalized policy analysis, capturing underlying SCMs without assuming specific functional forms and handling mixed discrete-continuous data through dequantization techniques [213]. Additionally, the integration of causal priors into scenario generation, as seen in Causal Autoregressive Flow (CausalAF), demonstrates how learning cause-and-effect mechanisms rather than mere correlations can significantly improve the efficiency and diversity of generating safety-critical scenarios in autonomous driving [214].

In summary, causal flows represent a powerful convergence of invertible generative modeling and causal theory, offering the rigorous identifiability guarantees that hierarchical VAEs strive for but often lack due to approximation errors. By leveraging autoregressive orderings, causal layers, and exact likelihood estimation, models like CausalVAE, CauF-VAE, and iFlow overcome the identifiability limitations of traditional deep generative models. They provide a mathematically grounded pathway to recover true non-linear mixing functions, disentangle causal factors, and perform reliable counterfactual reasoning. This establishes a solid foundation for causal discovery, setting the stage for the subsequent exploration of diffusion models, which, as discussed in the following section, offer a complementary pathway by maximizing mutual information to achieve causal interpretability in high-fidelity generative tasks.

4.3 Diffusion Models and Information-Maximizing Latent Spaces

4.3 Diffusion Models and Information-Maximizing Latent Spaces

While the preceding subsection highlighted how flow-based architectures achieve identifiability through invertible mappings and exact likelihoods, diffusion models offer a complementary yet distinct pathway for causal representation learning. The advent of diffusion models has revolutionized generative modeling, delivering unprecedented fidelity in synthesizing images, audio, and video. However, a critical limitation persists in their direct application to causal reasoning: the standard diffusion process, which gradually corrupts data with Gaussian noise, often yields latent spaces that lack explicit semantic structure or causal interpretability. In traditional formulations, latent variables function primarily as mechanisms for stochastic sampling rather than as disentangled factors representing high-level causal mechanisms. To bridge this gap between high-fidelity generation and causal disentanglement—addressing challenges similar to those faced by non-invertible VAEs but with different architectural tools—recent research has pivoted toward architectures that explicitly maximize mutual information between observed data and latent variables or employ multi-stage reconstruction pipelines to enforce semantic consistency. These approaches transform diffusion models from purely generative engines into powerful tools for discovering and manipulating causal factors of variation.

A foundational advancement in this domain involves regularizing the learning objective to prevent "posterior collapse," a phenomenon where expressive decoders ignore the latent code, thereby undermining causal disentanglement. The [215] paper addresses this by proposing an algorithm that augments diffusion models with low-dimensional latent variables designed to capture high-level factors of variation. By incorporating a learning objective regularized with the mutual information between observed and hidden variables, [215] ensures that the latent space retains semantic meaning rather than being bypassed by the expressive diffusion decoder. This mutual information maximization is crucial for causal representation learning, as it forces the model to isolate independent generative factors, thereby facilitating interventions and counterfactual reasoning. Empirical results from [215] demonstrate that this approach yields disentangled, human-interpretable representations that are competitive with state-of-the-art contrastive methods while maintaining the superior sample quality inherent to diffusion processes. This capability allows for the precise manipulation of attributes in generated samples, a prerequisite for testing causal hypotheses in synthetic environments.

Beyond general image domains, the integration of information-maximizing principles into diffusion models has yielded transformative results in neuroscience, particularly in decoding complex brain activities. The reconstruction of perceived natural images from functional Magnetic Resonance Imaging (fMRI) signals represents a quintessential causal discovery problem: inferring the external visual stimulus (cause) from neural responses (effect). The [216] study introduces "Brain-Diffuser," a two-stage framework that exemplifies the power of multi-stage reconstruction pipelines. In the first stage, a Very Deep Variational Autoencoder (VDVAE) reconstructs low-level properties and layouts from fMRI signals. In the second stage, a latent diffusion model is conditioned on predicted multimodal features to generate high-fidelity final images. This hierarchical approach effectively separates coarse causal structures from fine-grained details, mirroring the processing hierarchy of the human visual cortex. Similarly, [217] proposes a two-stage model that aligns semantic concepts and structural information separately. By utilizing VQ-VAE latent representations and CLIP text embeddings in the first stage, followed by structural alignment via CLIP visual features in the second, [217] achieves a cohesive reconstruction that surpasses previous methods in both semantic accuracy and structural fidelity. These works illustrate how decomposing the generative process into distinct causal stages—semantics versus structure—enables more robust and interpretable recovery of latent causes.

The challenge of ensuring faithfulness and consistency in these reconstructions has led to further innovations in guiding the diffusion process with domain-specific priors, addressing the tension between top-down generation and bottom-up perceptual constraints. For instance, [218] identifies that standard latent diffusion models often suffer from a lack of detail-driven bottom-up perception, leading to unfaithful reconstructions despite high semantic alignment. To rectify this, [218] introduces primary visual feature guidance in the form of gradients, effectively extending the bottom-up processing stream to complement the top-down generative prior. This ensures that the generated images not only match the high-level semantic intent but also adhere to the specific low-level causal constraints imposed by the neural data. Furthermore, the issue of analytical variability in neuroimaging has been addressed by [219], which treats different processing pipelines as stylistic variations and uses diffusion models to translate statistic maps across them, thereby isolating the invariant causal signals from pipeline-specific artifacts.

The versatility of these information-maximizing and multi-stage diffusion architectures extends to other complex domains requiring causal disentanglement, demonstrating their broad applicability beyond vision. In the realm of dialogue generation, [220] tackles the one-to-many problem by introducing continuous latent variables into the diffusion process, allowing for the generation of diverse responses that share the same underlying causal context. Similarly, in sequential recommendation, [221] represents items as distributions rather than fixed vectors, using the diffusion process to inject uncertainty and capture the multifaceted nature of user preferences, which can be viewed as latent causal factors driving behavior. Moreover, the application of diffusion models to biomedical signal synthesis, as seen in [222], demonstrates their capacity to generate high-fidelity, non-stationary multivariate signals, providing a robust platform for simulating causal interventions in data-scarce medical contexts.

Collectively, these advancements signify a paradigm shift where diffusion models are no longer viewed merely as samplers but as foundational architectures for learning causal and disentangled representations. By explicitly maximizing mutual information, as advocated by [215], and by employing sophisticated multi-stage reconstruction strategies evident in [216] and [217], researchers are endowing these models with the ability to discern and manipulate the underlying causal mechanisms of data. While Section 4.2 established identifiability through invertible flows and Section 4.4 will explore how geometric priors enforce disentanglement in object-centric settings, this subsection highlights how diffusion models leverage information maximization and hierarchical conditioning to achieve causal interpretability. This evolution is critical for developing AI systems that can reason about counterfactuals, generalize across domains, and provide interpretable explanations for their generative capabilities, ultimately bridging the gap between perception and causal reasoning.

4.4 Object-Centric, Spatial, and Geometric Disentanglement

4.4 Object-Centric, Spatial, and Geometric Disentanglement

While Section 4.3 demonstrated how diffusion models leverage information maximization and hierarchical conditioning to extract causal factors from high-dimensional data, a distinct yet complementary challenge arises in the domain of static visual scenes: the rigorous separation of intrinsic object properties from extrinsic spatial configurations. Standard generative models often fail to enforce this separation, resulting in latent spaces where semantic content (such as shape, texture, and identity) is entangled with geometric transformations (such as pose, rotation, and translation). This entanglement hinders robust reasoning and out-of-distribution generalization. To address this, a specialized class of architectures has emerged that explicitly incorporates geometric priors, group equivariance, and Riemannian manifold structures. These approaches move beyond the statistical independence assumptions prevalent in earlier sections, instead leveraging the physical symmetries of the data generation process to isolate causal factors. This focus on geometric structure provides a critical bridge between the high-fidelity generative capabilities discussed previously and the physics-guided temporal dynamics explored in the subsequent section, grounding causal representations in the fundamental symmetries of space.

A foundational strategy in this domain involves modifying Variational Autoencoders (VAEs) to explicitly parameterize spatial transformations within the generative process. The Spatial-VAE framework exemplifies this by formulating the generative model as a function of spatial coordinates, allowing the reconstruction error to be differentiable with respect to latent translation and rotation parameters [223]. By constraining specific latent variables to represent only rotation and translation, this architecture ensures that the remaining unstructured latent factors encode purely semantic content. This method has proven effective in domains requiring precise pose estimation, such as modeling continuous 2-D views of proteins in electron microscopy and analyzing galactic structures in astronomical imagery, where distinguishing orientation from intrinsic morphology is critical.

Building upon the concept of explicit spatial parameterization, recent advances have integrated group theory to enforce equivariance directly into the encoder and decoder architectures. The TARGET-VAE introduces a translation and rotation group-equivariant framework that jointly infers latent rotation, translation, and a rotation-translation-invariant semantic representation [224]. This approach addresses the pathologies of previous methods by ensuring that the semantic latent space remains stable regardless of the object's pose, significantly improving clustering performance even when training data is heavily corrupted by geometric noise. Similarly, the Polar Transformer Network (PTN) combines spatial transformer modules with canonical coordinate representations to achieve invariance to translation and equivariance to both rotation and scale [225]. By predicting a polar origin and applying a learned transformation, PTN effectively normalizes input variations, demonstrating state-of-the-art performance on datasets with complex clutter and perturbations.

Beyond Euclidean assumptions, sophisticated geometric disentanglement requires modeling latent spaces as curved manifolds to capture the continuous nature of physical transformations. The VTAE (Variational Transformer Autoencoder) addresses the limitations of linear projections in standard generators by minimizing geodesics on a Riemannian manifold [226]. By expanding the latent variable model to operate on a Riemannian manifold and utilizing a geodesic interpolation network, VTAE ensures smooth and plausible transitions between object states. This geometric perspective is further extended by Mixed-curvature Variational Autoencoders, which generalize latent spaces to products of constant curvature manifolds, allowing the model to adapt to data biases that Euclidean spaces cannot capture [227]. Furthermore, incorporating Riemannian Brownian motion priors instead of standard Gaussian priors enhances model capacity and provides a more principled geometric view of latent codes [228].

The integration of attention mechanisms with geometric algebra has also yielded powerful architectures for handling complex 3D data, extending these disentanglement principles to higher dimensions. The Geometric Algebra Transformer (GATr) represents inputs and hidden states within projective geometric algebra, offering a unified 16-dimensional vector-space representation that respects E(3) symmetries [229]. This allows GATr to process diverse geometric types, including points, vectors, and rotations, within a single scalable architecture, outperforming non-geometric baselines in tasks ranging from n-body modeling to robotic motion planning. For point cloud data, Tensor Field Networks utilize filters built from spherical harmonics to guarantee local equivariance to 3D rotations, translations, and permutations, eliminating the need for data augmentation to identify features in arbitrary orientations [230]. Complementary to this, Dual Quaternion representations have been employed to jointly describe rotations and translations of rigid motions, preserving the correlation between coordinate entities that real-valued networks often lose [231].

In scenarios where the transformation group is unknown or complex, self-supervised strategies offer a viable path to disentanglement without requiring explicit parameterization. The CARE framework introduces an equivariance objective in contrastive learning, theoretically proving that minimizing this objective forces input augmentations to correspond to rotations in the spherical embedding space [232]. This ensures that the embedding space retains sensitivity to important geometric variations while maintaining semantic consistency. Additionally, methods like Learning Symmetric Embeddings for Equivariant World Models propose encoding input spaces into feature spaces that transform in known manners under operations, facilitating the application of equivariant networks even when the exact effect of transformations on raw data is difficult to characterize [233].

Finally, the disentanglement of geometric deformation spaces extends to non-rigid shapes and volumetric data, pushing the boundaries of object-centric learning. Recent generative models factorize object geometry into rigid orientation, non-rigid pose, and intrinsic shape using spectral geometry and diffeomorphic flows, enabling tasks like pose transfer without supervision [234]. For volumetric patterns, architectures like ILPO-Net utilize Wigner matrix expansions to achieve inherent invariance to local pattern orientations, significantly reducing parameter counts while maintaining high accuracy in 3D recognition tasks [235]. Collectively, these architectures demonstrate that embedding geometric inductive biases—whether through group equivariance, manifold learning, or algebraic representations—is paramount for achieving causal, robust, and interpretable object-centric representations. While this subsection has established how static spatial symmetries enforce disentanglement, the following section will extend these causal principles to the temporal domain, exploring how physics-guided sequence modeling uncovers the dynamic laws governing system evolution over time.

4.5 Temporal Dynamics and Physics-Guided Sequence Modeling

4.5 Temporal Dynamics and Physics-Guided Sequence Modeling

While the previous section established how geometric priors and group equivariance enable object-centric disentanglement in static domains, extending causal representation learning to sequential data introduces the critical dimension of time. The transition from static image analysis to sequence modeling represents a fundamental leap, as it requires distinguishing genuine causal mechanisms driving system evolution from spurious temporal coincidences. To address this, recent architectures have increasingly integrated principles from dynamical systems theory, ordinary differential equations (ODEs), and physical laws into deep generative frameworks. These physics-guided sequence models aim not merely to predict the next frame in a video but to uncover the underlying governing equations and causal factors that dictate temporal progression. By imposing structural inductive biases derived from physics, these models achieve superior disentanglement, interpretability, and robustness to distribution shifts compared to purely data-driven recurrent approaches, setting the stage for the self-supervised regularization techniques discussed in the subsequent section.

A pivotal advancement in this domain is the development of frameworks capable of identifying causal factors from temporal sequences where interventions may occur. Traditional unsupervised methods often struggle to separate static content from dynamic causes without explicit supervision. However, the **CITRIS** framework addresses this by leveraging temporal sequences of images where underlying causal factors are subject to interventions. Unlike previous literature that often assumes scalar causal factors or full observability, **CITRIS** exploits the temporality of data and the observation of intervention targets to identify both scalar and multidimensional causal factors, such as 3D rotation angles. Theoretical proofs provided within this work establish identifiability even in general settings where only specific components of a causal factor are affected by interventions. Furthermore, by incorporating normalizing flows, **CITRIS** demonstrates the ability to extend these capabilities to pretrained autoencoders, facilitating sim-to-real generalization and proving that temporal interventions are a powerful signal for recovering true latent causal variables [236].

Parallel to intervention-based approaches, another significant strand of research focuses on embedding continuous-time dynamics directly into the latent space of variational autoencoders. Recurrent Neural Networks (RNNs), while popular, operate in discrete time steps and often fail to capture the smooth, continuous nature of physical processes, leading to error accumulation over long horizons. To overcome these limitations, models like **ODE2VAE** introduce a deep generative second-order ODE framework. **ODE2VAE** explicitly decomposes the latent space into momentum and position components, solving a system of second-order differential equations that inherently respect Newtonian physics. This physics-guided inductive bias allows the model to infer continuous-time latent dynamics from high-dimensional sequential data, such as motion capture or bouncing balls, without requiring supervision. The hierarchical organization of the latent space in **ODE2VAE** not only improves long-term motion prediction but also ensures that the learned representations align with meaningful physical states, distinguishing it from black-box ODE techniques [237]. Further analysis of **ODE2VAE** confirms its efficacy in learning meaningful latent representations across various physical motion datasets, validating the hypothesis that embedding physical laws into the generative process enhances the semantic quality of the learned dynamics [238].

Beyond specific ODE formulations, the broader paradigm of predictive coding offers a biologically inspired mechanism for uncovering causal dynamics. Deep predictive coding networks operate on the principle that the brain constantly generates top-down predictions and updates its internal model based on the prediction error (the difference between prediction and observation). **Deep Predictive Coding Networks** extend this concept by dynamically altering priors on latent representations in a context-sensitive manner. These networks capture temporal dependencies by using top-down information to modulate lower-layer representations, effectively filtering out noise and focusing on invariant features. Empirical results on natural video data demonstrate that this architecture can learn high-level visual features and exhibits robustness to structured noise, suggesting that the minimization of prediction error is a viable objective for learning causal representations in time-varying signals [239]. However, critical reviews of specific implementations like PredNet indicate that while the theoretical foundation is sound, practical implementations must carefully address how future predictions are conditioned to fully realize the benefits of predictive coding theory in complex real-world scenarios [240].

The integration of physical laws extends further into the discovery of explicit governing equations from raw visual data. Frameworks such as those proposed for uncovering closed-form governing equations combine encoder-decoder networks with sparse regression modules to distill analytical models directly from videos. These architectures learn a mapping between pixel coordinates and latent physical states, subsequently using numerical integrators to uncover parsimonious equations that govern the system's dynamics [241]. Similarly, **PhyDNet** introduces a two-branch architecture that explicitly disentangles physical dynamics, described by partial differential equations (PDEs), from unknown complementary information. By employing a recurrent physical cell inspired by data assimilation techniques, **PhyDNet** performs PDE-constrained prediction in the latent space, demonstrating superior performance in video prediction tasks and highlighting the importance of separating known physical constraints from unmodeled dynamics [242].

Moreover, the capability to handle non-Markovian dynamics, where the future state depends on the history of the system rather than just the current state, has been addressed by Neural Integro-Differential Equations (NIDE). This framework learns integral operators alongside differential components, enabling the modeling of complex systems like brain dynamics which exhibit memory effects. **Neural Integro-Differential Equations** demonstrate that decomposing dynamics into Markovian and non-Markovian constituents provides deeper insights into the underlying causal mechanisms, particularly in biological contexts where simple differential equations fail to capture the full temporal dependency [243]. Additionally, approaches like **Simple Video Generation using Neural ODEs** investigate time-continuous dynamics over continuous latent spaces, showing that trajectories learned via differential equations can be extrapolated to generate video frames beyond the training horizon, thereby enhancing the model's ability to reason about future causal outcomes [244].

Collectively, these architectures illustrate a paradigm shift from treating time as a mere sequence of indices to modeling it as a continuous, physically constrained dimension. Whether through the explicit identification of intervention targets in **CITRIS**, the second-order dynamics of **ODE2VAE**, the error-minimization of predictive coding, or the discovery of PDEs, these methods provide a robust foundation for learning causal representations that are interpretable, generalizable, and grounded in the fundamental laws governing the observed data. While these physics-guided approaches impose strong structural biases to uncover causal mechanisms, the following section explores how contrastive learning and self-supervised regularization can further refine these representations, offering flexible tools to enforce disentanglement even when explicit physical laws are unknown or difficult to parameterize.

4.6 Contrastive, Prototypical, and Self-Supervised Regularization

4.6 Contrastive, Prototypical, and Self-Supervised Regularization

While the preceding section demonstrated how embedding physical laws and temporal dynamics provides strong inductive biases for causal discovery, such explicit knowledge is not always available or easily parameterizable in complex real-world domains. Consequently, the field has increasingly turned to self-supervised regularization strategies—specifically contrastive learning and prototypical networks—to enforce causal disentanglement in the absence of explicit governing equations. These methods address a fundamental limitation of traditional Variational Autoencoders (VAEs), where the likelihood objective alone proves insufficient to separate independent generative factors without strong supervision. By leveraging data augmentations and structural constraints, these frameworks aim to isolate salient causal features by maximizing mutual information between consistent data views while penalizing dependencies between distinct latent subspaces. This approach effectively bridges the gap between statistical independence and causal mechanism separation, offering a flexible alternative to the rigid physics-guided models discussed previously and laying the groundwork for the modular intervention techniques explored in the subsequent section.

A prominent direction within this paradigm involves utilizing prototypical networks to impose geometric constraints on the latent manifold, thereby encouraging semantic consistency. The [245] framework exemplifies this by integrating a deep metric learning architecture trained via self-supervision. Unlike standard VAEs that rely solely on reconstruction loss and KL-divergence, [245] introduces a prototypical mechanism that constrains the mapping from the representation space to the data space. This ensures that controlled perturbations in latent variables correspond to semantically meaningful changes in data factors, achieving state-of-the-art disentanglement on benchmark datasets without requiring prior knowledge of the number of factors. Building on this, further refinements in prototypical contrastive learning have been proposed to mitigate the "coagulation" of examples in the embedding space, where intra-prototype diversity often collapses. The [246] study advocates for a balanced optimization strategy involving alignment, uniformity, and correlation losses. By revising the standard ProtoNCE loss to include a correlation term that increases diversity among prototypical features, this approach enhances the discriminability of latent representations, which is crucial for isolating distinct causal mechanisms in complex data distributions.

Complementing geometric constraints, contrastive regularization has emerged as a powerful tool for separating coupled representations, particularly in scenarios where factors are inherently dependent. The [247] model introduces a novel architecture that simultaneously factorizes the posterior using total correlation-driven decomposition while extracting dependencies via a neural Gaussian copula. A key innovation in [247] is the use of a self-supervised contrastive classifier that differentiates between disentangled and coupled representations, regularizing the learning process to eliminate entangled factors. Similarly, extending these principles to sequential settings, the [248] proposes a method to separate static (time-invariant) and dynamic (time-variant) factors. By maximizing the mutual information between inputs and latent factors while penalizing the mutual information between static and dynamic components, [248] utilizes contrastive estimations and data augmentation to introduce necessary inductive biases for temporal causal discovery, serving as a data-driven counterpart to the physics-based sequence models discussed earlier.

Beyond pure contrastive objectives, hybrid approaches combining contrastive learning with causal intervention principles have shown promise in removing spurious correlations where background information acts as a confounder. The [249] addresses background interference in visual representation learning by modeling the background as a confounder within a Structural Causal Model (SCM). This method employs a backdoor adjustment-based regularization to perform causal interventions, thereby alleviating background distractions and ensuring that the learned representations focus on foreground semantic information. In the realm of fairness and counterfactual reasoning, the [250] disentangles exogenous uncertainty into variables independent of interventions and those correlated without causality. This allows for the generation of counterfactual examples that preserve latent structures while flipping sensitive attributes, demonstrating how regularization can enforce causal consistency in fairness-aware models—a capability that directly informs the advanced counterfactual generation techniques detailed in the next section.

Furthermore, self-supervised strategies have been adapted to handle grouped observations and domain shifts, which are critical for robust causal representation in specialized fields. The [251] separates latent representations into semantically meaningful parts by operating at both group and observation levels, enabling the anchoring of semantics within specific data groupings. For medical imaging and other domains with limited annotations, methods like [252] utilize contrastive analysis to separate common factors from salient, dataset-specific patterns. [252] introduces regularization losses that enforce disentanglement between common and salient spaces, proving effective in distinguishing pathological features from healthy backgrounds. Additionally, the [253] framework demonstrates how weak supervision in the form of high-level labels can serve as an inductive bias to learn alternative disentangled representations, such as polar coordinates, without refined ground-truth factors.

Finally, the integration of these regularization techniques extends to addressing critical optimization challenges such as posterior collapse and dimensional collapse, which can otherwise undermine causal identifiability. The [254] introduces inference critics connected to contrastive learning objectives to increase the mutual information between observations and latents, thereby preventing the latent variables from becoming irrelevant. Similarly, [255] proposes local feature learning and feature decorrelation to counteract the high inter-image similarity in medical data, ensuring that the learned representations remain rich and informative. Collectively, these advancements illustrate that combining contrastive objectives, prototypical constraints, and self-supervised regularization provides a robust pathway to learning causal and disentangled representations. By reducing reliance on extensive labeled data and explicit physical models, these methods enhance the interpretability and robustness of deep generative models, setting the stage for the transition from passive representation learning to active modular interventions and counterfactual reasoning discussed in the following section.

4.7 Modular Interventions, Counterfactuals, and Robust Generation

4.7 Modular Interventions, Counterfactuals, and Robust Generation

Building upon the regularization strategies and self-supervised mechanisms discussed in the previous section, this subsection explores how deep generative architectures are evolving from passive observers of data distributions into active agents capable of performing modular interventions and generating robust counterfactuals. While contrastive and prototypical methods provide the inductive biases necessary to disentangle latent factors, the next critical step involves explicitly manipulating these factors to reason about “what-if” scenarios, address spurious correlations, and ensure out-of-distribution (OOD) generalization. By embedding causal mechanisms directly within generative frameworks, researchers have developed methods that not only isolate invariant causal factors from environmental noise but also enable precise manipulation of latent variables to synthesize data under hypothetical conditions.

A central theme in this domain is the development of modular architectures that allow for the independent manipulation of causal mechanisms. Traditional monolithic models often fail to adapt to distribution shifts because they entangle all predictive features, making it difficult to isolate specific causal pathways. In contrast, modular neural causal models factorize the data-generating process into distinct modules corresponding to the conditional distributions of variables given their causal parents. This decomposition enables adaptation to new environments by updating only a subset of parameters associated with the shifted mechanisms, thereby achieving robust zero-shot and few-shot adaptation [256]. Such modularity is critical for scaling causal reasoning to high-dimensional domains like imagery, where pre-trained foundation models can be leveraged.

For instance, sequential training algorithms have been proposed to adapt pre-trained conditional generative models into deep causal generative models. Approaches such as Modular-DCM utilize adversarial training to ensure the model can provably sample from identifiable interventional and counterfactual distributions, even in the presence of latent confounders, effectively bridging the gap between large-scale representation learning and rigorous causal inference [16].

Beyond structural modularity, significant advancements have been made in utilizing generative models for active causal reasoning through counterfactual generation. Counterfactuals provide a powerful lens for explainability and fairness, allowing practitioners to query how an outcome would change had a specific input variable been different. However, generating valid counterfactuals for high-dimensional data remains challenging due to the complexity of the underlying causal mechanisms. Recent frameworks have addressed this by employing diffusion models, which iteratively refine noise to generate samples. Diff-SCM, for example, builds on energy-based models to perform abduction and intervention steps within a diffusion process, producing realistic and minimal counterfactual images that adhere to known causal structures [257]. Similarly, other diffusion-based causal models encode nodes into latent representations that act as proxies for exogenous noise, enabling direct sampling under interventions and improving the fidelity of counterfactual estimation compared to prior state-of-the-art methods [17]. These approaches demonstrate that modern generative paradigms can be meticulously aligned with the three layers of Pearl's causal hierarchy to support complex reasoning tasks.

In scenarios involving noisy or corrupted data, generative models are increasingly employed to learn robust representations by explicitly modeling treatment effects. The Treatment Learning Causal Transformer (TLT) exemplifies this by incorporating binary information regarding the existence of noise as a treatment variable. By jointly estimating treatment effects and utilizing a latent generative model, TLT assigns inputs to specific inference networks based on the estimated noise level, significantly enhancing prediction accuracy in the presence of spurious correlations and domain shifts [258]. Furthermore, generative interventions can be synthesized to actively break spurious correlations during training. By steering generative models to manufacture interventions on features caused by confounding factors, researchers have created frameworks that learn representations consistent with underlying causal relationships, leading to state-of-the-art performance in OOD generalization tasks such as transferring from ImageNet to ObjectNet [259].

The application of these generative causal frameworks extends critically to the realm of algorithmic fairness and bias mitigation. Spurious correlations in training data often lead to attribute amplification, where unrelated attributes are inadvertently altered during interventions, exacerbating biases against protected groups. To counter this, soft counterfactual fine-tuning techniques have been introduced to mitigate attribute amplification in medical imaging, ensuring that generated counterfactuals remain faithful to the intended intervention without introducing unintended biases [260]. Moreover, generative adversarial networks (GANs) have been adapted to enforce counterfactual fairness directly. The Generative Counterfactual Fairness Network (GCFN) learns the counterfactual distribution of descendants of sensitive attributes, using this knowledge to regularize predictions and guarantee fairness even when latent variables are correlated with sensitive attributes [261]. Complementing this, data augmentation strategies guided by causal structures simulate interventions on spurious features, improving the generalization of text classifiers and reducing reliance on biased shortcuts [262].

Finally, the challenge of confounding in counterfactual generation has been addressed through methods that quantify and minimize correlations between generative factors. By characterizing the adverse effects of confounding and proposing methods to modify attributes independently, these frameworks generate counterfactual images that regularize downstream classifiers, ensuring representation consistency across various generative factors [263]. Additionally, specialized architectures like DR-VIDAL combine variational autoencoders with information-theoretic GANs to achieve doubly robust estimation of individual treatment effects, effectively handling selection bias and unobserved confounders in real-world observational data [264]. Collectively, these advancements illustrate a maturing field where generative models serve as the engine for modular, robust, and fair causal reasoning. While these methods establish the theoretical and algorithmic foundations for intervention and counterfactuals, the following section examines how these principles are adapted into advanced architectures specifically designed to handle the sparse, confounded, and domain-specific complexities of real-world scientific data.

4.8 Advanced Architectures for Sparse, Confounded, and Domain-Specific Data

4.8 Advanced Architectures for Sparse, Confounded, and Domain-Specific Data

Building on the theoretical foundations of modular interventions and counterfactual generation established in the previous section, this subsection examines how these principles are operationalized within advanced architectures designed to confront the messy realities of real-world scientific data. While Section 4.8 highlighted the capacity of generative models to perform "what-if" reasoning in idealized settings, practical application in high-stakes domains such as healthcare, genomics, and scientific imaging demands structures capable of navigating unobserved confounders, extreme data sparsity, and rigorous out-of-sample generalization requirements. To bridge the gap between causal theory and domain-specific practice, recent advancements have introduced specialized architectural paradigms that explicitly model confounding structures, leverage vector quantization for sparse representations, and enforce conditional invariance.

A primary obstacle in translating causal representation learning to practice is the pervasive presence of latent confounders that induce statistical dependencies between otherwise distinct causal mechanisms. Traditional variational autoencoders (VAEs), which often assume unconfoundedness, risk capturing spurious correlations rather than true causal semantics. To mitigate this, the framework of Confounded-Disentanglement (C-Disentanglement) was developed to explicitly introduce an inductive bias regarding confounders [265]. By utilizing labels derived from domain expertise, C-Disentanglement identifies causally disentangled factors even in the presence of common causes, demonstrating competitive performance in obtaining robust features under domain shifts. This approach underscores the necessity of integrating prior knowledge to break the identifiability barriers inherent in purely unsupervised settings. Furthermore, in scenarios involving continuous treatment variables where linear dependence assumptions fail, de-confounding representation learning frameworks utilize generative adversarial networks to disentangle covariate representations from treatment variables, effectively eliminating both linear and nonlinear confounding biases for accurate counterfactual inference [266].

Beyond addressing confounding, the challenge of generating reliable samples under unseen conditions—critical for tasks like drug response prediction—requires models that can generalize beyond the training distribution by treating interventions as sparse mechanism shifts. The transformer VAE (trVAE) addresses the limitations of standard conditional VAEs by matching distributions using maximum mean discrepancy (MMD) in the decoder layer [267]. This architectural innovation introduces strong regularization that not only improves reconstruction within known conditions but also enables accurate transformation across conditions, proving particularly effective in predicting cellular perturbation responses in high-dimensional single-cell gene expression data.

Similarly, the Sparse Additive Mechanism Shift Variational Autoencoder (SAMS-VAE) models the latent state of perturbed samples as a sum of local variations and sparse global intervention effects [268]. By sparsifying global latent variables, SAMS-VAE successfully identifies disentangled, perturbation-specific subspaces, offering a composable framework for understanding biological mechanisms of action in drug discovery contexts. These methods highlight a strategic shift towards architectures that exploit sparsity to facilitate better out-of-domain generalization.

In domains characterized by extreme sparsity, high dimensionality, and discrete underlying states, such as cancer subtyping and volumetric medical imaging, vector-quantized approaches have emerged as powerful tools to enforce discrete, interpretable representations. The Vector Quantized Variational AutoEncoder (VQ-VAE) has been successfully applied to transcriptomic data to extract informative latent features that retain only information relevant to reconstruction, thereby avoiding overfitting to spurious correlations common in omics data [269]. This capability allows for superior clustering performance and more robust prognosis compared to prevalent subtyping systems. In the realm of 3D medical imaging, VQ-VAE-inspired networks have demonstrated the ability to encode full-resolution brain volumes into a tiny fraction of their original size while preserving critical neuromorphological characteristics, enabling efficient storage and analysis without introducing bias during fine-tuning [270]. Extending this further, supervised vector quantized models like S-VQ-VAE combine supervised and unsupervised learning to obtain unique, interpretable global representations for each data class, revealing mechanism correlations between different perturbagens in gene expression data [271]. Additionally, dynamic vector quantization techniques have been proposed to encode image regions into variable-length codes based on information density, addressing the inefficiencies of fixed-length coding in autoregressive generation tasks [272].

The integration of these advanced architectures also extends to handling domain-specific constraints in medical imaging and fairness, ensuring that learned representations remain robust and equitable. For instance, models designed for single domain generalization in medical image classification leverage channel-wise contrastive disentanglement and style regularization to extract distinct feature representations despite distribution shifts [273]. In the context of fairness, the Disentangled Causal Effect Variational Autoencoder (DCEVAE) resolves the limitation of single latent variables absorbing all exogenous uncertainty by disentangling information caused by interventions from information merely correlated with them, thereby enabling the generation of counterfactually fair datasets [250]. Moreover, conditionally invariant representation learning approaches place distinct conditional priors on latent features to separate invariant biological signals from noise, facilitating data integration across different laboratories and conditions in single-cell genomics [274].

Collectively, these advanced architectures represent a significant leap forward in adapting causal representation learning to the complex constraints of scientific and medical data. By explicitly modeling confounders, enforcing sparsity through vector quantization, and designing mechanisms for robust out-of-sample generation, these models provide the necessary foundation for trustworthy AI in domains where data is often sparse, incomplete, and fraught with hidden biases. The synergy between domain-specific inductive biases and deep generative modeling ensures that learned representations are not only statistically sound but also causally meaningful and actionable, paving the way for the next generation of causal discovery systems capable of operating in open-world environments.

5 Causal Reasoning in Sequential Decision Making and Dynamic Environments

5.1 Nonlinear and Interpretable Extensions of Granger Causality

5.1 Nonlinear and Interpretable Extensions of Granger Causality

Granger causality has long served as a cornerstone for analyzing directional interactions in time-series data, traditionally relying on linear Vector Autoregressive (VAR) models to determine if the past of one variable aids in predicting the future of another. However, the assumption of linearity often fails to capture the complex, nonlinear dynamics inherent in real-world systems such as neuroscience, genomics, and econometrics. Consequently, recent advancements have shifted toward deep learning architectures that extend classical Granger causality to handle nonlinear relationships while maintaining a critical focus on interpretability. This shift directly addresses the "black-box" nature of standard neural networks by developing frameworks that not only detect causal links but also reveal the signs, magnitudes, and time-varying nature of these interactions, laying the groundwork for more robust causal analysis in dynamic environments.

A primary avenue for enhancing interpretability involves the adaptation of self-explaining neural networks (SENNs) to the temporal domain. Traditional deep learning approaches often obscure the specific contribution of input variables to the prediction output. In contrast, [275] proposes a novel framework that infers multivariate Granger causality under nonlinear dynamics by extending SENNs. This approach allows researchers to detect not only the existence of relational dependencies but also the signs of Granger-causal effects and their variability over time. By decomposing the prediction into interpretable components, this method offers a transparent alternative to sparse-input neural networks, achieving performance on par with powerful baselines while providing superior insight into interaction signs. Similarly, [276] introduces Neural Additive VAR (NAVAR) models, which utilize deep neural networks to extract additive Granger causal influences. This architecture successfully discovers nonlinear relationships in multi-variate time series, achieving state-of-the-art results on various benchmarks while ensuring that the mapped causal relations remain clearly interpretable through their additive structure.

Another significant direction involves leveraging the Jacobian matrix of neural networks to quantify variable importance and infer causal structure. The Jacobian provides a local measure of sensitivity, indicating how changes in input variables affect the output. [277] introduces JGC, an approach that uses the Jacobian as a measure of variable importance coupled with a thresholding procedure to infer Granger causal variables. This method consistently identifies causal variables, associated time lags, and interaction signs. Crucially, by incorporating a time variable, JGC can learn temporal dependencies in non-stationary systems where the underlying Granger causal structures evolve, offering a capability often lacking in static models and bridging the gap toward the non-stationary challenges discussed in subsequent sections. Complementing this, [278] builds upon earlier work by introducing the Learned Kernel VAR (LeKVAR) model, which learns a kernel parameterized by a neural network to handle underlying non-linearity. This work further demonstrates how decoupling lags and individual time series importance via decoupled penalties allows for better scaling and the embedding of lag selection directly into Recurrent Neural Networks (RNNs), facilitating the analysis of complex datasets like EEG signals in epilepsy studies.

The challenge of latent confounders, which can induce spurious causal conclusions in observational time series, has also been addressed through deep recurrent modeling. [108] harnesses the expressive power of RNNs to model nonlinear Granger causality while directly accounting for latent confounders. By employing a dual-decoder setup to conduct Granger tests, this approach effectively handles scenarios where a latent factor influences both cause and effect with different time lags, outperforming existing benchmarks in non-linear stochastic time series. Furthermore, [107] attempts to “recover” unobserved confounders by using a generative model with latent variables. This method estimates the posterior distribution of latent confounders using a Variational Autoencoder (VAE) combined with RNNs to build temporal relationships, offering a more robust estimation than unreliable proxy variables in semi-synthetic settings and foreshadowing the specialized treatments of hidden states in irregular environments.

Beyond recurrent architectures, attention mechanisms and point processes have been adapted to capture instance-level and event-based causality. [279] explores the utility of transformer models, demonstrating that the cross-attention module can effectively capture causal relationships among neurons in networks with complex nonlinear dynamics, matching or exceeding the accuracy of traditional Granger causality methods. For asynchronous event sequences, [280] proposes Instance-wise Self-Attentive Hawkes Processes (ISAHP). This framework is the first neural point process model to meet the rigorous requirements of Granger causality, leveraging self-attention to infer instance-level causal structures unsupervised, thereby handling complex interdependencies that classical models miss. Similarly, [281] introduces a framework that fits a neural point process and extracts Granger causality statistics using axiomatic attribution methods, achieving superior performance across diverse event interdependency scenarios.

Finally, innovations in computational efficiency and large-scale analysis have enabled the application of these nonlinear methods to high-dimensional data. [282] presents lsNGC, which models interactions via nonlinear state-space transformations to infer causal relations from short, high-dimensional recordings without explicit assumptions on functional interdependence. Additionally, [283] introduces a Koopman-inspired framework that lifts nonlinear dynamics to a higher dimension where they can be modeled linearly, simultaneously learning the lifting function and sparse lag matrices to reliably infer Granger causality. These diverse architectural advancements collectively signify a maturation of Granger causality, transitioning from rigid linear statistical tests to flexible, interpretable, and deep learning-driven frameworks. While these methods successfully relax linearity assumptions, they often presuppose regular sampling and stationary regimes; the following section extends this discourse by addressing the compounded complexities of non-stationarity, irregular sampling, and hidden states in dynamic environments.

5.2 Handling Non-Stationarity, Irregular Sampling, and Hidden States

5.2 Handling Non-Stationarity, Irregular Sampling, and Hidden States

While Section 5.1 successfully extended Granger causality to capture nonlinear dynamics and improve interpretability, those methods often presuppose regular sampling intervals and stationary regimes. Real-world sequential data, however, frequently deviates from these idealized assumptions, characterized by evolving data-generating processes, sporadic observations, and unobserved confounders. Addressing these compounded complexities requires a paradigm shift from static causal discovery to frameworks capable of explicitly modeling distribution shifts, handling irregular temporal granularity, and inferring latent structures from incomplete information. This transition lays the essential groundwork for the active decision-making scenarios discussed in Section 5.3, where agents must navigate non-stationary environments with hidden states.

Non-stationarity, traditionally viewed as a nuisance that degrades model performance, has been reframed by recent theoretical advancements as a valuable source of identification signal. By exploiting the principle that causal mechanisms often change independently, researchers have developed state-space models that utilize distribution shifts to disentangle causal structures that remain unidentifiable in stationary settings. For instance, approaches utilizing state-space models have demonstrated that allowing changes in both causal strengths and noise variances within nonlinear frameworks can render both the causal structure and model parameters identifiable [284]. This perspective is further supported by frameworks that treat non-stationary behavior as stationarity conditioned on a set of hidden state variables, effectively recovering underlying causal dependencies even when the global process appears volatile [285]. Moreover, the "sparse mechanism shift" hypothesis posits that distribution shifts are driven by changes in only a small subset of causal conditionals, allowing for the development of score-based approaches like the Mechanism Shift Score (MSS) that can provably identify the entire causal structure under these conditions [286]. Building on this, methods such as CD-NOD leverage independent changes in data distributions to detect causal skeletons and orientations without requiring window segmentation, proving that heterogeneity can benefit causal identification even in the presence of certain confounders [287].

Beyond identifying static structures in changing environments, modern frameworks aim to learn identifiable latent states that adapt to non-stationarity for improved forecasting and representation. The IDEA model, for example, formalizes the causal process by separating environment-irrelated stationary variables from environment-related nonstationary ones, using an autoregressive hidden Markov model to detect precisely when distribution shifts occur [288]. Similarly, the concept of Temporally Disentangled Representation Learning (TDRL) establishes identifiability theories for nonparametric latent causal processes, factorizing unknown distribution shifts into transition distribution changes and observation changes to recover time-delayed latent variables [289]. These principles are extended in frameworks like LiLY and NCTRL, which demonstrate that time-delayed latent causal influences can be reliably identified from nonlinear mixtures even under unknown nonstationarity, enabling efficient model correction with only a few samples from new environments [290; 291]. Furthermore, deep generative models such as the Deep Switching State Space Model (DS^{3}M) employ discrete latent variables to represent irregular regimes and continuous variables for random driving factors, successfully capturing complex nonlinear dependencies and regime-switching behaviors in diverse sectors ranging from healthcare to meteorology [166].

A second critical challenge in dynamic environments is irregular sampling, where observations are collected at non-uniform time intervals, rendering standard discrete-time models inadequate. To address this, novel architectures have been proposed that operate directly in continuous time or adapt to sparse inputs. The LipCDE framework, for instance, leverages Neural Controlled Differential Equations (CDEs) combined with Lipschitz regularization to model dynamic causal relationships and estimate treatment effects from irregular time series observations, even in the presence of hidden confounders and anomaly noise [292]. Complementing this, the Heteroscedastic Temporal Variational Autoencoder (HeTVAE) introduces a specific input layer to encode observation sparsity and a heteroscedastic output layer to reflect variable uncertainty, offering superior probabilistic interpolation for irregularly sampled data compared to homoscedastic baselines [293]. Alternative approaches include the Time-Parameterized Convolutional Neural Network (TPCNN), which parameterizes convolutional kernels as explicit functions of time to capture continuous-time hidden dynamics efficiently [294], and scalable Gaussian process adapters that connect irregularly sampled sequences to black-box classifiers while quantifying uncertainty [295]. Additionally, frequency-domain based estimation frameworks have been developed to naturally handle irregular sampling and long-range dependencies, eliminating the need for interpolation and enabling distributed causal analysis at scale [296].

Finally, the presence of hidden states and latent confounders necessitates methods capable of mining causal structures from sparse longitudinal sequences without inducing bias. Variable-lag causality methods address the limitation of fixed time-delay assumptions by utilizing Dynamic Time Warping (DTW) to infer causal relations where causes influence effects with arbitrary delays, a technique particularly useful in collective behavior and financial markets [297]. In scenarios involving subsampled time series where the measurement frequency is lower than the causal influence, proxy-based algorithms exploit the temporal structure itself—using future observations as proxies for hidden variables at unobserved time steps—to achieve full causal identification without parametric constraints [298]. Furthermore, constraint-based algorithms have been refined to handle autocorrelated and non-stationary data by optimizing conditioning sets and introducing surrogate variables to represent time dependency, thereby detecting both lagged and contemporaneous relationships alongside changing modules [35; 160]. Collectively, these advancements signify a maturing field where non-stationarity, irregularity, and hidden complexity are no longer barriers but rather informative features. By correctly modeling these dynamics, we unlock robust causal reasoning in passive observational settings, providing the fundamental theories of latent state recovery and dynamic adaptation required for the active intervention and planning strategies explored in the subsequent section on Reinforcement Learning.

5.3 Causal Representation Learning in Reinforcement Learning and Planning

5.3 Causal Representation Learning in Reinforcement Learning and Planning

Building upon the foundations for handling non-stationarity, irregular sampling, and latent states established in dynamic time-series analysis, the integration of causal representation learning into Reinforcement Learning (RL) represents a critical paradigm shift from statistical association to structural understanding. While Section 5.2 focused on recovering causal structures from passive, often noisy observational streams, this subsection addresses the active setting where agents must navigate complex, non-stationary environments through interaction. Traditional RL algorithms frequently struggle with distribution shifts or sparse interactions because they treat the environment as a black box of correlations. In contrast, causal RL seeks to uncover the underlying factored causal mechanisms that govern state transitions and rewards, leveraging the ability to intervene—a capability distinct from the purely observational constraints discussed previously.

A primary challenge in this domain remains non-stationarity, where environmental dynamics or objectives evolve over time. Recent approaches address this by modeling non-stationarity not as a monolithic shift but as the result of individual latent change factors interacting through a causal graph, extending the state-space perspectives from the preceding section into the decision-making realm. For instance, the Factored Adaptation for Non-Stationary RL (FANS-RL) framework learns a factored Markov Decision Process (MDP) alongside time-varying change factors, theoretically proving that the causal graph representing transition and reward functions can be recovered under standard assumptions [299]. Similarly, the Causal-Origin REPresentation (COREP) algorithm posits that non-stationarity propagates through causal relationships during state transitions; by learning a stable "causal-origin" representation, agents can trace the root causes of environmental changes, thereby exhibiting superior resilience compared to methods that merely model surface-level distribution shifts [300]. These methods demonstrate that disentangling invariant causal structures from transient environmental factors is crucial for maintaining policy performance in dynamic settings.

Beyond adapting to changing environments, a significant frontier is the emergence of causal reasoning through meta-reinforcement learning (meta-RL). Rather than hard-coding causal constraints, researchers are exploring whether agents can learn to perform causal inference and intervention planning end-to-end. Meta-RL agents trained on a diverse range of problems containing causal structures have been shown to develop the ability to select informative interventions, draw causal inferences from observational data, and even make counterfactual predictions in novel situations [152]. This emergent behavior suggests that causal reasoning can arise naturally from the pressure to maximize rewards in structured environments. Building on this, the Hypothesis Network Planned Exploration (HyPE) framework introduces a generative hypothesis network that forms potential models of state transition dynamics and actively designs experiments to eliminate incorrect models, significantly accelerating adaptation speeds in rapidly evolving tasks [301]. Furthermore, specific frameworks like "Learning by Doing" formalize a virtuous cycle where agents alternate between using interventions for causal structure learning during exploration and leveraging the learned causal graph to guide policy exploitation, providing theoretical guarantees for performance improvement [151].

Central to these advancements is active intervention planning, which allows agents to efficiently reconstruct causal graphs to inform their decision-making processes. While Section 5.2 discussed inferring causality despite irregular sampling and hidden confounders, active RL agents can strategically intervene to disambiguate causal directions that passive observation cannot resolve. Deep reinforcement learning has been successfully applied to the problem of experiment design, where an agent learns to sequentially intervene on variables to recover the causal structure with a minimum number of actions [302]. The CORE framework exemplifies this by learning to sequentially reconstruct causal graphs while simultaneously planning informative interventions, demonstrating scalability to larger graphs and superior sample efficiency compared to existing approaches [154]. Additionally, meta-reinforcement learning algorithms have been developed specifically to learn sequences of ego-centric actions that correspond to targeted manipulations of the environment, effectively bridging the gap between low-level action spaces and high-level causal interventions in latent structural settings [155].

The utility of these causal representations extends to improving the interpretability and generalization of RL policies in complex, multi-objective tasks. By learning factored causal representations, agents can decompose the state space into independent mechanisms, facilitating transfer learning and compositionality. The DECAF framework, for example, detects which causal factors learned in one environment can be reused or adapted in a new environment, enabling accurate representation learning with only a few samples [303]. In goal-conditioned RL, augmenting the agent with a causal graph allows for variational likelihood maximization that separates the relations between objects and events, leading to more generalizable models and interpretable policies [304]. Moreover, in domains with combinatorial action spaces, such as healthcare, leveraging factored action spaces induced by causal structures allows for linear Q-function decomposition, which improves sample efficiency and enables accurate inferences in under-explored regions of the state-action space [305].

Finally, the intersection of causal discovery and planning has led to novel algorithms that explicitly utilize causal graphs for decision optimization. Q-Cogni redesigns Q-Learning by integrating an autonomous causal structure discovery method, allowing the agent to query a pre-learned structural causal model to infer cause-and-effect relationships embedded in the state-action space, resulting in improved learning efficiency and policy optimality in high-dimensional problems like vehicle routing [306]. Similarly, approaches utilizing Temporal-Logic-based Causal Diagrams (TL-CD) capture temporal causal relationships to reduce the exploration space required for accomplishing temporally extended goals, leading to faster convergence to optimal policies [307]. These advancements collectively highlight that embedding causal representation learning into the core of RL agents is not merely an additive improvement but a fundamental requirement for achieving robust, adaptable, and interpretable autonomous intelligence. As we move to the next section, these principles of hierarchical and structured causal reasoning will be further expanded to address the complexities of multi-agent systems and asynchronous event dynamics.

5.4 Hierarchical, Multi-Agent, and Event-Based Dynamic Systems

5.4 Hierarchical, Multi-Agent, and Event-Based Dynamic Systems

Building upon the principles of active intervention and factored causal representations established in Reinforcement Learning (Section 5.3), this subsection addresses the escalation of complexity found in real-world sequential data, which often transcends simple linear temporal dependencies or single-agent interactions. Real-world systems frequently exhibit nested hierarchies, involve multiple interdependent agents, and operate through asynchronous event streams across varying timescales. In domains ranging from social coordination and robotics to epidemiological modeling and neuroscience, causal reasoning must therefore operate across multiple levels of abstraction. This section reviews specialized approaches that integrate hierarchical generative models, graph-based multi-agent systems, and neural point processes to uncover causal mechanisms in these dynamic environments, setting the stage for the broader generative forecasting frameworks discussed in the subsequent section.

A primary challenge in extending causal representation learning to these complex systems is the organization of latent representations across varying timescales. Real-world processes often possess a nested structure where high-level goals or contexts evolve slowly, while low-level actions or observations change rapidly. Hierarchical generative models have emerged as a powerful solution, enabling the disentanglement of spatiotemporal features and the detection of event boundaries without explicit supervision. For instance, the Variational Predictive Routing (VPR) framework organizes latent video features into a temporal hierarchy based on their rates of change, modeling continuous data as a hierarchical renewal process [308]. By employing an event detection mechanism driven solely by latent representations, such models can dynamically adjust internal states to promote an optimal organization of representations, facilitating accurate time-agnostic rollouts of the future. Similarly, the concept of “jumpy imagination” has been formalized through Variational Temporal Abstraction (VTA), a hierarchical recurrent state-space model that infers latent temporal structures to perform stochastic state transitions hierarchically [309]. These approaches allow agents to compress sensory streams into distinct event-like contexts, as seen in architectures utilizing surprise-gated recurrent networks that develop compact compressions of ongoing contexts to resource-efficiently model event-predictive cognition [310]. Furthermore, deep dynamic generative models, such as Deep Temporal Sigmoid Belief Networks, construct hierarchies of temporal belief networks to capture complex dependencies between subsequences, achieving state-of-the-art predictive performance in tasks like polyphonic music and motion capture [311].

In multi-agent systems, causal discovery becomes significantly more intricate due to the interdependence of entities and the emergence of group-level dynamics that cannot be reduced to individual behaviors. Effective modeling requires distinguishing between individual-level causality and collective system-level influences, extending the factored representations of single-agent RL to interactive settings. Hierarchical switching-state models have been developed to simultaneously explain both system-level and individual-level dynamics by employing a latent system-level discrete state Markov chain that drives latent entity-level chains [312]. This allows for the interpretation of group dynamics and improved forecasting of future entity behavior. Extending this to continuous time and space, the Trajectron utilizes dynamic spatiotemporal graphs to predict potential futures for multiple agents in highly dynamic scenarios, combining recurrent sequence modeling with variational deep generative modeling [313]. To explicitly capture the interactive nature of team behaviors, recent formulations introduce interactive hierarchical latent spaces that model group-level consensus alongside individual interactions, substantially improving trajectory prediction in team sports and pedestrian navigation [314]. Moreover, relational state-space models (R-SSM) leverage Graph Neural Networks (GNNs) to simulate joint state transitions of multiple correlated objects, incorporating relational information directly into the modeling of multi-object dynamics [315]. For scenarios involving continuous-time counterfactuals in multi-agent settings, such as vaccine distribution, models like CounterFactual GraphODE (CF-GODE) treat the system as a graph and use treatment-induced GraphODEs to estimate outcomes while removing confounding bias caused by inter-dependencies between units [316]. Additionally, latent variable sequential set transformers, known as AutoBots, generate scene-consistent multi-agent trajectories by alternately processing temporal and social dimensions, effectively approximating the true joint distribution of contextual and social information [317].

Parallel to these spatial and hierarchical concerns, the modeling of asynchronous event streams requires robust methods to discover causal graphs from irregularly sampled data, addressing limitations often encountered in passive observational settings. Neural point processes have become the standard framework for this task, yet traditional methods often struggle to uncover the underlying causal influence across diverse nodes. To address this, Graph Regularized Point Process (GRPP) models characterize event interactions across nodes by inductively learning node representations and uncovering influence strengths through graph regularization, thereby enhancing model interpretability [318]. Similarly, Graph Biased Temporal Point Processes (GBTPP) leverage structural information from graph representation learning to model both direct influence between nodes and indirect influence from event history, proving effective in modeling retweeting and news transmission [319]. For discovering instance-level causal structures in unsupervised manners, Instance-wise Self-Attentive Hawkes Processes (ISAHP) utilize self-attention mechanisms to align with Granger causality principles, enabling the discovery of complex causal structures at the event instance level [280]. Furthermore, hierarchical nonparametric point processes have been introduced to model content diffusion over social media, adapting their temporal and topical complexity to the data through a mixture of intensities and hierarchical dependent nonparametric approaches for event marks [320]. In scenarios involving missing events, unsupervised models based on marked temporal point processes can jointly learn the generative processes of observed and missing events, effectively imputing data and identifying optimal positions for missing events [321]. Finally, the integration of graph structures into latent variable models allows for the learning of dependency graphs within sub-intervals, removing non-contributing influences of past events to improve prediction accuracy in continuously observed event occurrences [322]. Together, these advancements highlight a paradigm shift towards unified frameworks that seamlessly integrate hierarchy, multi-agent interaction, and event-based causality. These structured representations provide the necessary foundation for the next section, which explores how such causal constraints are embedded into general generative modeling architectures to enhance forecasting and counterfactual reasoning.

5.5 Generative Modeling and Forecasting with Causal Constraints

5.5 Generative Modeling and Forecasting with Causal Constraints

Building upon the specialized architectures for hierarchical, multi-agent, and event-based systems discussed in the previous subsection, this section examines how causal constraints are integrated into general generative modeling frameworks to enhance sequential decision-making and dynamic environment analysis. While the aforementioned approaches address specific structural complexities, a broader challenge remains: traditional deep learning methods for time-series forecasting often rely on statistical correlations, rendering them susceptible to spurious relationships and distribution shifts. In contrast, generative models infused with causal priors offer a robust mechanism to disentangle true driving forces from environmental noise. This integration improves forecasting accuracy, enables valid counterfactual reasoning, and ensures out-of-distribution generalization. Here, we survey recent advancements where structural causal constraints are explicitly enforced during the training of recurrent variational autoencoders, diffusion models, and flow-based architectures.

A primary avenue of research involves adapting Variational Autoencoders (VAEs) to respect causal structures within temporal data. Standard VAEs often fail to capture the directional dependencies inherent in dynamical systems, leading to latent representations that mix causal factors. To address this, the Causal Recurrent Variational Autoencoder (CR-VAE) introduces a multi-head decoder architecture designed to learn a Granger causal graph directly from multivariate time series [323]. By imposing sparsity-inducing penalties on the decoder weights, CR-VAE ensures that the generation of each variable dimension is strictly conditioned only on its causal parents, effectively embedding the principles of Granger causality into the data generation process. This approach not only enhances the transparency of the generative model but also yields faithful causal graphs that outperform traditional Granger causality inference methods in complex domains like EEG and fMRI analysis [323]. Similarly, the concept of modular world models has been advanced through Variational Causal Dynamics (VCD), which exploits the invariance of causal mechanisms across different environments [324]. By jointly optimizing a representation model and a transition model structured as a causal graphical model, VCD can identify reusable components and adapt rapidly to unseen intervened environments, demonstrating that causally factorized transition models significantly extend the capabilities of latent world models in reinforcement learning contexts [324].

Beyond VAEs, diffusion models have emerged as powerful tools for enforcing causal constraints due to their ability to model complex, high-dimensional distributions through iterative denoising processes. The application of diffusion models to time-series forecasting allows for the generation of diverse and realistic scenarios while adhering to underlying physical or causal laws. For instance, the D3VAE framework combines a bidirectional VAE with a coupled diffusion probabilistic model to augment time-series data without increasing aleatoric uncertainty [325]. Crucially, D3VAE integrates multiscale denoising score matching and disentangles latent variables by minimizing total correlation, ensuring that the generated series move toward the true target while maintaining interpretability [325]. In the realm of constrained generation, guided diffusion models such as GuidedDiffTime have been proposed to tackle the problem of generating time series that satisfy specific numerical constraints, which is critical for creating valid counterfactual stress scenarios in finance and energy sectors [326]. Unlike rejection sampling or retraining-based approaches, GuidedDiffTime utilizes a constrained optimization framework to efficiently sample realistic time series that adhere to user-defined constraints, significantly reducing the computational carbon footprint while ensuring the realism of the generated data [326]. Furthermore, diffusion models have been adapted for spatiotemporal imputation through frameworks like C^2TSD, which incorporates disentangled temporal representations (trend and seasonality) as conditional information to guide the generative process, thereby improving stability and generalizability in missing data scenarios [327].

The generation of valid counterfactuals represents another critical application of these causally constrained generative models, particularly for explaining forecasting outcomes and assessing model robustness. Counterfactual explanations provide insights into how slight perturbations in input time series could lead to different forecasted outcomes, offering a level of interpretability absent in black-box models. The ForecastCF algorithm addresses this by applying gradient-based perturbations to original time series under specific constraints to generate meaningful counterfactual explanations for deep forecasting models [328]. This method ensures that the generated counterfactuals remain close to the data manifold while effectively altering the prediction, outperforming baselines in both validity and closeness [328]. On a broader scale, tools like CounterfacTS have been developed to probe the robustness of forecasting models against concept drift by allowing users to visualize and quantify the impact of hypothetical transformations on time-series data [329]. Such tools facilitate the creation of counterfactual datasets that cover regions of the data distribution not present in the original training set, thereby enhancing the model's ability to generalize to new regimes [329].

Moreover, the synergy between generative modeling and causal learning extends to improving long-range forecasting accuracy by balancing bias and variance. The Generative Forecasting (GenF) strategy employs a conditional Wasserstein GAN to generate synthetic data for intermediate time steps, which are then used alongside observed data to make long-range predictions [330]. This approach theoretically proven to better balance forecasting variance and bias compared to direct or iterative forecasting strategies, resulting in significantly lower prediction errors [330]. Similarly, the integration of causal constraints into autoregressive flows, as seen in Causal Autoregressive Flow (CausalAF), enables the efficient generation of safety-critical driving scenarios by learning the cause-and-effect mechanisms rather than mere correlations [214]. By utilizing causal masking operations, CausalAF ensures that the generated scenarios follow true causal relationships, significantly improving learning efficiency and the robustness of autonomous driving algorithms trained on such synthetic data [214].

Finally, the challenge of confounding in sequential data is being addressed through advanced generative frameworks that explicitly model and debias latent factors. Methods that leverage counterfactual data augmentation and causal regularization have shown promise in breaking spurious correlations caused by observed or unobserved confounders [263]. By learning to modify specific attributes in the generative process while holding others constant, these models can produce debiased datasets that facilitate the training of downstream classifiers robust to distribution shifts [331]. The convergence of these techniques—ranging from causal VAEs and diffusion models to flow-based generators—signals a maturing field where generative modeling is no longer just about mimicking data distributions but about understanding and replicating the underlying causal mechanisms that govern dynamic systems.

6 Enhancing Robustness, Generalization, Fairness, and Explainability

6.1 Causal Invariance for Out-of-Distribution Generalization and Domain Adaptation

6.1 Achieving Robust Generalization via Causal Invariance and Transportability

The pursuit of robust machine learning systems capable of operating reliably in non-stationary environments has positioned causal invariance as a cornerstone principle for Out-of-Distribution (OOD) generalization and domain adaptation. The fundamental premise driving this paradigm is that while statistical correlations between inputs and labels may fluctuate across different domains due to environmental changes, the underlying causal mechanisms generating the data often remain invariant. By identifying and leveraging these stable causal structures, models can disentangle semantic factors that are truly predictive of the target variable from spurious correlations that are specific to the training environment. This section surveys the theoretical foundations and algorithmic advancements that exploit causal invariance to achieve robust transfer learning, ranging from risk minimization frameworks to causal transportability theories.

A primary approach to enforcing causal invariance is through Invariant Risk Minimization (IRM) and its various extensions, which seek predictors that perform optimally across multiple training environments. The core intuition is that a predictor relying on causal features will maintain low risk regardless of the domain shift, whereas a predictor exploiting spurious correlations will suffer performance degradation when the environment changes. However, the practical application of IRM faces significant challenges, particularly in non-linear settings where the optimization landscape can be treacherous. Critical analysis reveals that IRM can fail catastrophically if the test distribution diverges too significantly from the training distribution or if the number of training environments is insufficient relative to the complexity of the spurious features [332]. Furthermore, in classification tasks where invariant features capture all information about the label, the invariance principle alone may be insufficient, necessitating the integration of information bottleneck constraints to filter out redundant or misleading information [333]. To address the sensitivity of IRM to geometric skews and pseudo-invariant features, recent work has proposed the Invariant Information Bottleneck (IIB), which formulates invariant causal prediction via mutual information, demonstrating superior performance over standard IRM baselines [334].

Beyond strict invariance, the concept of partial invariance has emerged to handle scenarios where the assumption of fully invariant mechanisms across all environments is too restrictive. In many real-world hierarchical or networked data settings, causal mechanisms may only be invariant within specific subsets of environments. The Partial Invariance framework addresses this by partitioning environments based on hierarchical differences and enforcing invariance locally, thereby mitigating the trade-off between fairness and risk that plagues global invariance methods [335]. Similarly, the notion of "distributional stability" has been introduced to quantify the stability of prediction mechanisms among sub-populations, leading to Stable Risk Minimization (SRM) algorithms that enhance model robustness without requiring the strong assumptions of strict invariance [336]. Another variant, Risk Extrapolation (REx), proposes reducing the variance of risks across training domains to extrapolate robustness to extreme distributional shifts, effectively recovering causal mechanisms even when input distributions contain both causal and anti-causal elements [337].

Causal transportability offers a complementary perspective, focusing on the ability to transfer causal knowledge from source to target domains even when the joint distributions differ. This is particularly relevant in unsupervised domain adaptation (UDA), where standard methods assuming covariate or conditional shifts often suffer from semantic loss due to unobserved confounders. The Transporting Causal Mechanisms (TCM) framework utilizes domain-invariant disentangled causal mechanisms to identify confounder strata in an unsupervised fashion, effectively removing confounding effects and achieving state-of-the-art performance on challenging UDA benchmarks [338]. In scenarios involving generalized target shifts, optimal transport maps can be learned to align pretrained representations while preserving the problem's structure, providing strong theoretical guarantees for recovery of target proportions and control of generalization risk [339]. Furthermore, the theory of hidden covariate shift posits a minimal assumption for domain adaptation, suggesting that learning a representation where the label distribution conditioned on the representation is domain-invariant is sufficient to handle both target shift and concept drift [340].

Multi-domain adaptation strategies have also evolved to explicitly separate semantic factors from environmental variations. The Causality Inspired Representation Learning (CIRL) algorithm enforces representations to be separated from non-causal factors, jointly independent, and causally sufficient, effectively simulating causal factors to improve generalization [341]. In settings where environment labels are unavailable, frameworks like Heterogeneous Risk Minimization (HRM) and its kernelized variant (KerHRM) jointly learn latent heterogeneity and invariant relationships, enabling stable predictions despite unobserved distribution shifts [342][343]. For time-series and evolving domains, methods such as Mutual Information-Based Sequential Autoencoders (MISTS) disentangle dynamic and invariant features to handle non-stationary tasks where new domains evolve continuously [344]. Additionally, approaches like Environment Inference for Invariant Learning (EIIL) directly infer environment partitions that are maximally informative for downstream invariant learning, outperforming methods that rely on pre-defined domain labels [345].

Finally, the integration of causal perspectives into graph learning and language modeling has expanded the scope of invariant learning. The Causality Inspired Invariant Graph LeArning (CIGA) framework extracts subgraphs containing maximal invariant intra-class information, ensuring immunity to distribution shifts in complex graph structures [346]. In natural language processing, Invariant Language Modeling adapts game-theoretic formulations of IRM to learn representations that ignore spurious correlations and generalize better across diverse textual environments [347]. Collectively, these methods underscore the transformative potential of causal invariance in building machine learning systems that are not only accurate but also robust, fair, and trustworthy in the face of the unpredictable shifts characterizing real-world deployments. While the aforementioned approaches focus on learning invariant representations through risk minimization and structural constraints, an alternative and increasingly prominent direction involves actively modifying the data distribution itself. The following section explores how causal interventions and data augmentation strategies are employed to explicitly decouple causal features from confounding artifacts, thereby debiasing models against spurious correlations.

6.2 Debiasing Spurious Correlations via Causal Interventions and Data Augmentation

6.2 Debiasing Spurious Correlations via Causal Interventions and Data Augmentation

While Section 6.1 established the theoretical imperative of causal invariance for robust generalization, relying solely on risk minimization frameworks often proves insufficient when spurious correlations are deeply entrenched in the training data. In many practical scenarios, the optimization landscape prevents models from naturally discarding non-causal shortcuts, necessitating a more proactive approach: actively modifying the data distribution itself. This section explores how causal interventions and data augmentation strategies are employed to explicitly decouple causal features from confounding artifacts. By leveraging the counterfactual framework to simulate interventions, these methods force models to learn representations that remain invariant across varying environmental conditions, thereby bridging the gap between the invariance principles discussed previously and the fairness objectives addressed subsequently.

A primary strategy in this domain is Counterfactual Data Augmentation (CDA), which involves generating synthetic training examples where specific causal or non-causal attributes are altered while holding others constant. The theoretical underpinning suggests that by exposing models to these "what-if" scenarios, the learning process is regularized to ignore spurious signals. For instance, research has demonstrated that augmenting training datasets with counterfactual examples can effectively break spurious correlations in image classifiers, provided the generation process accounts for the underlying confounding structure [263]. However, the efficacy of CDA is contingent upon the quality of the generated counterfactuals; if the generation process itself is biased or fails to respect causal constraints, it may inadvertently reinforce existing biases. To mitigate this, recent works have proposed formal analyses of how confounding biases impact downstream classifiers, arguing that removing these biases is essential for learning invariant features that generalize beyond the observed distribution [348].

In the realm of natural language processing (NLP), where manual annotation of counterfactuals is prohibitively expensive, automated and implicit augmentation techniques have gained prominence. One notable approach involves implicitly generating counterfactual features in the latent space rather than explicitly creating new data points, deriving a surrogate loss that approximates the effect of infinite augmentation [349]. This method enhances training efficiency while maintaining the robustness benefits of explicit augmentation. Furthermore, for text classification tasks where labels are spuriously correlated with specific attributes, strategies guided by causal structure have been employed to simulate interventions on these features. By matching examples based on auxiliary data and utilizing large language models to represent conditional probabilities, researchers have achieved superior OOD accuracy compared to standard invariant learning algorithms [262]. Similarly, automated generation of counterfactuals by substituting causal features with antonyms and flipping labels has proven effective in training robust text classifiers that emphasize causal features over non-causal ones [350].

Beyond simple augmentation, advanced frameworks integrate causal reasoning directly into the training objective through regularization and adversarial learning. The concept of Counterfactual Maximum Likelihood Estimation (CMLE) proposes performing maximum likelihood estimation on the interventional distribution rather than the observational one, thereby theoretically grounding the reduction of spurious correlations caused by observed confounders [351]. Complementing this, regularization approaches have been developed to separately penalize causal and spurious features, allowing practitioners to tune the model's reliance on each type of feature to enhance both robustness and fairness [352]. In multimodal settings, where biases can arise from complex interactions between modalities, causal information minimization has emerged as a powerful tool. By minimizing the information content of features derived from biased pretrained models, these methods learn confounder representations that can be subtracted to remove bias without sacrificing in-distribution performance [353].

Generative models, particularly Variational Autoencoders (VAEs) and Diffusion Models, play a pivotal role in synthesizing high-fidelity counterfactuals for debiasing. However, naive application of generative models can lead to "attribute amplification," where unrelated attributes are spuriously affected during interventions. To counter this, soft counterfactual fine-tuning techniques have been introduced to mitigate attribute amplification while maintaining the realism of generated images, a critical advancement for medical imaging applications [260]. Furthermore, frameworks like the Causal Counterfactual Generative Model (CCGM) incorporate partially trainable causal layers to learn causal relationships explicitly, enabling the generation of de-biased datasets that lie outside the domain of the original training data [331]. In conditional generation, spurious causality can lead to imbalanced label-conditional distributions; a two-step strategy involving fairness intervention and corrective sampling has been proposed to resolve these issues across varying levels of supervision [354].

The challenge of identifying which correlations are spurious without extensive human annotation has also been addressed through novel detection mechanisms. Methods such as "Counterfactual Alignment" allow for the detection of spurious correlations in black-box classifiers by analyzing the consistency of model responses to counterfactual inputs across different classifiers [355]. Additionally, techniques that amplify correlations before slicing the data have shown promise in discovering bias-conflicting slices, informing downstream mitigation strategies with greater precision [356]. For scenarios involving unobserved confounders, variational learning models have been developed to infer the posterior distribution of latent confounders, enabling more accurate counterfactual inference and debiasing even when the unconfoundedness assumption is relaxed [357].

Ultimately, the convergence of causal theory and generative modeling provides a robust toolkit for debiasing. Whether through the active generation of pairwise counterfactuals to reduce labeling costs [358], the use of diffusion models to diversify ensemble training and remove shortcut cues [359], or the application of causal bootstrapping during pre-training [360], these methods collectively shift the paradigm from passive statistical fitting to active causal reasoning. By ensuring that models rely on true causal features rather than fleeting statistical artifacts, causal interventions and data augmentation not only enhance robustness but also lay the groundwork for equitable AI. This focus on removing bias through intervention naturally transitions to the next critical dimension: the rigorous enforcement of counterfactual fairness to ensure individual equity in heterogeneous spaces, as detailed in the following section.

6.3 Achieving Counterfactual Fairness and Individual Equity in Heterogeneous Spaces

6.3 Achieving Counterfactual Fairness and Individual Equity in Heterogeneous Spaces

While the previous section detailed how causal interventions and data augmentation can debias models by breaking spurious correlations, this section shifts focus to the rigorous enforcement of **counterfactual fairness** as a specific equity objective. Counterfactual fairness demands that a predictor's outcome for an individual remains invariant even if their sensitive attributes (e.g., race or gender) were different in a counterfactual world. Achieving this standard in heterogeneous data spaces—ranging from tabular records to complex graphs and images—presents unique challenges regarding identifiability, model specification, and the disentanglement of causal factors. Recent advancements address these challenges by developing frameworks that explicitly disentangle sensitive attributes from causal representations, utilizing causal graph constraints, specialized fair metrics, and regularization techniques to ensure individual equity.

A fundamental prerequisite for counterfactual fairness is the generation of valid counterfactuals, which traditionally relies on a fully known Structural Causal Model (SCM). However, in real-world scenarios, the true causal graph is rarely known, and misspecification can lead to biased predictors. To address this, researchers have developed methods that operate under partially known causal structures. For instance, frameworks have been proposed to achieve counterfactual fairness using Partially Directed Acyclic Graphs (PDAGs), leveraging domain knowledge to identify ancestral relations without requiring a complete graph [361]. Similarly, other approaches formulate constrained optimization problems to balance fairness and accuracy when the causal graph is only partially known, ensuring that interventions on sensitive attributes do not propagate unfair effects through the network [362]. These methods are crucial for heterogeneous spaces where causal dependencies between diverse features are complex and often only partially understood.

To mitigate the reliance on explicit causal graphs, deep learning architectures have emerged that learn fair representations through disentanglement and variational inference. Variational Autoencoders (VAEs) have been adapted to separate sensitive information from task-relevant features. The Disentangled Causal Effect Variational Autoencoder (DCEVAE), for example, resolves limitations in prior latent causal models by disentangling exogenous uncertainty into variables independent of interventions and those correlated with them, allowing for the generation of counterfactually fair datasets while preserving original information [250]. Furthermore, the Counterfactual Fairness Variational AutoEncoder (CF-VAE) imposes structured representations based on domain knowledge to theoretically guarantee downstream counterfactual fairness [363]. In scenarios where sensitive attributes are unobserved due to privacy regulations, frameworks like SRCVAE infer a proxy for sensitive information within a causal variational autoencoding structure to perform adversarial bias mitigation [364].

In heterogeneous data environments such as social networks and recommendation systems, fairness must account for topological dependencies where a node's neighbors influence its outcome. Standard counterfactual fairness definitions often fail to capture these relational biases. To tackle this, novel notions like graph counterfactual fairness have been introduced, which consider the causal impact of neighbors' sensitive attributes on a target node [365]. Frameworks such as CAF select realistic counterfactuals from training data to learn fair node representations in Graph Neural Networks (GNNs), avoiding the generation of non-realistic perturbations [366]. Additionally, explainability in GNNs has been enhanced through counterfactual reasoning that perturbs topological structures to uncover user unfairness explanations, providing transparency in recommendation systems [367].

Beyond representation learning and graph structures, regularization techniques and metric learning play a pivotal role in enforcing individual equity across diverse modalities. The concept of a "causal fair metric" has been formulated to bridge causality, individual fairness, and adversarial robustness, defining a distance metric that reflects counterfactual proximity rather than mere statistical similarity [368]. This metric is instrumental in heterogeneous data spaces, enabling the creation of causal adversarial perturbations that regularize classifiers to be robust against attacks while maintaining fairness [369]. Moreover, regularization approaches have been developed to penalize features based on their causal role, emphasizing causal features while de-emphasizing spurious ones to build models that are both robust and fair [352]. Specific regularizations targeting the Controlled Direct Effect (CDE) have also been proposed to remove the direct impact of unprivileged group variables on outcomes, applicable across various model architectures [370].

The intersection of counterfactual fairness and individual fairness is further explored through methods that generate "antidote data" to remedy individual unfairness. Unlike traditional Distributionally Robust Optimization that may optimize on off-manifold samples, these approaches generate on-manifold antidote data that respects inherent feature correlations, ensuring that similar individuals are treated similarly even in the presence of sensitive attribute variations [371]. Additionally, the operationalization of individual fairness has been advanced by learning pairwise fair representations that capture data-driven similarity alongside side-information from fairness graphs, eliminating the need for human-specified distance metrics [372].

Despite these advancements, challenges remain regarding the sensitivity of counterfactual fairness to unmeasured confounding. Tools have been developed to assess how misspecification of the causal model, particularly through unobserved confounders, impacts fairness measures, highlighting the need for robust sensitivity analysis in deployed systems [373]. Furthermore, the relationship between counterfactual fairness and simpler group fairness metrics like demographic parity continues to be a subject of theoretical debate, with some studies suggesting equivalence under certain conditions while others emphasize the distinct individual-level guarantees provided by counterfactual approaches [374]. Ultimately, achieving counterfactual fairness in heterogeneous spaces requires a synergy of causal discovery, representation disentanglement, and robust regularization. This foundation naturally leads to the next frontier: integrating these causal fairness principles directly into adversarial training frameworks to simultaneously secure models against malicious perturbations and ensure equitable outcomes, as discussed in the following section.

6.4 Integrating Causality, Robustness, and Fairness through Adversarial Training

Building on the foundation of counterfactual fairness and individual equity established in the previous section, this subsection explores the critical convergence of causality, adversarial robustness, and fairness within unified training frameworks. While Section 6.3 detailed how to enforce equity through representation disentanglement and specialized metrics in heterogeneous spaces, it also highlighted the persistent vulnerability of these systems to model misspecification and the need for robust defenses. Historically, causality, robustness, and fairness have been investigated in isolation, often leading to fragmented solutions where optimizing for one property compromises the others. However, recent theoretical and empirical advancements reveal a deep synergy: causal reasoning provides the structural inductive biases necessary to develop models that are simultaneously robust to adversarial perturbations and fair across diverse data distributions. The core intuition driving this integration is that adversarial vulnerability frequently stems from a model's reliance on spurious correlations rather than true causal mechanisms. By explicitly modeling the causal structure of data generation—particularly regarding sensitive attributes identified in prior fairness discussions—researchers can design adversarial training frameworks that eliminate non-causal shortcuts, thereby enhancing both security and equity.

A fundamental challenge in unifying these domains is defining a metric that captures semantic similarity while respecting both causal structures and fairness constraints. Traditional adversarial metrics, such as ℓ_p norms, often fail to account for the underlying data distribution or the presence of sensitive attributes, potentially generating perturbations that are semantically invalid or discriminatory. To address this, recent work has proposed the development of causal fair metrics that bridge causality, individual fairness, and adversarial robustness [368]. These metrics utilize structural causal models (SCMs) to define a "fair distance" between individuals, ensuring that adversarial perturbations do not violate the counterfactual proximity principles discussed in Section 6.3 or exacerbate biases against protected groups. By formulating fairness through the lens of causal intervention, these approaches ensure that similar individuals—defined by their causal features rather than superficial correlations—are treated equitably even under attack.

Leveraging these metrics, novel adversarial training strategies have emerged that incorporate causal priors directly into the optimization objective. One prominent approach involves the use of causal adversarial perturbations, where the adversary is constrained to generate examples that respect the underlying causal graph, particularly concerning discrete sensitive attributes [369]. This method introduces a regularizer that jointly optimizes for classification accuracy, individual fairness, and robustness. The rationale is that by forcing the model to be invariant to causal adversarial perturbations, the learning algorithm is compelled to discard spurious features that are often the source of both adversarial fragility and algorithmic bias. This is particularly crucial in heterogeneous data spaces, where standard uniform perturbations may not reflect realistic threats or may inadvertently target specific subgroups [375].

Furthermore, integrating causality into adversarial training has proven effective in mitigating reliance on non-robust features. Theoretical analyses suggest that the origin of adversarial vulnerability lies in the model's focus on spurious correlations that do not generalize across distributions [376]. Methods like CausalAdv exploit this insight by aligning the natural and adversarial distributions through causal interventions, effectively removing the statistical artifacts that attackers exploit. This distribution alignment perspective extends to frameworks utilizing adversarial distribution balancing for counterfactual reasoning, which directly estimates potential outcomes to remove spurious causal relations [377]. By treating the adversarial setting as a domain adaptation problem where the "domain shift" is induced by causal interventions, these methods ensure that learned representations remain stable and fair regardless of the intervention applied to sensitive variables.

The synergy between these concepts is also evident in anti-causal prediction tasks, where the input is generated by the target label and protected attributes. In such settings, there is an explicit connection between the fairness criterion of separation and the robustness notion of risk invariance [378]. This connection implies that robustness-motivated approaches, such as those enforcing invariance across different environments or perturbations, can naturally enforce fairness criteria without requiring explicit fairness constraints. Conversely, fairness interventions can enhance robustness by preventing the model from overfitting to biased, non-robust features that are susceptible to adversarial manipulation.

Moreover, the complexity of real-world data necessitates moving beyond simple norm-bounded attacks to more sophisticated, semantically preserving perturbations. Approaches that combine adversaries with anti-adversaries in training have demonstrated that considering varied perturbation directions and bounds can lead to better trade-offs between robustness and generalization, while simultaneously improving fairness between classes [379]. This is complemented by methods that evaluate robustness through the lens of fairness confusion matrices, identifying perturbations that cause both accuracy drops and biased predictions [380]. Such holistic evaluation frameworks are essential for verifying that a model's robustness does not come at the cost of increased disparity among demographic groups.

Finally, the practical deployment of these integrated systems requires addressing the unique challenges of heterogeneous data, where features possess complex dependencies and semantic meanings. Non-uniform perturbations that respect feature correlations and importance have been shown to yield more realistic adversarial examples and stronger certification bounds in domains like finance and malware detection [381]. By embedding causal knowledge into the generation of these perturbations, training procedures ensure that resulting models are resilient to attacks that exploit the intricate structure of real-world data. In summary, the integration of causality, robustness, and fairness through adversarial training offers a unified pathway toward AI systems that are not only secure against malicious inputs but also equitable and consistent with underlying causal mechanisms. This robust, fair, and causally grounded foundation is a prerequisite for the next critical application: generating valid, actionable counterfactuals for explainable AI and recourse, as discussed in the following section.

6.5 Generating Valid Counterfactuals for Explainable AI and Recourse

6.5 Generating Valid Counterfactuals for Explainable AI and Recourse

Building upon the integration of causality, robustness, and fairness in adversarial settings, the generation of valid counterfactual explanations emerges as a critical application of causal representation learning for Explainable Artificial Intelligence (XAI). While the previous subsection focused on defending models against malicious perturbations, this section addresses the constructive generation of semantically meaningful changes to interpret black-box predictions and provide actionable recourse. Unlike feature attribution methods that merely highlight input importance, counterfactual explanations answer the pivotal "what-if" question by identifying minimal perturbations to an input instance that would alter the model's decision. However, the utility of these explanations hinges on their validity, plausibility, and actionability. Recent advancements have shifted focus from simple proximity-based optimization to sophisticated frameworks that incorporate causal reasoning, multi-objective trade-offs, and probabilistic guarantees, ensuring that generated counterfactuals are not only mathematically optimal but also feasible within the real-world causal mechanisms discussed in the context of robustness and fairness.

A primary challenge in counterfactual generation is balancing competing desiderata such as sparsity, proximity, diversity, and plausibility. Traditional approaches often collapse these objectives into a single weighted loss function, which requires difficult a priori tuning and may fail to capture the nuanced trade-offs required for high-quality explanations. To address this, researchers have proposed translating the counterfactual search into a multi-objective optimization problem. For instance, the Multi-Objective Counterfactuals (MOC) method generates a diverse set of explanations that explicitly visualize the trade-offs between different objectives, allowing users to select solutions that best fit their specific context [382]. Similarly, other frameworks utilize multi-objective genetic algorithms to handle mixed-type data while preserving underlying causal relationships, thereby avoiding the parameter tuning burdens associated with gradient-based methods [383]. Furthermore, exploring the trade-off between plausibility, change intensity, and adversarial power through multi-objective optimization has been shown to unveil counterfactuals that comply with human intuition, increasing user trust and aiding in bias detection [384]. This approach mirrors the adversarial strategies discussed previously, yet here the goal is not to maximize loss but to find the most interpretable path across the decision boundary.

Beyond optimization strategies, ensuring the probabilistic validity and robustness of counterfactuals is paramount, particularly in high-stakes domains like healthcare and finance. Uncertainty estimation has historically been neglected in counterfactual generation, leading to potentially misleading explanations when models face distributional shifts. Novel solutions now integrate techniques such as Monte Carlo Dropout and trust scores to quantify the uncertainty of generated explanations, ensuring they are well-grounded in the training data [385]. Moreover, the concept of probabilistic contrastive counterfactuals has been introduced to quantify direct and indirect influences of variables, providing a causality-based approach that transcends purely associational concepts. Systems like LEWIS leverage this framework to compute provably effective explanations and recourse at local, global, and contextual levels without requiring access to the internal mechanics of the black-box model [386].

The issue of actionability is intrinsically linked to causal consistency, forming a direct bridge to the structural causal models emphasized in prior sections. A counterfactual suggestion is useless if it violates the causal mechanisms governing the data, such as suggesting a change in a variable that is an effect rather than a cause. While full knowledge of the Structural Causal Model (SCM) is ideal, it is rarely available in practice. Consequently, recent work focuses on providing causal recourse under imperfect causal knowledge. Probabilistic approaches have been developed to select optimal actions that achieve recourse with high probability by capturing uncertainty over structural equations or by computing average effects on similar subpopulations, thus offering reliable recommendations even when the true SCM is unknown [387]. Additionally, methods leveraging partial SCMs or learning feasibility constraints through user interaction have been proposed to ensure that generated counterfactuals respect logical causal relations [388].

Advanced generative models are also reshaping the landscape of counterfactual synthesis, extending the capabilities of deep learning beyond static classification tasks. Diffusion models, known for their high-fidelity generation capabilities, are being adapted to create counterfactuals in structured and image data. The Structured Counterfactual Diffuser (SCD) framework, for example, utilizes diffusion models to generate counterfactuals that exhibit superior plausibility and diversity compared to state-of-the-art methods [389]. Similarly, Latent Diffusion Counterfactual Explanations (LDCE) harness foundation latent diffusion models to focus on semantic parts of the data while filtering out adversarial noise, demonstrating versatility across diverse learning paradigms [390]. In the realm of graphs, frameworks like CLEAR leverage graph variational autoencoders to generate counterfactuals that maintain causality without prior knowledge of the causal model, addressing the unique challenges of discrete and disorganized graph spaces [391]. These generative advances lay the groundwork for the complex multimodal reasoning tasks explored in the subsequent section.

Finally, the security and stability of counterfactual explanations remain critical concerns, echoing the robustness themes from Section 6.4. Explanations can be vulnerable to data poisoning attacks that increase the cost of recourse, necessitating robust generation methods [392]. Furthermore, the risk of generating "unjustified" counterfactuals—those disconnected from ground-truth data manifolds—has prompted the development of energy-constrained conformal counterfactuals that guarantee faithfulness to the black-box model's behavior [393]. As the field progresses, the integration of interactive elements allows users to personalize explanations by adjusting conditional statements, bridging the gap between automated generation and human-centric understanding [394]. Together, these advancements signify a maturation of counterfactual XAI from theoretical constructs to robust, actionable tools. By ensuring that explanations are causally sound and actionable, this domain provides the necessary interpretability layer that enables the deployment of the large-scale multimodal and language models discussed in the following section, where causal reasoning must scale to handle complex, dynamic environments.

6.6 Causal Mechanisms in Multimodal Systems and Large Language Models

6.6 Causal Mechanisms in Multimodal Systems and Large Language Models

Building on the foundation of actionable counterfactuals established in the previous section, the integration of causal representation learning into multimodal systems and Large Language Models (LLMs) marks a pivotal transition from statistical pattern matching to genuine reasoning. While Section 6.5 focused on generating valid recourse for individual predictions, this section addresses the challenge of scaling causal mechanisms to handle the complexity of foundation models. As these models grow in scale, their reliance on spurious correlations within massive datasets has become a critical bottleneck for robustness, fairness, and interpretability. Causal mechanisms now offer a theoretical framework to disentangle true generative factors from environmental noise, thereby enhancing the reliability of vision-language systems and text-based reasoning agents in dynamic environments.

In the realm of multimodal systems, particularly Vision-Language Models (VLMs), a primary challenge is the mitigation of biases inherited from training data. These biases often manifest as "decision shortcuts" where models rely on superficial correlations rather than invariant causal features. To address this, researchers have developed strategies that explicitly model confounders and apply causal adjustments. For instance, the Causal-Context Generation (Causal-CoG) framework introduces a prompting strategy that generates contextual information to enhance precise visual question answering, utilizing causality filtering to select samples where context is genuinely helpful [395]. Furthermore, debiasing efforts have moved beyond simple adversarial training to include causal information minimization. By treating specific features as confounder representations and minimizing their information content, models can learn simpler predictive features that capture the underlying data distribution more robustly, significantly improving out-of-distribution performance without sacrificing in-distribution accuracy [353].

Parallel to visual debiasing, the estimation of causal effects in textual data via counterfactual language models has seen substantial advancement. Traditional explainability tools often fail to distinguish between correlation and causation, particularly when dealing with high-level language concepts. The CausaLM framework addresses this by fine-tuning deep contextualized embedding models with auxiliary adversarial tasks derived from a causal graph, effectively learning counterfactual representations for specific concepts [396]. This allows for the estimation of a concept's true causal effect on model predictions, providing a rigorous method for bias detection and mitigation. Similarly, in tasks involving cause-effect relation classification, augmenting pretrained language models with commonsense knowledge graphs has proven effective. By continually pretraining models on verbalized commonsense knowledge, researchers have achieved superior performance on causal reasoning benchmarks, demonstrating that explicit knowledge integration is crucial for sound causal inference [397].

Beyond debiasing and effect estimation, integrating commonsense causal knowledge is essential for improving the interpretability and reasoning capabilities of AI systems. Humans naturally employ causal models to navigate the world, yet standard deep learning models often lack this implicit understanding. The iReason framework attempts to bridge this gap by inferring visual-semantic commonsense knowledge from both videos and natural language captions, integrating a causal rationalization module to aid interpretability and error analysis [398]. This approach highlights the necessity of juxtaposing visual and linguistic modalities to mine causal relationships that are implicit in temporal events but explicit in text. Moreover, the development of frameworks like CausalKG leverages hyper-relational graph representations to support interventional and counterfactual reasoning within knowledge graphs, enabling AI systems to provide domain-adaptable explanations that mimic human retrospection [399].

The evaluation of causal reasoning in LLMs has revealed both promising capabilities and significant limitations, echoing the need for robustness discussed in earlier sections on adversarial settings. While models like GPT-4 show accuracy in predicting intervention effects, they remain sensitive to distracting factors in prompts, suggesting that their reasoning is not yet fully robust [400]. Investigations into whether LLMs possess inherent causal reasoning abilities or merely rely on memorized knowledge suggest that "knowledge is, indeed, what LLMs principally require for sound causal reasoning" [401]. To enhance these capabilities, novel architectures such as CARE CA combine explicit causal detection modules with implicit reasoning through LLMs, utilizing counterfactual statements to accentuate the model's understanding of causality [402].

Furthermore, the application of causal mechanisms extends to improving the factual consistency of generated content, a direct extension of the plausibility constraints emphasized in counterfactual generation. In abstractive text summarization, factual inconsistencies often arise from language and irrelevancy biases. The CoFactSum framework constructs causal graphs to identify these intrinsic causes and employs counterfactual estimation strategies to debias the generation process, resulting in more factually consistent summaries [403]. In the domain of code, Code-LLMs have demonstrated enhanced causal reasoning abilities compared to text-only models, likely due to the explicit conditional structures inherent in programming languages [404].

Despite these advances, challenges remain in scaling these methods to complex, real-world scenarios, underscoring the gap between current model performance and human-level causal understanding. The creation of specialized datasets, such as ACQUIRED for multimodal counterfactual reasoning in real-life videos, highlights the need for more rigorous evaluation standards [405]. Similarly, benchmarks like C-VQA reveal substantial performance drops in multi-modal models when faced with counterfactual presuppositions, highlighting the need for more robust causal training paradigms [406]. Future research must focus on developing universal causal foundation models that can seamlessly integrate diverse modalities, leverage external knowledge bases, and perform reliable counterfactual reasoning across dynamic environments, paving the way for the next generation of autonomous, causally-aware agents.

7 Frontiers in Foundation Models, Multimodal Systems, and Scientific Applications

7.1 Large Language Models as Knowledge-Guided Causal Architects

7.1 Large Language Models as Knowledge-Guided Causal Architects

The integration of Large Language Models (LLMs) into causal discovery marks a paradigm shift, transforming these models from passive query tools into active "causal architects" capable of synthesizing vast repositories of human knowledge with empirical data. Traditionally, causal structure learning has been hindered by the necessity of expert-defined priors or the computational intractability of searching vast graph spaces without guidance. LLMs address this bottleneck by encoding common sense and domain-specific causal mechanisms, effectively acting as virtual domain experts that can guide, validate, and refine data-driven algorithms. This symbiotic relationship facilitates the construction of more robust Structural Causal Models (SCMs), particularly in complex domains where data is sparse or noisy, such as healthcare, psychology, and industrial systems [407; 408].

A primary function of LLMs in this architecture is serving as expert auditors and hypothesis generators. By leveraging training on extensive scientific literature, LLMs can extract causal relation pairs and construct specialized causal graphs that mirror expert-level insights. For instance, in psychology, frameworks combining LLMs with causal knowledge graphs have successfully generated novel hypotheses regarding well-being that align closely with those conceived by doctoral scholars, significantly outperforming LLM-only approaches [409]. Similarly, in biomedical research, LLMs have demonstrated the capacity to propose untrained yet validated hypotheses by analyzing background knowledge from literature, effectively breaking down information barriers across disciplines [410]. These capabilities suggest that LLMs can automate the initial stages of scientific discovery, proposing causal structures for human validation and thereby accelerating research in fields ranging from medicine to social science.

Beyond hypothesis generation, LLMs play a critical role in refining causal graph structures through knowledge-guided constraints. Purely data-driven methods often struggle with identifiability issues, especially when faced with unobserved confounders or limited sample sizes. To mitigate these challenges, researchers have developed frameworks where LLMs provide prior knowledge constraints to algorithmic solvers. A notable approach involves using LLMs to infer the topological order of variables, which is often more tractable for both humans and models than determining specific edge existence. Once the causal order is established by the LLM acting as a virtual expert, it can be fed into traditional constraint-based or score-based algorithms to significantly improve accuracy [411]. Furthermore, iterative frameworks have been proposed where LLMs supervise causal structure learning by refining Directed Acyclic Graphs (DAGs) based on feedback loops, ensuring that the resulting graph adheres to known physical or logical laws [71]. This integration allows for the correction of spurious correlations that purely statistical methods might otherwise accept as causal links.

The application of LLMs as causal architects is particularly transformative in high-stakes domains like healthcare, where accurate causal understanding is vital for treatment strategies. In studies involving Non-Small Cell Lung Cancer (NSCLC), LLMs have been employed to determine the directionality of edges in causal graphs derived from electronic health records and genomic data. These models have shown the ability to accurately predict causal directions, outperforming state-of-the-art statistical methods by leveraging implicit medical knowledge encoded during pre-training [412]. Moreover, LLMs can act as auditors for existing causal networks, analyzing edge directionality and identifying potential confounders or mediators that automated algorithms may have missed. This human-AI collaborative approach, where LLMs provide insights edge-by-edge, enables analysts to derive holistic and comprehensive causal models even when full domain expertise is not immediately available [413].

However, relying on LLMs for causal architecture presents challenges, primarily concerning the reliability of the knowledge they extract. LLMs can suffer from hallucinations or overconfidence, leading to erroneous prior knowledge that, if blindly integrated, could bias the causal discovery process. Recent research highlights that LLM-based causal reasoning is highly dependent on prompt design and context, carrying risks of false predictions if not carefully managed [414]. To address this, advanced frameworks incorporate mechanisms to detect and mitigate prior errors. For example, strategies have been developed to classify prior errors into types such as order-reversed or irrelevant, using theoretical insights about "quasi-circles" in graph structures to identify and correct mistakes without human intervention [72]. Additionally, Retrieval-Augmented Generation (RAG) techniques allow LLMs to ground their causal deductions in specific scientific literature, reducing the reliance on parametric memory and enhancing the fidelity of the extracted causal graphs [415].

Furthermore, the role of LLMs extends to discovering hidden variables from unstructured data, a task traditionally reserved for human experts. Frameworks like COAT (Causal representatiOn AssistanT) utilize LLMs as factor proposers to extract potential causal factors from raw text, which are then parsed into structured data for rigorous causal learning modules [416]. This capability is crucial for transitioning from structured tabular data to the messy, unstructured reality of real-world observations. By acting as a bridge between natural language descriptions and formal causal models, LLMs enable the discovery of causal laws in scenarios where variables are not predefined, effectively expanding the scope of causal discovery to include "dark" common sense reasoning [407].

In conclusion, LLMs are redefining the landscape of causal discovery by serving as knowledge-guided architects that enhance the accuracy, efficiency, and interpretability of causal models. Whether through generating novel hypotheses, providing topological constraints, auditing graph structures, or extracting latent variables from text, these models complement traditional statistical methods. While challenges regarding robustness and hallucination persist, ongoing advancements in error mitigation and hybrid frameworks promise a future where LLMs and causal algorithms work in tandem to unlock deeper understanding of complex systems. This foundational capability sets the stage for the next frontier: integrating these semantic reasoning powers with structural modeling to address the complexities of multimodal data and graph-based integration, as explored in the following section. [417].

7.2 Multimodal Causal Reasoning and Graph-Based Integration

Building upon the semantic reasoning and knowledge-guided architectural capabilities of Large Language Models (LLMs) discussed in the previous section, the frontier of causal AI expands into the realm of multimodal systems. While LLMs excel at synthesizing textual knowledge and proposing causal structures, real-world phenomena—from intuitive physics to social cognition—are inherently multimodal, involving a complex interplay between visual, textual, and structural data. Addressing the fundamental limitation of current deep learning models that often rely on spurious correlations rather than true mechanistic understanding, this section explores how Graph Neural Networks (GNNs) serve as the architectural backbone for integrating these diverse modalities. By fusing heterogeneous data streams into a unified graph structure, these systems can disentangle confounding factors and reason about counterfactual scenarios that are inaccessible to purely statistical or text-only models, thereby setting the stage for the autonomous closed-loop discovery systems explored subsequently.

A primary challenge in multimodal causal learning is the effective fusion of heterogeneous data sources where modalities may be unaligned or possess different temporal resolutions. Traditional fusion methods often fail to capture the nuanced causal interactions between modalities, leading to suboptimal performance in dynamic environments. Recent advancements have proposed graph-based architectures that explicitly model intra- and inter-modal dynamics to overcome these limitations. For instance, the Multimodal Graph framework utilizes a hierarchical structure where graph convolutional networks learn intra-modal dynamics, while a graph pooling fusion network automatically discovers associations between nodes from different modalities, effectively handling unaligned sequences [418]. This approach moves beyond simple concatenation, allowing the model to infer causal links even when visual and acoustic events do not occur simultaneously, a critical capability for understanding complex temporal systems.

In the domain of social cognition and human-computer interaction, multimodal causal reasoning is essential for tasks such as emotion recognition in conversation. Standard approaches that directly fuse modalities often suffer from information redundancy and the loss of diverse causal signals due to the heterogeneity gap. To address this, directed graph-based frameworks have been developed to model cross-modal feature complementation. The GraphCFC module, for example, constructs a directed graph to extract various types of edges, enabling GNNs to accurately encode crucial contextual and interactive information through message passing [419]. By treating different modalities as distinct subspaces and employing pair-wise complementary strategies, such models can isolate the causal drivers of emotional states, distinguishing between genuine affective cues and environmental noise.

The application of GNNs extends significantly into the realm of intuitive physics and scientific discovery, where understanding the causal structure of physical systems from observational data is paramount. Causal discovery from video data, which combines visual perception with dynamic inference, has been formulated as an end-to-end task using graph-based modules. These models typically consist of a perception module for extracting keypoint representations, an inference module for determining the graph distribution of interactions, and a dynamics module for prediction [420]. By assuming access to data from unknown interventions, these systems can recover the underlying causal graph of multi-body interactions, enabling long-term prediction and counterfactual reasoning about unseen physical configurations. Furthermore, the integration of physical inductive biases into GNNs enhances their ability to generalize across different system instances. Approaches that relax strict equivariance to subequivalence allow models to account for external fields like gravity, thereby capturing the true causal laws governing object interactions more accurately than standard simulators [421].

Beyond physical systems, the fusion of structural and functional data is transforming biomedical analysis. In neuroscience, understanding how dynamic functional interactions emerge from static structural backbones requires a multimodal causal perspective. GNN frameworks have been successfully deployed to describe functional interactions based on anatomical layouts, combining structural information from diffusion tensor imaging with temporal neural activity profiles [422]. These models provide a multi-modal measure of causal connectivity strength, capturing long-term dependencies that traditional vector auto-regression methods miss. Similarly, in oncology, the integration of radiology, pathology, and genomics data via GNNs and Transformers allows for the extraction of relational information residing at varying scales, improving diagnosis by encapsulating the heterogeneous nature of biological data [423].

Crucially, the role of LLMs as knowledge guides, introduced in Section 7.1, further amplifies the capabilities of these graph-based multimodal systems, creating a synergistic loop between semantic reasoning and structural modeling. In time-series analysis, where sensor data alone may introduce bias due to limited training samples, LLMs can encode extensive general knowledge to construct "Knowledge-Link" graphs. These graphs capture semantic relationships between sensors based on physical principles, which are then aligned with data-derived graphs to improve representation learning [424]. Moreover, in visual-semantic reasoning, frameworks like iReason blend causal relationships with input features from videos and natural language to infer commonsense knowledge, utilizing a causal rationalization module to enhance interpretability and detect biases [398]. This integration demonstrates how the semantic priors from LLMs can ground the structural learning of GNNs, bridging the gap between abstract knowledge and concrete multimodal observations.

Despite these advances, challenges remain in scaling these models and ensuring their trustworthiness. The black-box nature of deep GNNs necessitates the development of explanation methods that are both faithful and consistent. Recent work emphasizes the need for explanations that align with causal rationales rather than spurious correlations, proposing alignment losses to optimize for faithfulness in multimodal settings [425]. Additionally, the concept of counterfactual learning on graphs is gaining traction as a means to alleviate bias and improve fairness in multimodal applications, providing a unified understanding of how graph-structured data can support causal interventions [426]. As the field progresses, the convergence of GNNs, LLMs, and causal inference theory promises to unlock new levels of cognitive AI capable of reasoning across diverse modalities with human-like robustness and insight. This robust multimodal foundation is a prerequisite for the next evolutionary step: the deployment of these causal engines within generative closed-loop systems that can autonomously hypothesize, experiment, and discover new scientific laws, as detailed in the following section.

7.3 Generative Closed-Loop Systems for Autonomous Scientific Discovery

7.3 Generative Closed-Loop Systems for Autonomous Scientific Discovery

Building upon the multimodal causal integration and graph-based reasoning discussed in the previous section, the paradigm of scientific discovery is undergoing a profound transformation, shifting from traditional, manual hypothesis-testing cycles to AI-driven, closed-loop systems capable of autonomous exploration. These emerging frameworks integrate generative models with causal reasoning to not only accelerate the pace of innovation but also to uncover fundamental laws that might remain obscured by human bias or the sheer complexity of high-dimensional search spaces. The core vision of this approach is to create agents that can self-drive the entire scientific process: generating hypotheses, designing optimal experiments, executing them (either in silico or via robotic platforms), and refining their internal causal models based on the observed outcomes. This integration addresses a critical limitation in current artificial intelligence, where deep learning models often excel at pattern recognition but struggle with the causal understanding necessary for genuine scientific insight [427].

A central component of these closed-loop systems is the utilization of generative models to navigate vast chemical and physical spaces. In domains such as material science and drug discovery, the candidate space is often too large for exhaustive search, and traditional trial-and-error methods are prohibitively slow and expensive. Generative models, particularly those designed for scientific discovery, offer a solution by learning the underlying distribution of viable candidates and proposing novel structures with desired properties. For instance, frameworks like the Generative Toolkit for Scientific Discovery (GT4SD) enable researchers to train and deploy state-of-the-art generative models to accelerate material design, effectively bridging the gap between data availability and hypothesis formulation [428]. Similarly, in the quest to replace harmful "forever chemicals," generative models are being employed within human-AI co-creation workflows to iteratively guide the exploration of molecular designs, leveraging domain-specific tacit knowledge to filter and refine generated candidates [429].

However, generation alone is insufficient for robust scientific discovery; it must be coupled with active learning and causal inference to ensure that the generated hypotheses are both novel and scientifically valid. The challenge lies in distinguishing spurious correlations from true cause-effect relationships, a pitfall that plagues purely data-driven approaches in fields like drug discovery where confounding factors are rampant [430]. To address this, advanced frameworks incorporate causal reasoning to ground the discovery process. By formulating problems in the language of causality, these systems can explicitly model interventions and counterfactuals, allowing them to predict the effects of modifying specific molecular features or physical parameters rather than merely associating them with outcomes. This causal grounding is essential for developing models that generalize across different environments and experimental conditions, moving beyond the limitations of static datasets [431].

The realization of fully autonomous closed-loop systems often relies on the concept of the "scientific autonomous reasoning agent." A prime example is the Scientific Autonomous Reasoning Agent (SARA), which demonstrates the power of hierarchical autonomous experimentation. SARA integrates robotic synthesis and characterization with a hierarchy of AI methods to efficiently map complex processing phase diagrams. By employing nested active learning cycles and incorporating the underlying physics of experiments, SARA can autonomously design lateral gradient laser spike annealing experiments and interpret optical spectroscopy data to identify phase transitions, achieving orders-of-magnitude acceleration in discovering metastable materials [432]. This exemplifies the shift towards level-five autonomy, where AI systems operate with no human intervention in the production of scientific knowledge, akin to the highest levels of autonomy in self-driving cars [433].

Furthermore, the integration of probabilistic frameworks such as GFlowNets has emerged as a pivotal technique for balancing exploration and exploitation in these closed loops. GFlowNets are uniquely suited for scientific discovery because they learn to sample diverse, high-reward candidates from a distribution defined by a reward function, which is crucial when dealing with expensive but accurate measurements versus cheap but inaccurate ones [434]. This capability allows the system to estimate epistemic uncertainty and design experiments that maximize information gain, effectively guiding the search towards regions of the hypothesis space that are most likely to yield fundamental insights. Such approaches are complemented by Bayesian co-navigation strategies, where theoretical models and experimental data are updated concurrently to minimize uncertainty and create dynamic digital twins of material structures [435].

The role of large language models (LLMs) and generative AI in these systems extends beyond mere data processing; they act as collaborative partners that can structure vast amounts of scientific knowledge and propose testable hypotheses. While early iterations of LLMs may exhibit high error rates, their ability to synthesize information and generate creative hypotheses suggests a future where swarms of "hypothesis machines" work in synergy with automated experimentation platforms [436]. Moreover, the integration of human expertise remains a vital element, particularly in validating causal graphs and refining search spaces. Human-in-the-loop frameworks, including those utilizing virtual reality to visualize AI-generated graph structures, enable researchers to understand and steer the discovery process, ensuring that the "alien" hypotheses generated by AI are grounded in physical reality [437].

Ultimately, the convergence of generative modeling, causal inference, and autonomous experimentation represents a frontier where AI does not merely automate existing workflows but fundamentally augments the scientific method. By combining the creativity of generative models with the rigor of causal analysis, these closed-loop systems hold the promise of unlocking discoveries in fields ranging from quantum optics to climate science, potentially solving challenges that have long eluded human-only efforts [438]. As these systems evolve, they will likely transition from specialized tools for specific domains to general-purpose "AI scientists" capable of skeptical learning and reasoning across diverse scientific disciplines [439]. This progression naturally leads to the deployment of such causal foundations in embodied agents, where the ability to reason about physical interventions and plan under uncertainty becomes paramount for robotics, as explored in the following section.

7.4 Causal Foundations for Robotics and Embodied Intelligence

7.4 Causal Foundations for Robotics and Embodied Intelligence

Building on the transition from specialized scientific tools to general-purpose autonomous agents discussed in the previous section, the deployment of causal foundations becomes paramount in the realm of embodied intelligence. While Section 7.3 highlighted how generative closed-loop systems can accelerate discovery in controlled or simulated environments, robotics presents a distinct challenge: agents must operate in unstructured, dynamic physical worlds where the cost of failure is high and distribution shifts are constant. Here, the integration of causal reasoning represents a paradigm shift from purely reactive, correlation-based control to proactive, model-based intelligence. Traditional robotic systems often rely on extensive training data to learn statistical regularities, yet these models frequently falter when encountering novel physical interactions or environmental changes absent from the training set. Causal representation learning addresses this limitation by empowering agents to discover the underlying structural mechanisms governing their environment, thereby enabling them to reason about interventions, predict counterfactual outcomes, and plan effectively under uncertainty. This subsection surveys the emerging landscape of causal foundations for embodied intelligence, focusing on three critical pillars: human-in-the-loop causal graph construction, vision-based discovery of physical dynamics, and the deployment of causal world models for robust task planning.

A primary challenge in deploying causal inference in robotics is the accurate construction of the causal graph itself, which defines the variables and their directional relationships. While fully automated discovery is the ultimate goal, the complexity of real-world physical environments often necessitates human expertise to bootstrap the process, echoing the human-AI co-creation workflows seen in scientific discovery. Recent advancements have introduced human-centered frameworks that leverage augmented reality and interactive interfaces to facilitate the creation of causal graphical models. For instance, systems have been developed where humans actively participate in selecting relevant variables, establishing relationships, and performing interventions to validate the resulting graph at every step [440]. This collaborative approach not only accelerates the modeling phase but also ensures that the causal structure aligns with domain-specific physical laws that might be obscure to purely data-driven algorithms. Furthermore, natural language interfaces are being explored to allow non-expert users to communicate causal knowledge directly to planners, enabling robots to generalize solutions to new environments based on human-guided causal models [441]. In complex industrial or scientific settings, such as mine surveying, probabilistic causal frameworks are being adapted to handle online structural adaptation and provide post-hoc counterfactual explanations, thereby enhancing the trustworthiness of autonomous agents operating in hazardous conditions [442].

Beyond manual construction, a significant frontier involves enabling robots to autonomously discover causal structures from high-dimensional sensory inputs, particularly video. The ability to infer physical interactions and dynamic dependencies from visual observations is crucial for agents operating in unstructured environments where pre-defined models are unavailable, mirroring the autonomous experimentation capabilities discussed previously but applied to real-time perception. End-to-end frameworks have been proposed that combine perception modules for extracting keypoint representations with inference modules capable of determining the graph distribution induced by detected objects [420]. These systems assume access to data from unknown interventions—such as varying environmental conditions or object configurations—to disentangle true causal dependencies from spurious correlations. Similarly, deep causal learning approaches aim to bridge the gap between psychological findings on human causal learning and robotic needs by utilizing deep neural architectures to infer causal graphs from sensor data [443]. In scenarios involving human-robot coexistence, causal discovery methods are applied to model spatial interactions and predict human behaviors by analyzing time-series sensor data, allowing robots to anticipate the outcomes of their own interventions in shared spaces [444]. To make these methods practical for real-time deployment, efficient algorithms like Filtered PCMCI have been developed to reconstruct causal models rapidly from robot sensor data, ensuring that computational constraints do not hinder the agent's ability to understand its environment [445].

Once a causal structure is identified or provided, it serves as the backbone for causal world models that drive robust decision-making and planning. Unlike standard world models that may suffer from compounding errors over long horizons, causal world models enforce structural constraints that preserve the validity of predictions under intervention. In reinforcement learning, causal representations allow agents to capture invariant mechanisms that can be transferred across tasks, significantly improving robustness against environmental changes [446]. Frameworks such as CausalCF integrate causal curiosity and counterfactual reasoning to enable agents to perform self-supervised experiments, discovering causal factors of variation with significantly less data than conventional planners [447]. For manipulation tasks, physics-informed causal inference frameworks integrate rigid-body dynamics with causal Bayesian networks to probabilistically reason about candidate actions, achieving high success rates in block-stacking tasks even under partial observability [448]. Moreover, generative models like GATSBI transform raw observations into structured latent representations that capture spatio-temporal contexts, allowing agents to predict physically plausible future states and generalize across diverse interaction scenarios [449].

The application of causal world models extends to complex sequential planning and multi-agent systems, laying the groundwork for the domain-specific applications detailed in the following section. Approaches leveraging graph-structured world models learn latent landmarks to facilitate temporally extended reasoning, overcoming the limitations of step-by-step rollouts in long-horizon planning [450]. In dynamic environments shared with humans, causal models are essential for predicting multimodal human behaviors and navigating crowded scenes safely. Methods utilizing spatial-temporal graphs and attention mechanisms enable robots to interpret the relationships between agents over time, leading to more informed and foresighted navigation decisions [451]. Additionally, causal induction from visual observations allows agents to contextualize goal-conditional policies, enabling them to complete tasks in novel environments with previously unseen causal structures [452]. As robotics moves toward more autonomous and intelligent systems, the integration of these causal foundations—from human-guided bootstrapping to vision-based discovery and causal planning—provides the necessary adaptability and reliability for real-world deployment. This progression naturally leads to the application of these robust causal frameworks in high-stakes domains such as healthcare, climate science, and industrial systems, where the consequences of decision-making are critical, as explored in the next section.

7.5 Domain-Specific Applications in Healthcare, Climate, and Industrial Systems

7.5 Domain-Specific Applications in Healthcare, Climate, and Industrial Systems

Building upon the foundational principles of causal discovery in embodied intelligence discussed in the previous section, the application of causal representation learning extends critically into high-stakes domains where data complexity and decision consequences are paramount. In sectors such as healthcare, climate science, and industrial cyber-physical systems, the transition from theoretical frameworks to real-world solutions is driven by the need for robustness, interpretability, and generalization—capabilities that purely associational models often fail to provide. These domains present unique challenges, including hidden confounders, non-stationary environments, and the absolute necessity for counterfactual reasoning, which recent advancements in deep causal models are beginning to address systematically.

In the realm of healthcare and precision medicine, the objective has evolved from mere prediction to elucidating the underlying cause-effect mechanisms driving disease progression and treatment efficacy. The complexity of biomedical data, encompassing high-dimensional genomic sequences, electronic health records (EHR), and medical imaging, necessitates approaches capable of disentangling true causal signals from spurious correlations induced by confounding variables. Recent developments have demonstrated the feasibility of using deep learning architectures to learn causal networks in large-scale biological problems, combining convolutional and graph neural networks within a causal risk framework to validate findings against unseen experimental data [453]. Furthermore, specialized frameworks like CausalEGM have been developed to estimate causal effects by encoding generative modeling, effectively mapping high-dimensional confounders to a low-dimensional latent space to decouple dependencies between treatments and outcomes [454]. This capability is particularly crucial in scenarios involving continuous-valued interventions, where scalable sensitivity and uncertainty analyses are required to bound causal estimates in the presence of unobserved confounders, as seen in studies linking human emissions to cloud properties [455].

Beyond static analysis, causal inference in healthcare increasingly addresses dynamic, time-series data and individualized treatment effects. Methods leveraging Lipschitz regularization and neural controlled differential equations have been proposed to estimate treatment effects from irregular time-series observations while accounting for hidden confounders, a common issue in clinical monitoring [292]. The push towards precision medicine is further accelerated by the application of causal machine learning to clinical decision support systems, which aim to quantify intervention effects to maintain robustness against confounders in conditions such as Alzheimer's disease [456]. Moreover, the interpretability of these models is paramount; techniques utilizing variable decorrelation regularizers and association rule mining have been introduced to handle nonlinear confounding in healthcare applications, ensuring that the derived causal relationships align with human-interpretable decision patterns [457]. The emergence of large AI models in health informatics also promises to transform fields ranging from medical diagnosis to public health by leveraging massive multi-modal datasets to uncover latent causal structures [458].

Parallel to biomedical applications, causal discovery provides a rigorous foundation for climate science and Earth systems, enabling the attribution of extreme weather events and the understanding of teleconnections within the global climate system. The primary challenge in this domain lies in the spatiotemporal nature of the data and the presence of complex, nonlinear interactions among atmospheric variables. Survey efforts have cataloged state-of-the-art time-series and spatiotemporal causality methods specifically tailored for Earth science, enabling researchers to answer critical questions regarding sea-level rise and extreme weather attribution [459]. To address the resolution gap in climate projections, multi-resolution deep learning frameworks have been deployed to correct biases in General Circulation Models (GCMs) and reconstruct finer-scale weather extremes, thereby improving probabilistic risk assessments for natural disasters [460]. Additionally, unsupervised physics-based methods, such as the DisCo workflow, have demonstrated the ability to decompose complex spatiotemporal climate simulations into coherent structures by identifying latent local causal state variables, processing terabytes of simulation data with high efficiency [461]. These approaches are complemented by quantitative causality analyses that have successfully identified anthropogenic causes of global warming and predicted phenomena like El Niño Modoki, showcasing the power of information flow theory in geoscience [462].

Industrial automation and cyber-physical systems (ICPS) represent another frontier where causal models are essential for fault diagnosis, prognosis, and health management. The increasing complexity of these systems requires a departure from purely data-driven correlation mining toward models that incorporate known physical constraints and component behaviors. A Uniform Causality Model has been proposed to bridge the gap between engineering practices, where systems are constructed from components with known behaviors, and data-driven approaches, facilitating better communication and data usage across industrial disciplines [463]. In the context of Prognostics and Health Management (PHM), large-scale foundation models are increasingly being recognized for their potential to enhance reliability and safety in ICPS, although consensus on their application is still forming [464]. Specific applications include the development of spatio-temporal frameworks for pandemic monitoring that utilize fog/edge computing to model disease spread based on human mobility, illustrating the versatility of causal analytics in managing large-scale industrial and societal systems [465]. Furthermore, causal testing frameworks are being introduced to validate scientific modeling software by inferring metamorphic test outcomes from reused, confounded data, ensuring the integrity of simulations used in critical policy decisions [466].

Collectively, these domain-specific applications underscore the transformative potential of causal representation learning. By moving beyond association to capture the invariant mechanisms governing healthcare outcomes, climate dynamics, and industrial processes, these methods enable more robust, fair, and actionable AI systems capable of operating effectively under distribution shifts and in the presence of unobserved confounders. However, as these models scale and permeate critical infrastructure, new challenges emerge regarding their reliability and cognitive depth. The transition from domain-specific success to generalizable, trustworthy cognitive AI necessitates addressing fundamental issues such as hallucination in large models, the explainability of graph-based representations, and the acquisition of "dark" common sense, topics which are explored in the following section.

7.6 Challenges in Scaling, Trustworthiness, and the Path to Cognitive AI

7.6 Challenges in Scaling, Trustworthiness, and the Path to Cognitive AI

Building on the domain-specific successes in healthcare, climate science, and industrial systems outlined previously, the trajectory of causal representation learning now faces three critical frontiers that define the future of artificial intelligence: mitigating hallucination risks in Large Language Model (LLM)-based causal inference, establishing trustworthiness and explainability in Graph Neural Networks (GNNs), and executing a fundamental paradigm shift from statistical correlation to "dark" common sense reasoning. While scaling laws have empowered models to process unprecedented data volumes, the transition from pattern recognition to genuine causal understanding remains obstructed by profound structural and epistemological challenges.

The most immediate barrier to trustworthy causal AI is the phenomenon of hallucination within LLMs, particularly when these models are tasked with causal reasoning or scientific discovery. Despite their impressive benchmark performance, LLMs frequently generate plausible yet factually incorrect causal graphs or counterfactual scenarios. This issue stems not merely from statistical noise but from deep mechanistic failures, such as insufficient subject attribute knowledge in lower layers and errors in selecting correct object attributes in upper attention heads [467]. When deployed as "knowledge-guided causal architects," as envisioned in recent frameworks, this propensity to hallucinate poses severe risks in high-stakes domains like healthcare and law. Without rigorous constraints, an LLM may invent non-existent causal links or legal precedents, a problem exacerbated by a reliance on spurious correlations rather than grounded causal mechanisms [468]. Recent studies indicate that while LLMs can outperform traditional algorithms on specific causal tasks, they exhibit unpredictable failure modes that necessitate novel interpretability techniques for diagnosis [408]. Furthermore, reliance on pre-trained knowledge without evidential closure means outputs are not necessarily synonymous with claims supported by evidence, leading to confident but erroneous causal assertions [469]. Addressing this requires moving beyond simple prompt engineering to develop metacognitive approaches that enable models to self-identify and correct mispredictions prior to deployment [470].

Parallel to language model challenges is the urgent need for trustworthy and explainable Graph Neural Networks (GNNs), which are increasingly pivotal for modeling complex causal structures in scientific and social systems. While GNNs excel at capturing dependencies in graph-structured data, their "black-box" nature raises significant concerns regarding robustness, fairness, and accountability. The integration of causal learning techniques into GNNs—termed Causality-Inspired GNNs (CIGNNs)—has shown promise in alleviating distribution shifts and biases by focusing on underlying data causality rather than superficial correlations [471]. However, scaling these models to massive, real-world graphs while maintaining interpretability remains an open challenge. The epistemic complexity of deep neural networks often hinders the estimation of their reliability, creating a gap between standard error analysis and the deeper epistemological understanding required for scientific validation [472]. To bridge this gap, future research must develop frameworks that combine technical explainability with legal and economic considerations, ensuring that the level of explanation provided matches the operational context and potential harm [473]. Moreover, the susceptibility of GNNs to adversarial attacks and their potential to perpetuate discrimination necessitates a holistic approach to trustworthiness encompassing robustness, privacy, and environmental well-being [474].

Ultimately, overcoming these scaling and trustworthiness hurdles requires a paradigm shift from the current "big data for small tasks" approach to a "small data for big tasks" paradigm centered on human-like common sense. Current deep learning systems, including advanced LLMs and GNNs, largely operate within a framework of statistical association, failing to grasp the invisible drivers of reality such as functionality, physics, intent, causality, and utility (FPICU). These core domains constitute the "dark matter" of vision and cognition; they are invisible in pixel data or text tokens yet drive the creation and maintenance of scenes and narratives [475]. True cognitive AI must move beyond detecting statistical patterns to understanding the "why" and "how" of the world. This shift is critical because, as some philosophers argue, the inability of computation to truly "understand" anything represents a category mistake that causal reasoning alone may not fix without a foundational change in architecture [476]. The path forward involves developing systems capable of reasoning under deep uncertainty, a capability currently lacking in most AI tools, which poses structural risks when deployed in critical decision-making roles [477]. Furthermore, achieving this level of cognition may require moving beyond vector-based representations to qualitative extensions, such as sphere neural networks, which can better model logical validity and rational reasoning, potentially escaping the "swamp of hallucination" that traps current models [478].

In conclusion, the journey toward cognitive AI is not merely a matter of scaling parameters but of fundamentally re-engineering how machines represent and reason about causality. It demands a concerted effort to eliminate hallucinations through evidential grounding, to imbue graph-based models with transparent causal mechanisms, and to cultivate the "dark" common sense that distinguishes human intelligence from statistical curve fitting. Only by addressing these interconnected challenges can the field transition from powerful correlational tools to trustworthy, reasoning agents capable of navigating the complexities of the real world.

8 Evaluation Benchmarks, Open Challenges, and Future Research Directions

8.1 Benchmarking Causal Capabilities in Large Language Models and Foundation Systems

8.1 Benchmarking Causal Capabilities in Large Language Models and Foundation Systems

The rapid ascent of Large Language Models (LLMs) and foundation systems has catalyzed a paradigm shift in how artificial intelligence approaches causal reasoning, necessitating a rigorous evolution in evaluation methodologies. While traditional causal discovery relied heavily on static structural metrics like Structural Hamming Distance (SHD) applied to synthetic numerical data, these measures prove insufficient for assessing the nuanced, context-dependent causal comprehension of modern foundation models. Consequently, the field has transitioned toward comprehensive benchmarks designed to probe the depth of causal understanding, hypothesis generation, and graph extraction capabilities within complex textual and multimodal contexts. This shift reflects a growing consensus that true causal intelligence requires navigating the full ladder of causation—from association to intervention and counterfactuals—rather than merely recognizing statistical patterns.

A pivotal development in this domain is the introduction of **CausalBench**, a comprehensive benchmark specifically engineered to evaluate the causality understanding capabilities of LLMs across diverse scenarios. Unlike previous efforts that were often straightforward or homogeneous, **CausalBench** integrates causal networks of varying scales and densities to explore the upper limits of model performance under increasing difficulty [479]. Crucially, this benchmark incorporates background knowledge and structured data, allowing researchers to assess how effectively LLMs can leverage long-text comprehension and prior information utilization to outperform classic causal learning algorithms. The findings from **CausalBench** reveal significant insights into the adaptability of LLMs to specific structural networks and highlight the existing gap between their capabilities in textual contexts versus numerical domains, underscoring the need for models that can seamlessly bridge these modalities [479]. Complementing this, specialized benchmarks such as **CLadder** have begun to systematically evaluate the three layers of Pearl's causal ladder, ensuring that models are not merely retrieving memorized correlations but are capable of genuine interventional and counterfactual reasoning [479; 480].

Parallel to general causal reasoning, the scientific community has recognized the necessity of evaluating LLMs on domain-specific scientific problem-solving, leading to the creation of **SciBench**. This benchmark suite addresses the limitation of existing evaluations that focus largely on high-school level algebra, instead presenting collegiate-level problems in mathematics, chemistry, and physics that require complex reasoning [481]. **SciBench** is instrumental in revealing that current LLMs, despite their impressive general capabilities, often fall short in delivering satisfactory performance on rigorous scientific tasks, with top models achieving scores barely exceeding 40% [481]. By categorizing errors into ten distinct problem-solving abilities, **SciBench** provides a granular view of where causal and logical reasoning breaks down, serving as a critical tool for guiding future improvements in scientific AI [481].

Beyond reasoning capabilities, the assessment of factual knowledge and its relationship to causal structures is addressed by benchmarks like **Head-to-Tail**, which investigates the extent to which LLMs internalize facts across the popularity spectrum (head, torso, and tail) [482]. This benchmark is essential for understanding whether LLMs can replace traditional Knowledge Graphs (KGs) or if they suffer from hallucinations when dealing with less frequent, tail-end facts that are often critical for constructing accurate causal graphs in specialized domains [482]. The evaluation metrics developed in **Head-to-Tail** approximate the knowledge an LLM confidently internalizes, providing a necessary corrective to over-optimistic claims about the completeness of LLM knowledge bases [482].

Furthermore, emerging benchmarks are shifting focus from static question answering to the dynamic aspects of causal discovery, such as graph extraction from unstructured text. For instance, **Text2KGBench** evaluates the ability of language models to generate Knowledge Graphs from natural language text while adhering to strict ontological constraints, measuring fact extraction performance and hallucination rates [483]. Similarly, **CRAB** (Causal Reasoning Assessment Benchmark) focuses on assessing the strength of causal relationships between real-world events in narratives, demonstrating that models often struggle with complex causal structures compared to simple linear chains [484]. These benchmarks collectively highlight a critical trend: the field is moving from evaluating isolated causal facts to assessing the robustness of causal graph construction and the ability to handle complex, interdependent event networks.

The integration of explicit graph structures into LLM evaluation is further exemplified by benchmarks like **NLGraph** and **GRBench**, which test the ability of models to solve graph-based problems and reason over text-attributed graphs, respectively [485; 486]. These efforts underscore the importance of structural reasoning, a capability that is often a bottleneck for pure language models when faced with explicit graphical data [487]. Collectively, these diverse benchmarks expose the current limitations of LLMs in handling complex causal structures and domain-specific reasoning, providing the necessary infrastructure for developing the next generation of foundation models capable of true causal cognition. However, while these benchmarks establish *what* to measure, they primarily rely on task-specific accuracy scores. To fully understand the functional validity of these models—particularly their ability to generalize interventions and counterfactuals beyond structural resemblance—we must look beyond simple benchmark scores to more granular evaluation metrics, as discussed in the following section.

8.2 Evaluation Metrics Beyond Graph Structure: The Ladder of Causal Distances and Qualitative Assessment

8.2 Evaluation Metrics Beyond Graph Structure: The Ladder of Causal Distances and Qualitative Assessment

While Section 8.1 established a rich landscape of benchmarks for assessing the causal capabilities of Large Language Models and foundation systems, a critical limitation persists: these benchmarks primarily rely on task-specific accuracy scores or structural resemblance metrics. As noted, such measures often fail to capture the *functional validity* of a model—specifically, its ability to generalize interventions and counterfactuals beyond mere structural correctness. To address this gap, the field must transition from evaluating *what* models know based on benchmark performance to rigorously quantifying *how* well they adhere to the principles of causal inference through advanced metrics. This shift necessitates moving beyond the historical dominance of Structural Hamming Distance (SHD), which counts edge additions, deletions, and reversals but offers no guarantee of semantic validity in terms of causal inference. A model may achieve a low SHD yet fail to accurately predict the outcomes of interventions or generate plausible counterfactuals, rendering it ineffective for downstream decision-making. Consequently, the evaluation paradigm must evolve to align with Pearl's Ladder of Causation, encompassing observational, interventional, and counterfactual capabilities through distributional and qualitative measures.

To bridge the disconnect between structural accuracy and functional utility, recent research advocates for the adoption of distributional distances that measure the divergence between true and estimated causal distributions at each rung of the ladder. The concept of "A Ladder of Causal Distances" proposes a hierarchy of three distinct distance metrics corresponding to the three levels of causation: observational, interventional, and counterfactual [125]. Unlike structural measures, these distances are derived from the causal distributions induced by the models, offering a more intuitive and efficient approximation of model performance. Empirical benchmarks using these causal distances have demonstrated that structural measures often correlate poorly with the accuracy of estimated interventional distributions, frequently leading to misleading conclusions regarding algorithm selection and parameter tuning [488]. Therefore, evaluating causal models by directly comparing interventional distributions is emerging as a new standard, as it captures the model's ability to generalize beyond passive observation to active manipulation of the system.

The challenge becomes even more pronounced at the third rung of the ladder: counterfactual reasoning. Counterfactuals require not only the correct graph structure but also the accurate identification of exogenous noise variables and functional mechanisms. Traditional metrics often fail to capture the nuances of counterfactual fidelity. To remedy this, frameworks have been developed to evaluate image counterfactuals based on Pearl's axiomatic definitions—composition, reversibility, and effectiveness [489]. By framing counterfactuals as functions of input variables and their parents, researchers have derived distance metrics that quantify the deviation between approximate and ideal counterfactual functions. This axiomatic approach allows for the comparison of different generative models, revealing trade-offs and shortcomings that structural metrics overlook. Furthermore, the development of Distribution-consistency Structural Causal Models (DiscoSCMs) highlights the limitations of standard frameworks in modeling individual-level heterogeneity and proposes new identifiable parameters, such as the probability of consistency, to better evaluate Layer 3 capabilities [12; 13].

Beyond quantitative distributional metrics, there is a critical gap in capturing fine-grained reasoning failures and providing actionable diagnostics for model improvement. Quantitative scalar metrics, while useful for ranking, often obscure the specific nature of a model's errors, leaving developers to rely on manual, hit-or-miss adjustments. To bridge this gap, qualitative evaluation frameworks like QualEval have been introduced [490]. QualEval augments quantitative benchmarks with automated qualitative analysis powered by Large Language Models (LLMs) and linear programming solvers. This approach generates human-readable insights and fine-grained visualizations that diagnose specific failure modes, effectively acting as a "data-scientist-in-a-box." By leveraging such qualitative tools, researchers can accelerate model development and achieve significant performance gains, as evidenced by improvements in dialogue tasks where qualitative insights led to absolute performance increases of up to 15% [490].

The necessity for robust evaluation extends to the realm of explainable AI, where counterfactual explanations (CEs) are increasingly used to provide recourse and interpretability. However, the generation of CEs often relies on models that do not account for the true underlying causal structure, leading to unjustified or unfeasible recommendations. Studies have shown that a significant portion of generated counterfactuals conflict with those computed using rigorous causal methods, highlighting the danger of blind reliance on post-hoc interpretability without causal validation [491; 492]. To ensure faithfulness, new metrics such as Correlational Explanatory Faithfulness (CEF) and the Correlational Counterfactual Test (CCT) have been proposed [493]. These metrics move beyond binary prediction changes to account for the total shift in the model's predicted label distribution, offering a more faithful assessment of whether an explanation truly captures the factors responsible for a prediction. Additionally, the evaluation of counterfactuals must consider their plausibility and actionability; frameworks like FACE (Feasible and Actionable Counterfactual Explanations) utilize density-weighted metrics to ensure that suggested recourses lie within high-density regions of the data manifold, avoiding the prescription of impossible goals [494].

Moreover, the evaluation of causal models must account for the robustness of explanations against adversarial perturbations and the stability of recourse under imperfect causal knowledge. Recent works demonstrate that counterfactual explanations can be fragile, where minor perturbations render the prescribed actions ineffective or overly costly [495]. Metrics that incorporate robustness to adverse perturbations and probabilistic approaches to handle uncertainty in structural equations are essential for deploying causal systems in high-stakes environments [387]. The integration of these diverse evaluation strategies—ranging from ladder-aligned distributional distances to qualitative diagnostic tools and faithfulness tests for explanations—represents a necessary evolution in the field. Moving forward, the community must adopt a holistic evaluation protocol that prioritizes functional causal validity over mere structural resemblance. However, establishing these rigorous metrics reveals a parallel and formidable challenge: the scarcity of data capable of supporting such evaluations in complex, real-world domains. As we shift focus from well-defined synthetic structures to the messy reality of human and natural systems, we encounter the problem of "indefinite data," where traditional assumptions break down, a topic explored in the following section.

8.3 The Challenge of Indefinite Data and Domain-Specific Benchmark Gaps

8.3 The Challenge of Indefinite Data and Domain-Specific Benchmark Gaps

While Section 8.2 established the necessity of moving beyond structural metrics to evaluate causal validity, a parallel challenge lies in the data itself: the scarcity of benchmarks capable of supporting such rigorous evaluation in complex, real-world domains. A critical frontier in causal representation learning is addressing the profound data voids associated with "indefinite data." Unlike traditional statistical datasets characterized by fixed-dimensional vectors and well-defined causal structures, indefinite data encompasses complex, non-statistical modalities such as natural language dialogue, unstructured video streams, and dynamic spatial systems. These domains are defined by multi-structure data forms and multi-value representations that fundamentally break the assumptions of conventional causal discovery algorithms, rendering many existing methods infeasible [496]. The primary challenge is not merely the volume of data but the inconsistency between the underlying causal structure and its high-dimensional representation, a gap that has led to a scarcity of robust benchmarks capable of evaluating true causal reasoning capabilities beyond pattern recognition [497].

The concept of indefinite data was explicitly formalized to highlight the unique difficulties posed by dialogue and video, where causal relationships are often latent, context-dependent, and entangled with spurious correlations [36]. To bridge the dataset gap in this emerging sphere, recent initiatives have introduced specialized benchmarks designed to capture the nuances of these complex modalities. A pivotal contribution in this domain is the release of **Causalogue** and **Causaction**, two high-quality datasets specifically engineered to address the lack of causal annotations in text dialogue and video action samples, respectively [496]. **Causalogue** serves as the first high-quality causal dialogue dataset, providing an essential testbed for evaluating how models handle causal inconsistencies between structures and representations in natural language interactions [497]. Similarly, **Causaction** offers video action samples annotated with causal ground truth, enabling researchers to move beyond descriptive video understanding toward explanatory and counterfactual reasoning [496]. These datasets are foundational because they incorporate the "indefinite" nature of real-world data, challenging models to establish causation conditions under non-fixed causal structures and to treat causal strength as a latent variable rather than relying on independent noise assumptions [36].

Beyond temporal and linguistic domains, spatial confounding presents another significant benchmark gap, particularly in scientific fields like climate science, epidemiology, and social sciences where unobserved spatial variables can induce spurious associations. Traditional benchmarks often fail to simulate the complex, continuous nature of spatial confounders, leading to over-optimistic performance estimates for causal inference methods. To rectify this, **SpaCE: The Spatial Confounding Environment** was introduced as the first comprehensive toolkit providing realistic benchmark datasets and evaluation tools specifically for spatial confounding [498]. **SpaCE** distinguishes itself by including training data alongside true counterfactuals, detailed spatial graphs with coordinates, and quantitative scores for smoothness and confounding effects [498]. By generating realistic semi-synthetic outcomes using state-of-the-art machine learning ensembles, **SpaCE** facilitates an automated, end-to-end pipeline for rigorously evaluating whether models can truly disentangle treatment effects from missing spatial confounders, a capability largely absent in previous synthetic benchmarks [498].

In the realm of physical reasoning and embodied intelligence, existing datasets have historically focused on elementary events, lacking the complexity required to test causal learning in realistic 3D environments. The **SPACE: A Simulator for Physical Interactions and Causal Learning in 3D Environments** addresses this scarcity by generating synthetic video datasets that depict complex daily object interactions, including containment, stability, and contact [499]. Unlike earlier benchmarks that focused on simple rolling or falling objects, **SPACE** provides a curriculum for evaluating physics-based models on tasks that require deep intuitive physics and causal abstraction [499]. Furthermore, the need for counterfactual reasoning in multimodal settings is addressed by datasets like **ACQUIRED**, which provides annotated videos encompassing physical, social, and temporal reasoning dimensions, revealing a significant performance gap between current multimodal models and human-level counterfactual understanding [405]. Similarly, **CLEVRER** has established a diagnostic standard for evaluating causal structures in videos through descriptive, explanatory, predictive, and counterfactual questions, highlighting that while models excel at perception, they struggle with the underlying dynamics of causal relations [500].

The challenge of indefinite data is further compounded by the need to handle irregular sampling and complex spatiotemporal dependencies, as seen in sensor networks and multi-agent systems. Methods that rely on fixed grid assumptions often fail in these settings, necessitating benchmarks that reflect real-world irregularities. Recent work on modeling dynamics in continuous spatial-temporal processes represented by irregular samples underscores the necessity for datasets that do not require voxelization or uniform grids, allowing for flexible prediction queries across arbitrary space-time points [501]. Additionally, the issue of missing data and non-causal shortcut edges in constructed sensor networks has led to the development of frameworks like **CASPER**, which revisits spatiotemporal imputation from a causal perspective to block confounders via frontdoor adjustment, emphasizing the need for benchmarks that explicitly test robustness against non-causal correlations [502].

Despite these advances, significant gaps remain. Many current benchmarks still rely heavily on synthetic data that may not fully capture the "dark common sense" or the intricate contextual dependencies found in real-world indefinite data. For instance, while **Causalogue** and **Causaction** provide crucial baseline results, the broader ecosystem of indefinite data benchmarks is still nascent compared to the mature landscape of image classification datasets [496]. There is an urgent need for more datasets that integrate multi-modal constraints, such as those combining visual and textual causal annotations, to support the development of models capable of joint video and text parsing for understanding events [503]. Moreover, the evaluation metrics must evolve alongside these datasets; relying solely on structural Hamming distance is insufficient for indefinite data where the causal graph may be dynamic or partially observed. Addressing these limitations requires more than just new data; it demands a shift in methodology where human expertise guides the definition of causal structures in these ambiguous domains. This necessity naturally leads to the integration of domain knowledge and human-in-the-loop validation mechanisms, which serve as the bridge between these complex, unstructured data environments and scientifically robust causal discovery, as discussed in the following section. Future research must prioritize the creation of diverse, large-scale indefinite data benchmarks that encompass a wider range of domains, from healthcare dialogue to industrial cyber-physical systems, ensuring that causal discovery methods can generalize beyond controlled laboratory settings to the messy, unstructured reality of human and natural systems [504]. Only by filling these domain-specific benchmark gaps can the field progress toward truly robust and generalizable causal artificial intelligence.

8.4 Integrating Domain Knowledge and Human-in-the-Loop Validation Mechanisms

8.4 Integrating Domain Knowledge and Human-in-the-Loop Validation Mechanisms

Addressing the limitations of "indefinite data" and the scarcity of robust benchmarks highlighted in the previous section necessitates a fundamental paradigm shift: moving from fully automated, data-driven algorithms to collaborative human-AI frameworks. In high-stakes domains such as healthcare, finance, and scientific research, purely statistical methods often exhibit brittleness when confronted with finite data, unobserved confounders, or violations of strict assumptions like causal faithfulness. Consequently, the integration of expert domain knowledge and the implementation of human-in-the-loop (HITL) validation mechanisms have emerged as critical strategies for ensuring scientific validity, reducing hallucination, and bridging the gap between abstract causal graphs derived from ambiguous data and actionable scientific insight. This approach directly responds to the call in Section 8.3 for methodology shifts where human expertise guides the definition of causal structures in complex, unstructured environments.

One of the primary motivations for integrating domain knowledge is the computational and statistical inefficiency of unconstrained search spaces, a problem that becomes exponentially more severe as one attempts to scale beyond the controlled settings of current benchmarks. Causal discovery is an ill-posed problem where multiple graphical structures can explain the same observational data. Incorporating prior knowledge acts as a powerful regularizer that drastically reduces this search space. For instance, research has demonstrated that integrating expert constraints into A*-based causal discovery methods can significantly accelerate the learning process while improving accuracy, even when only small amounts of domain knowledge are available [505]. Similarly, studies utilizing the NOTEARS framework have shown that biasing models with known causal relations—particularly constraints on active edges—can lead to statistically significant improvements in graph recovery, correcting mistakes that purely data-driven approaches might make due to spurious correlations or sampling noise [506]. These findings underscore that human expertise is not merely a supplementary add-on but a fundamental component for stabilizing inference in complex, high-dimensional environments, laying the groundwork for the scalable systems discussed in the following section.

Beyond static constraints, dynamic human-in-the-loop frameworks are essential for addressing uncertainty and refining causal hypotheses iteratively, particularly when dealing with the latent and context-dependent relationships characteristic of indefinite data. Traditional causal discovery algorithms often output a single point estimate of a graph, failing to convey the epistemic uncertainty inherent in the inference process. To mitigate this, novel approaches have been proposed that sample from distributions over ancestral graphs, allowing experts to interactively probe relations among variables and reduce uncertainty through optimal experimental design [507]. This collaborative exploration enables the system to leverage human intuition to resolve ambiguities that statistical tests alone cannot distinguish, particularly in the presence of latent confounders where causal sufficiency assumptions are violated. Furthermore, in scenarios involving relational data or complex social systems, declarative languages like CaRL have been developed to allow experts to specify causal background knowledge and queries, facilitating reasoning about interventions in heterogeneous domains where flat-table assumptions fail [508].

The necessity of human validation becomes even more pronounced when evaluating the outputs of Large Language Models (LLMs) in causal tasks, especially given the reliance of many indefinite data benchmarks on textual and multimodal inputs. While LLMs have demonstrated remarkable capabilities in generating causal graphs and identifying background context from natural language, they are prone to hallucinations and overconfidence, especially when relying solely on text-based descriptions without grounding in empirical data [414]. To address these limitations, hybrid frameworks are emerging where LLMs act as "virtual domain experts" to propose causal orders or meta-knowledge, which are then validated and refined using rigorous statistical causal discovery algorithms [509]. For example, the RealTCD framework leverages LLMs to extract meta-knowledge from textual system descriptions to guide temporal causal discovery in industrial settings, effectively combining the semantic understanding of LLMs with the structural rigor of score-based methods [41]. Such integrations ensure that the generative capabilities of foundation models are anchored in statistical reality, reducing the risk of propagating errors into downstream decision-making processes.

Moreover, the evaluation of these hybrid systems requires moving beyond automated metrics like Structural Hamming Distance to include human-centered assessment protocols, echoing the broader critique of evaluation metrics in Section 8.3. The reliability of causal models in critical applications depends heavily on how well they align with human judgment and domain-specific constraints. Frameworks like OpenHEXAI highlight the importance of standardized human-centered benchmarks to evaluate the effectiveness of explanations and the trust calibration between humans and AI [510]. In clinical and industrial settings, the workflow design—specifically whether AI inferences are presented before or after human review—can significantly impact decision quality and the willingness of experts to seek second opinions [511]. Therefore, future research must focus on developing adaptive interfaces that facilitate seamless collaboration, where AI systems can explicitly communicate their uncertainty and defer to human experts when confidence is low, as seen in deferral collaboration frameworks designed to handle unobserved confounding [512].

Ultimately, the path toward trustworthy causal AI lies in recognizing that causality is not just a statistical property of data but a reflection of underlying mechanisms that often require human insight to unravel. By synthesizing data-driven discovery with expert constraints and interactive validation loops, the community can develop systems that are not only computationally efficient but also scientifically robust and aligned with human reasoning. This symbiotic relationship ensures that causal models serve as reliable tools for hypothesis generation and policy formulation, mitigating the risks of automation bias and hallucination in an era of increasingly complex data landscapes. Crucially, these human-in-the-loop mechanisms provide the necessary stability and semantic grounding required to transition from domain-specific solutions to the universal, scalable causal foundation models explored in the next section, where maintaining causal identifiability amidst super-exponential search spaces will depend heavily on the structural priors and validation principles established here.

8.5 Scal

8.5 Scalability Challenges and the Path to Universal Causal Foundation Models

Building upon the collaborative human-AI frameworks discussed in the previous section, the transition from validated, domain-specific causal models to universally applicable foundation models represents one of the most formidable frontiers in artificial intelligence. While the integration of expert knowledge stabilizes inference in high-stakes environments, scaling these approaches to the complexity of real-world systems—characterized by high-dimensional data, dynamic non-stationarity, and intricate feedback loops—remains an open challenge. The core difficulty lies not merely in computational throughput but in maintaining causal identifiability and structural consistency as the dimensionality of the state space expands. As noted in the exploration of scalable benchmark frameworks for supercomputers, verifying whether a system can achieve its desired goals becomes a tedious and complex task when the design involves a multi-parameter search space [513]. This observation is particularly pertinent to causal discovery, where the search space of potential Directed Acyclic Graphs (DAGs) grows super-exponentially with the number of variables, necessitating novel algorithmic paradigms that can balance computational efficiency with the theoretical guarantees established through human-in-the-loop validation.

A primary bottleneck in scaling causal representation learning is the computational cost associated with exploring the space of causal structures, especially when dealing with large-scale datasets or continuous optimization objectives. In the realm of reinforcement learning and sequential decision-making, which often serve as testbeds for causal agents, algorithms must efficiently navigate unknown environments without incurring prohibitive regret. Recent advancements in efficient exploration-exploitation strategies, such as those utilizing bias-span constraints, suggest that focusing on specific structural properties of the value function rather than the entire diameter of the Markov Decision Process (MDP) can yield significant improvements in scalability [514]. Extending this logic to causal discovery implies that future scalable models must leverage intrinsic structural sparsity or low-rank assumptions to prune the search space dynamically, rather than relying on exhaustive search or brute-force constraint testing. Furthermore, the development of variants that reduce computational complexity while maintaining theoretical regret bounds highlights the potential for designing causal learners that are both statistically sound and computationally tractable in high-dimensional settings [515].

Another critical aspect of scalability involves the integration of causal reasoning into large-scale generative models and foundation systems. The deployment of Large Language Models (LLMs) and other massive neural architectures has highlighted the tension between model size and deployment efficiency, particularly regarding memory requirements and computational ability. Techniques such as fine-grained post-training quantization have emerged as essential tools for compressing these models, allowing for the deployment of state-of-the-art performance on resource-constrained hardware [516]. For causal foundation models to scale effectively, similar compression and optimization strategies must be adapted to preserve the delicate causal relationships encoded within the model's weights. Simply quantizing a causal model without considering the sensitivity of causal mechanisms to precision loss could lead to the collapse of counterfactual reasoning capabilities. Therefore, future research must develop "causal-aware" quantization and pruning techniques that prioritize the preservation of invariant causal mechanisms over mere statistical fidelity.

Moreover, the scalability of causal discovery is intrinsically linked to the ability to handle sparse and structured data efficiently. In communication systems, the management of spectral efficiency through sparse code multiple access (SCMA) demonstrates how sparsity can be leveraged to enable near-optimal detection with manageable complexity, even when the number of multiplexed layers exceeds the dimension of codewords [517]. Analogously, scalable causal discovery algorithms must exploit the inherent sparsity of real-world causal graphs, where each variable typically interacts with only a small subset of others. By adopting message-passing algorithms and sparse attention mechanisms similar to those used in efficient decoding and multiple access schemes, causal learners can potentially scale to thousands of variables while maintaining robustness against noise and confounding factors [518]. The adaptation of such sparse processing techniques to the domain of causal graph learning could unlock the ability to model complex systems like climate dynamics or whole-brain connectivity, which are currently beyond the reach of dense matrix operations.

The challenge of scaling also extends to the evaluation and validation of causal models in indefinite and evolving data regimes. As systems grow in complexity, the assumption of static causal structures often fails, requiring models that can adapt to non-stationary distributions and regime shifts. The development of adaptive decoding strategies in communication theory, which adjust list sizes and decoding paths based on channel conditions, offers a conceptual blueprint for adaptive causal learners [519]. A scalable causal foundation model must similarly possess the agility to re-evaluate and update its structural hypotheses in real-time as new data streams in, without requiring a complete retraining from scratch. This necessitates the creation of lifelong learning frameworks that can accumulate causal knowledge across domains while mitigating catastrophic forgetting.

Finally, the path toward universal causal foundation models requires a paradigm shift that synthesizes algorithmic scalability with the robust knowledge-integration principles outlined previously. Fully automated discovery at scale is prone to hallucinations and spurious correlations, particularly in high-stakes domains like healthcare and scientific discovery. The success of automated side-channel analysis frameworks, which combine heuristic search with rigorous statistical testing to identify leakage points efficiently, underscores the value of hybrid approaches that guide automated search with structural priors [520]. Future scalable causal AI must therefore evolve into a collaborative entity, capable of incorporating expert constraints and validating its discoveries against established scientific principles. By bridging the gap between data-driven scalability and knowledge-guided robustness, the next generation of causal foundation models will be able to tackle the "dark common sense" reasoning required for true cognitive intelligence, moving beyond pattern recognition to genuine understanding of the mechanisms governing our world. The journey toward this goal demands not only algorithmic innovations in optimization and sparsity but also a fundamental rethinking of the architecture of intelligence itself, ensuring that as models grow larger, their causal reasoning capabilities grow deeper and more reliable.

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